Tag Functions and Their Applications to Lattice-based Signatures and IBEs — Compact Designs and Tighter Security

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Abstract. The existing lattice-based signature and IBE schemes suffer from the non-compactness of public keys or larger reduction loss in the security analysis. Thus we solve and improve those deficiencies as follows:

– First, we construct a lattice-based short signature scheme with a compact verification key in the standard model based on the ring short integer solution (RSIS) assumption. Under the same compactness, the ring modulus of our signature scheme is significantly smaller than the compact signature scheme of Alperin-Sheriff (PKC 2015). More importantly, our signature scheme achieves better reduction loss than all the previous confined guessing-based signatures. In other words, our signature scheme achieves better security and efficiency simultaneously.
– Secondly, we further design a short signature scheme with a nearly compact public key size and an even smaller reduction loss. Our second signature scheme achieves even better reduction loss than our first signature scheme yet at the cost of increasing the public key to a super-constant number of ring vectors.
– Last but not least, we construct an adaptively secure compact IBE scheme from the lattice assumptions and the truncation collision-resistant hash functions (TCRHF) introduced by Jager and Kurek (ASIACRYPT 2018). Note that the previous TCRHF-based IBE schemes are not even close to compactness.

The above improvements mainly benefited from our compact design of the tag functions and their more compact homomorphic evaluations. We also believe that our newly designed tag function may find new applications in designing other cryptographic schemes, like ABE and others.

Keywords: Lattices · Signatures · IBE

1 Introduction

To achieve stronger security, the heuristic cryptographic ideal model—random oracle (RO) is introduced. However, there is no concrete instantiation of RO in the real world. Nonetheless, hash functions are used as a plausible instantiation of RO to obtain a better efficiency, yet there exists a cryptographic scheme for which any such instantiation yields security collapse [15]. Therefore, security in the standard model is more plausible for cryptographic schemes. In this paper, we focus on construction of two types of cryptographic schemes in the standard model: signature and identity-based encryption (IBE). Since the adaptive security is a more realistic security notion, we focus on adaptive security of the above two schemes.

Although there are many classic constructions [10, 26, 30, 31, 48, 50, 53] (and many others) from classic number-theoretic hard problems, the vulnerability for the quantum attacks [51] causes serious concern. However, many plausible facts show that lattice-based schemes have many advantages over other post-quantum candidates (e.g., hash-based and multivariate-based) if considering both theoretical security and practical efficiency. Thus we focus on constructing lattice-based schemes.

Signature constructions. Theoretically, cryptographic signature schemes can be constructed from any one-way functions [49], yet this kind of lattice-based constructions [41] suffer from exceptionally large parameters. The mainstream lattice-based signature constructions are based on two types of hard problems: Short Integer Solution (SIS) problems initiated from the seminal work of Ajtai [5] and NTRU Problems [28].

Although the NTRU-based signature schemes [27, 29] are comparable with the classic scheme upon efficiency, yet the security is a great concern [23, 25, 44, 52]. Meanwhile, the SIS-based signature schemes try to balance efficiency and security. From the practical point of view, the SIS-based signature schemes can be divided into two categories: the Full Domain Hash (FDH) paradigm and the Fiat-Shamir (F-S) paradigm. The FDH type signature schemes [24,43] strongly rely on the efficiency of lattice trapdoors [6,8,17,22,43]. Although the F-S type signature schemes [19,20,39–41,46] avoided the inefficiency of the lattice trapdoors, yet these schemes still suffer from the inefficiency of rejection sampling. Nonetheless, the security analyses of the above two types of constructions are based on the random oracle, an ideal cryptographic model to achieve better practical efficiency.

One way to construct adaptively secure signature in the standard model is to use the lattice-based IBE schemes [1–3,12,36,55]. However, this approach suffers from large verification key sizes and computation complexities. Another approach is first to use the generalized confined guessing technique [10, 30, 31, 50] to construct a non-adaptively secure signature scheme, then use the lattice-based chameleon hash function [21,24] to construct fully secure signature schemes. Following the research line of confined guessing technique [10, 21], the work of [7] constructed a lattice-based short signature scheme with compact public keys in the standard model. Yet, due to its inherent structure, it suffers from a larger reduction cost and large public key size as follows:

- From the construction point of view, following the research line of confined guessing technique [10,21], the short signature scheme presented in [7] is the most compact scheme so far. However, the work of [7] left the construction of short signatures with short public keys with less trapdoor growth as an open problem. Note that smaller trapdoor growth leads to smaller parameter sizes in the scheme, and thus improves the practicality of the signature schemes. To our knowledge, there has been no significant improvement/construction on this research line for many years until this work.
- From the reduction point of view, all the confined guessing-based constructions suffer from a constant parameter. More precisely, the reduction loss of confined guessing-based lattice signature schemes [7, 10, 21] is $O\left((Q^2/\epsilon)^c\right)$ for some constant $c > 1$. From the previous literature, it is a great challenge to remove the constant c from the reduction cost if using the confined guessing technique. This problem has seen no significant progress for nearly a decade.

On one hand, to achieve a short signature with compact public keys, we seemingly need a confined guessing technique [7]. On the other hand, using the confined guessing technique leads to a larger security reduction loss. In general, it is challenging to construct a short signature scheme with short public keys, with less trapdoor growth and a smaller reduction cost.

Identity-Based Encryption. The first lattice-based IBE scheme is constructed in [24] through the pre-image sampling technique, yet the construction is in the random oracle model. Although the works [3, 16] firstly constructed the lattice based IBE in the standard model, the master public key of these schemes consists of $O(\ell)$[1] random matrices, and thus the large public key size restricts the wider applications of the IBE scheme. Although there are a series of sequential works [1, 9, 36, 37, 54–56] using the partition technique to reduce the public parameter size of lattice-based IBE schemes, the most recent work [2] in this research line presented a lattice-based short IBE scheme in the ring setting, yet their IBE scheme still needs at least $\omega(1)$ ring vectors in master public keys, which is super constant! On one hand, the complexity leveraging argument or the bootstrapping technique [14, 18] to transform selectively secure IBE into an adaptively secure one. However, they either increase the reduction lost of 2^{ℓ} or depends on strong cryptographic tools, like, garbled circuits. On the other hand, the work of [34] attempts to construct a compact lattice-based IBE scheme from TCRHF introduced in [33]. However, their construction need $\omega(log(\lambda))$ matrices in the mpk. This result (even using the TCRHF) still more worse than the construction of [2] (which is in the standard model). Note that the IBE scheme of [2] achieves almost compactness[2], thus the natural question comes to mind is:

Can we construct a *compact* adaptively secure IBE scheme from standard lattice assumptions and TCRHF ?

1.1 Our Contributions

Contributions in Signature. Motivated by the above challenges, we construct a lattice-based short signature scheme with short public keys and with a smaller reduction cost. We overcome the above elusive dilemma by designing a new tag function. Benefiting from the compact design of our tag function, our main contributions to lattice-based digital signatures are as follows.

- From the construction point of view, we design a short signature scheme with short public keys and smaller public parameters. Benefiting from the creative design and compact homomorphic evaluation of our tag function, we significantly improve the trapdoor growth (ring modulus q) in [7], and thus obtain smaller public keys.
- From the reduction point of view, our scheme enjoys tighter security analysis. we remove the seemingly unavoidable constant (which is a trades off between the public key size and the reduction loss in confined guessing-based signatures) in the security loss and thus

[1] ℓ denotes the length of the identity
[2] We call an adaptively secure IBE scheme is almost compact if the master public key size of the underlying scheme is about $\omega(1)$ times the mpk size of the selectively secure IBE scheme.

obtained a significant improvement over all previously constructed confined guessing-based lattice-based short signature schemes. More precisely, the reduction loss of our signature scheme is $\Omega(Q^2/\epsilon)$, which is significantly better than the reduction loss of $\Omega((Q^2/\epsilon)^c)$ achieved by Ducas and Micciancio (CRYPTO 2014) and all other confined guessing-based signatures
- We further design a short signature scheme which achieves better security analysis than our first scheme at the cost of increasing the public key size. Our second signature scheme achieves an even significantly smaller reduction loss than our first construction, yet the public key includes an additional $\omega(1)$ ring vectors.

In Table 1, we summarize our results and compare our results with previous typical constructions. It is easy to see from Table 1 that our first signature scheme achieves better reduction loss and the better SIS parameters simultaneously compared with previous compact short signature of [7]. For the schemes based on plain SIS problems, to be fair, we considered the ring variant of them. Due to the cost of large public key sizes (need for super-polynomially large modulus and a large number of matrices in the public keys), we didn't list those (almost) tight constructions of lattice-based signatures in the standard model [13,38]. In addition, the signature scheme of Zhang et al. [56] having the reduction loss $O(Q^2 \cdot \lambda)$ and SIS parameter $O(\lambda^{5.5})$ only achieves Q-bounded security, i.e., that the security of the scheme is not guaranteed any more if the number of signature queries that the adversary makes exceeds Q, where Q is a parameter that *must* be determined at the setup phase of the scheme. Thus, we didn't list it in Table 1. The one-time signature schemes, e.g., [41], are also not included in Table 1.

Notice that asymptotically comparing the ring vectors is plausible. Since the ring modulus q is polynomial, any ring vector in $R_q^k = (\mathbb{Z}[x]/(x^n+1))^k$ is of bit length $kn \cdot \lceil \log(q) \rceil$ which is asymptotically $O(kn \log(n))$. Therefore, in Table 1, we asymptotically compare the number of vectors in different aspects.

Contributions in IBE. Following the research line of partition-based constructions, we obtain an adaptively secure compact IBE scheme from ring learning with errors (LWE) problems and TCRHF. Namely, the mpk in our IBE scheme only includes a constant number of ring vectors, which is comparable with the selectively secure IBE construction of [4]. Compared with the previous TCRHF-based IBE scheme of Jager et al [34], in which the master public key includes $\tilde{O}(\log(\lambda))$ matrices (or ring vectors if considering in the ring setting), our scheme only needs 3 ring vectors in the master public key. Furthermore, our IBE scheme shares relatively small asymptotic parameters than the previous almost compact IBE construction of [2]. To achieve this compact IBE scheme, we combine the more compact homomorphic equality test function given by Abla et al. [2], prefixing idea of [16] and the TCRHF introduced by [33].

In Table 2, we summarize our results and compare our results with the previous partition based *typical* constructions. For the IBE schemes based on plain LWE problems, to be fair, we considered the ring variant of them. Due to the cost of large public key sizes (need for super-polynomially large modulus and a large number of matrices in the master public

Table 1. Lattice–based Adaptively Secure Signature Schemes in the Standard Model.

Scheme	# of ring vectors in vk	# of ring vectors in sk	# of ring vectors in signatures	Reduction loss	SIS Param β
[16]	n	n	n	Q	$\tilde{\Omega}(n^{3/2})$
[12, 43]	n	n	1	Q	$\tilde{\Omega}(n^{7/2})$, $\tilde{\Omega}(n^{5/2})$
[11]	1	1	$d^{\dagger}$	$(Q^2/\epsilon)^{c^{\sharp}}$	$\tilde{\Omega}(n^{5/2})$
[21]	d	1	1	$(Q^2/\epsilon)^c$	$\tilde{\Omega}(n^{7/2})$
[35]	n	1	1	(Q/n)	$\tilde{\Omega}(n^{7/2})$
[35]	$\log(n)$	1	1	$(Q^2/n)^c$	$\tilde{\Omega}(n^{7/2})$
[7]	1	1	1	$(Q^2/\epsilon)^c$	$\tilde{\Omega}(d^{2d} \cdot n^{11/2})$
Ours I	1	1	1	Q^2/ϵ	$\tilde{\Omega}(n^{9/2})$
Ours II	$\omega(1)$	1	1	$Q/\omega(\log(n))$	$\tilde{\Omega}(n^{9/2})$

Notations: pk, sk denotes the verification key and secret key of signature schemes. β and n denotes SIS parameter and ring dimension. Q, ϵ denotes the number of signature queries and the advantage of the adversary against the signature scheme.

($\sharp$) $c > 1$ is a parameter which trade-offs between public key size and the reduction loss.

($\dagger$) d is a value satisfying $\frac{2Q^2}{\epsilon} < 2^{\lfloor c^d \rfloor}$ for an arbitrary constant $c > 1$

keys), we didn't list those (almost) tight constructions of lattice-based IBEs in the standard model [13,38].

Table 2. Comparison with Lattice Based IBE Schemes in the Ring Setting.

Scheme	# of ring vectors in mpk	# of ring vectors in $\mathsf{ct}/\mathsf{sk}_{\mathsf{id}}$	R-LWE Param $\frac{1}{\alpha}$
[3]	$O(\lambda)$	$O(1)$	$\tilde{O}(n^{3.5})$
[37]	$O(\lambda^{1/\mu})^{\dagger}$	$O(1)$	$O(n^{0.5+2\mu})$
[55] I + [36]	$\omega(\log^2(\lambda))$	$O(1)$	$\tilde{O}(n^{5.5})$
[55] II	$\omega(\log(\lambda))$	$O(1)$	$\mathsf{poly}(n)^{*}$
[34]	$O(\log(\lambda))$	$O(1)$	$\tilde{O}(n^{6})$
[1]	$\omega(\log(\lambda)/\log\log(\lambda))$	$O(1)$	$O(n^{8.5})$
[2] A	$\omega(\log(\lambda))$	$O(1)$	$O(n^{4.5+\frac{4}{\kappa}})^{\dagger\dagger}$
[2] B	$\omega(1)$	$O(1)$	$\tilde{O}(n^{7.5+\frac{4}{\kappa}})$
Ours	3	$O(1)$	$\tilde{O}(n^{5.5})^{\ddagger}$

Notations: $\mathsf{mpk}, \mathsf{ct}, \mathsf{sk}_{\mathsf{id}}$ denotes the master public key, ciphertext and secret key of identity id. α, λ and n denotes R-DLWE parameter, security parameter and the ring dimension.

(‡) This value can be reduced to $\tilde{\mathsf{O}}(\mathsf{n}^{4.5})$ if we consider ours IBE scheme in the *crs* model as in [2] A.

$*$ $\mathsf{poly}(n)$ denotes some fixed but large polynomial. It is hard to determine an explicit bound for comparison due to the implicit construction of the work.

† $\mu \in \mathbb{N}$ is a constant that can be chosen arbitrary. Since the reduction cost is exponential in μ, this value typically set very small (e.g., $\mu = 2$ or 3).

†† $\kappa \geq 1$ can be any constant that satisfies $n^{\frac{1}{\kappa}} > 3 + \kappa$, e.g., 2 or 4, depending on how we set parameters of the underlying error correcting code.

1.2 Overview of Our Techniques

In this section, we overview our high-level construction ideas both from the point of design approach and security reduction approach.

The Signature Design

Research Frontlines. First, we recall the non-adaptive lattice-based signature scheme of [21]. The public parameters are $\mathsf{vk} = \{\mathsf{a}, \mathsf{a}_0, \mathsf{a}_1, \cdots, \mathsf{a}_d, \mathsf{u}, v\}$, where $\mathsf{a}_0, \mathsf{a}_1, \cdots, \mathsf{a}_d, \mathsf{u}$ are random vectors sampled over R_q^k and $\mathsf{a} \in R_q^k$ is generated by trapdoor generation algorithm with the trapdoor $\mathsf{R} \in R_q^{k\times k}$. For any tag $t \in \mathcal{T}_d$ [3], the public parameters implicitly define the tag matrix $\mathsf{a}_t := [\mathsf{a}^\top | \mathsf{a}_0^\top + \sum_{i=1}^{d} t_{[i]} \mathsf{a}_i^\top]^\top$. The signature algorithm uses the trapdoor of a to generate the corresponding signature (Gaussian ring vector over a q-ary lattice) by a ring variant of the pre-image sampling algorithm [43]. In the non-adaptive security proof,

[3] The tag space $\mathcal{T}_i$ is the set of bit strings defined by two real constants $c > 1$ and $\alpha > \frac{1}{c-1}$ as $\mathcal{T}_i := \{0,1\}^{c_i}$ for $i \in \{1, 2, \cdots, d\}$, where $c_i := \lfloor \alpha \cdot c^i \rfloor$ and $d := \lfloor \log_c(\frac{n}{2\alpha}) \rfloor$. For each tag $t = \{t_0, t_1, \cdots, t_{c_d - 1}\} \in \mathcal{T}_d$, we denote the prefix of t of length c_i as $t_{\leq i}$, and identify it as ring elements with corresponding binary coefficients, furthermore let $t_{[i]} := t_{\leq i} - t_{\leq (i-1)}$

the simulator will replace each a_i with $\mathsf{a}^\top \cdot \mathsf{R}_i + h_i(t^*, i^*) \cdot \mathsf{g}$, where the function $h_i(t^*, i^*)$, indexed by $i \in [d]$, is defined as follows:

$$h_i(t^*, i^*) = \begin{cases} -t^*_{\leq i^*} & i = 0 \\ 1 & 0 < i \leq i^* \\ 0 & i^* < i \end{cases}.$$

Therefore, for a tag $t \in \mathcal{T}$, the tag matrix $\mathsf{a}_t = [\mathsf{a}^\top | \mathsf{a}^\top \cdot \mathsf{R}^* + F_t(t^*, i^*) \cdot \mathsf{g}]^\top$ for $\mathsf{R}^* := \mathsf{R}_0 + \sum_{i=1}^{d} t_{[i]} \cdot \mathsf{R}_i \in R_q^{k \times k}$ and $F_t(t^*, i^*) := \sum_{i=1}^{d} h_i(t^*, i^*) \cdot t_{[i]} - t^*_{\leq i^*} \in R_q$. It is not hard to see that the simulator can generate a legal signature using the trapdoor information R^* if the message tag t doesn't have the same i^*-prefix as the tag t^*. Namely, if $F_t(t^*, i^*) \neq 0$, the simulator is able to generate a valid signature. For the case of $F_t(t^*, i^*) = 0$, the simulator can embed SIS challenges by the uniformity of the ring vector u and the ring element v. It is obvious from the above construction that the public key consists of $d + 2$ ring vectors. However, to satisfy the confined guessing stage of the security proof, the value d is about $d = \lfloor \log_c(\frac{n}{2\alpha}) \rfloor = O(\log(n))$, which is super constant! Thus, the underlying scheme is not a compact signature.

Following the idea of above confined guessing construction, Alperin-Sheriff [7] reduced the public parameters via re-formulating the function $F_t(t^*, i^*)$ as $F_t(t^*, i^*) = \sum_{i=1}^{d} p_{i^*}(i) \cdot t_{\leq i} - t^*_{\leq i^*}$, where $p_i(\cdot) : \mathbb{Z}_q \to \mathbb{Z}_q$ is a polynomial of degree $d-1$ such that $p_i(x) := \begin{cases} 1 & 1 \leq x \leq i \\ 0 & i < x \leq d \end{cases}$. Since the Z_q is a field for some prime modulus q, the polynomial $p_i(\cdot)$ can be realized as unique Lagrange interpolation. Therefore, there are d^2 coefficients for all the d polynomials $p_i(\cdot)$'s. Thus, as a part of the public parameters, this increases the public parameters about d^2 elements in $\mathbb{Z}_q$ meanwhile decreasing the random matrix number to $O(1)$.

Challenges. Although the work [7] decreased the number of random matrices to constant, the construction increases the R-SIS parameter[4] β as large as $\Omega\left(d^{2d} \cdot n^{5.5}\right)$, where d is the value satisfying $\frac{2Q^2}{\epsilon} < 2^{\lfloor c^d \rfloor}$. In the ring setting, as the work [7] stated, if set $d = \omega(\log(n))$, then we come up with super polynomial modulus in the scheme, which results in larger public key sizes in bits compared with the short signature with non-short public keys. Therefore, the work [7] suggested to set $d = O(\frac{\log(n)}{\log\log(n)})$. However, in this case, the modulus still suffers from the seemingly unavoidable factor d^{2d} due to the design difficulty of tag functions or their homomorphic evaluations. Below, we show how we use our new design of tag functions to avoid this multiplicative factor in the R-SIS parameter and obtain an even smaller R-SIS parameter β.

Our Techniques. In order to overcome the above challenges, we reformulated the tag function more intuitively via the equality test functions. The equality test function Equal_i : $[n] \to \{0, 1\}$ is a two-valued function that outputs 1 if the function index i equals the input

[4] In the SIS problems, β evaluates the hardness of SIS problems. The larger the beta, the smaller the bit security of SIS problems.

value and outputs 0 otherwise. Following the confined guessing techniques from [7, 21], we define the tag function $F_t(t^*, i^*)$ as follows:

$$F_t(t^*, i^*) := -t^*_{\leq i^*} + \sum_{j=1}^{d} \mathsf{Equal}_j(i^*) \cdot t_{\leq j}. \tag{1}$$

It is not hard to see that the function value $F_t(t^*, i^*)$ is equal to 0 if and only if the tag t has the same prefix, of length c_{i^*}, with the tag t^*. Furthermore, the previous work [2] showed that the expanding factor of the homomorphic evaluations of the equality test functions is independent of the value d as long as $d < n$. Therefore, from the fact that $\mathsf{deg}(t_{\leq j}) \leq d$, we know that the expanding factor of homomorphic evaluation of $\mathsf{Equal}_j(i^*) \cdot t_{\leq j}$ includes a multiplicative factor d (comparing with multiplicative d^d factor in the work of [7]). Hence the above description of equation (1) shows that the expanding factor of the homomorphic evaluation of the tag function F_t includes a multiplicative d^2 factor. Thus we can set $d = \omega(\log(n))$ without incurring super polynomial modulus, even without increasing the modulus asymptotically !

Note that if we use the common reference model as stated in the work of [2], the R-SIS parameter β of our scheme is as small as $\tilde{\Omega}(n^{9/2})$ which is even asymptotically, significantly smaller than the explicit part of the previous work [7]. Therefore, our reconstruction of tag functions not only allows us to construct a short lattice-based signature scheme with short public keys and avoid the multiplicative d^d factor in the R-SIS parameter β, but also results in less trapdoor growth in the scheme.

The Signature Reduction

Research Frontlines. Let $\mathcal{T}_i$ be the tag space defined by two real constants $c > 0$ and $\alpha > \frac{1}{c-1}$ as $\mathcal{T}_i := \{0,1\}^{c_i}$ for $c_i := \lfloor \alpha \cdot c^i \rfloor$, and let $\mathcal{T}$ be the full tag space $\mathcal{T}_d$. For a full tag $t \in \mathcal{T}$, $t_{\leq i} \in \mathcal{T}_i$ denotes the prefix of t of length c_i. Below we show how the constant c trade-offs the security loss and the size of public parameters.

In the security proof of non-adaptive confined guessing-based signature schemes [7, 21], the simulator chooses Q random tags $\{t^1, \cdots, t^Q\}$ from $\mathcal{T}$ for corresponding Q messages $\{\mu^1, \cdots, \mu^Q\}$, and chooses an index i^* with the hope that the probability of any two different tags in $\{t^1, \cdots, t^Q\}$ have the same prefix of length c_{i^*} is no larger than $\epsilon/2$. From the randomness of the tags, it is enough to set $i^* \in [d]$ as the smallest index such that $|\mathcal{T}_{i^*}| \geq \frac{2Q^2}{\epsilon}$, in this case, we have that $\frac{2Q^2}{\epsilon} > 2^{c_{i^*-1}}$

In addition, if there is no collision in the previous tag sampling stage, the simulator chooses another random tag $t^*_{\leq i^*} \in \mathcal{T}_{i^*}$ with the hope that the adversary outputs a valid signature with a tag $t^f \in \mathcal{T}$ such that $t^*_{\leq i^*} = t^f_{\leq i^*}$. In this case, the simulator can, with overwhelming probability, solve the corresponding RSIS problems. Note that from the randomness of c_{i^*} prefix of $t^*_{\leq i^*}$, the probability of $t^*_{\leq i^*} = t^f_{\leq i^*}$ is at least $\frac{1}{\mathcal{T}_{i^*}}$. Thus the success probability of the simulator is at least $\frac{\epsilon}{2} \cdot \frac{1}{\mathcal{T}_{i^*}}$.

From the definition of c_i above, we have that $c_{i^*} \leq \alpha \cdot c^{i^*} = c \cdot (\alpha \cdot c^{i^*-1}) \leq c \cdot (c_{i^*-1} + 1)$, and thus we have followings for the cardinality of $\mathcal{T}_{i^*}$

$$|\mathcal{T}_{i^*}| = 2^{c_{i^*}} \leq 2^{c \cdot (c_{i^*-1}+1)} \leq \left(\frac{4Q^2}{\epsilon}\right)^c.$$

Thus the reduction loss of the reduction is at most $\Omega\left(\left(\frac{4Q^2}{\epsilon}\right)^c\right)$.

Challenges. Note that for larger constant c, the value $|\mathcal{T}_{i^*}|$ is larger than $\frac{2Q^2}{\epsilon}$ for smaller index i^*. Since $i^* \in [d]$, we have smaller d for larger constant c. Therefore, it results in shorter public keys, yet increases the reduction cost. To decrease the reduction loss, one can set c be a small number. However, for the smaller constant number c, the value $|\mathcal{T}_{i^*}|$ grows very slowly. Therefore, to satisfy $|\mathcal{T}_{i^*}| \geq \frac{2Q^2}{\epsilon}$, the simulator have to choose larger index i^*. As a result, smaller c increases the full tag size d, and thus increases the public keys. From the reduction point of view, the major challenge in confined guessing-based signature schemes is to remove the constant c from the reduction cost without affecting the other parameters. Below, we show how we overcame this dilemma.

Our Techniques. An intuitive observation is that the length of tags in the i-th sub-tag space $\mathcal{T}_i$ is approximately the c times larger than the tags in $\mathcal{T}_{i-1}$. In particular, if set $c \gtrapprox 1$ [5], then the tag length of $\mathcal{T}_i$ is approximately one bit larger than the tag length of $\mathcal{T}_{i-1}$[6] . In other words, for the constant $c \gtrapprox 1$, the sub-tag space $\mathcal{T}_i \approx \{0,1\}^i$. An important advantage of this type of setting is that we can upper bound the volume $|\mathcal{T}_i|$ only in terms of its lower bound. Namely, if we have $B \lessapprox |\mathcal{T}_i|$ for some integer B, then we have $|\mathcal{T}_i| \lessapprox B \cdot 2$.

More formally, if we set $\mathcal{T}_i = \{0,1\}^i$ in the confined guessing-based lattice-based signature scheme, the choice of i^* tells that $\frac{2Q^2}{\epsilon} \leq |\mathcal{T}_{i^*}| = 2^{i^*} \leq 2 \times \frac{2Q^2}{\epsilon}$, and thus the success probability of the simulator, denoted ϵ_{SIS}, is

$$\epsilon_{\mathsf{SIS}} \geq \frac{\epsilon}{2} \cdot \frac{\epsilon}{4Q^2} = \Omega\left(\frac{\epsilon^2}{Q^2}\right),$$

except with negligible probability. The setting of $\mathcal{T}_i = \{0,1\}^i$ may incur large public parameters due to the fact that the tag functions in the previous literature [21] strongly rely on the number of sub-tag spaces. As we discussed in the construction idea of our tag function, our design is robust to the number of sub-tag spaces. Thus above setting does not affect the number of public keys of our short signature scheme. In general, we have a short signature scheme with short public keys with smaller parameters and, more importantly, with tighter security.

The Revised Signature Scheme

From the above construction idea, we observe that the success probability of the simulator is $\epsilon_{\mathsf{Sim}} \geq (\epsilon - \mathsf{Pr}[\mathsf{coll}]) \cdot \frac{1}{|\mathcal{T}^*|}$, where $\mathsf{Pr}[\mathsf{coll}]$ is the random tag collision probability. Thus, if

[5] We use the notion $a \gtrapprox b$ to denote that $a > b$ and a is very close to b.
[6] Note that tags in $\mathcal{T}_i$ should be at least one bit larger than the tags in $\mathcal{T}_{i-1}$.

we set $(\epsilon - \mathsf{Pr}[\mathsf{coll}]) \geq \epsilon/2$, then the reduction loss becomes approximately $\frac{1}{|\mathcal{T}^*|}$. Therefore, to achieve better reduction loss, we need to set the tag space size $|\mathcal{T}^*| = 2^{i^*}$ (cardinality of $\mathcal{T}^*$, and thus the index i^*) as small as possible on condition on $\mathsf{Pr}[\mathsf{coll}] \leq \epsilon/2$.

We observe that increasing the collision capacity can achieve our goal. Consider a $\mathsf{L}+1$ collision of i^*-prefixes of random tags, we have the following for the collision probability

$$\mathsf{Pr}[\mathsf{coll}] = \binom{Q}{\mathsf{L}+1} \cdot \frac{1}{(|\mathcal{T}^*|)^{\mathsf{L}}} \leq \frac{Q^{\mathsf{L}+1}}{(\mathsf{L}+1)!} \frac{1}{(|\mathcal{T}^*|)^{\mathsf{L}}} \leq \frac{Q^{\mathsf{L}+1} e^{\mathsf{L}}}{(\mathsf{L}+1)^{\mathsf{L}+1}} \left(\frac{1}{|\mathcal{T}^*|}\right)^{\mathsf{L}},$$

where the constant e is the base of the natural logarithm. Thus by setting i^* be the largerst integer such that $i^* < \log(\frac{eQ}{\mathsf{L}+1} 2^{\frac{\mathsf{L}+\log(Q)}{\mathsf{L}}})$, we have $|\mathcal{T}^*| \leq \frac{2Q}{\mathsf{L}+1}$ for $\mathsf{L} \geq \omega(\log(n))$, and thus $\mathsf{Pr}[\mathsf{coll}] \lesssim \mathsf{negl}(\lambda) < \epsilon/2$. Furthermore, the corresponding reduction loss would become $\frac{eQ}{\mathsf{L}+1}$ (which is asymptotically $\Omega(\frac{Q}{\omega(\log(n))})$, and significantly better than $\frac{Q^2}{\epsilon}$ of the above scheme).

However, achieving the above better reduction loss is not for free. Note that increasing the number of i^*-prefix collisions results in a crux that there are at most L tags of which i^*-prefixes are matched with the challenge tag's i^*-prefix. Therefore the simulator has to generate corresponding signatures for such tags (corresponding messages). However, the simulators in the existing schemes are only able to simulate one signature for such tags.

We observe that one possible way to simulate multiple signatures is to increase the public ring elements. We modify the design framework of previous construction [21], such that the resulting scheme still shares a short signature and nearly compact property. For the detailed construction, please refer to the main body of this paper.

The IBE Design

Research Frontlines. Recall that in the adaptive security game of IBE scheme, the adversary has been allowed to make key extraction queries to KeyGen for any non-challenge identity id at any stage of the security game, thus the simulating algorithm should be able to respond with a corresponding legal secret key $\mathsf{sk}_{\mathsf{id}}$. Meanwhile, to reduce the security of the IBE scheme to the associated R-DLWE problem, the simulating algorithm needs to embed the R-DLWE challenges into the challenge ciphertexts of the IBE scheme. In general, the identity space is divided into two parts: for one part, the simulating algorithm can generate secret keys; and for the other part, it can generate challenge ciphertexts. If there is a keyed function $F(K, \cdot)$ which can split the identity space as desired, then the simulating algorithm may use this function to do the reduction. We call this function $F(K, \cdot)$ as a partition function. More compact (need fewer inputs) partition functions result in more compact IBE schemes. Partition-based IBE schemes, as its name indicates, use the above intuitive observation. Thus the most challenging part of this approach is to design a more compact partition function.

While Considering the above intuition in lattice settings as in [3], the simulating algorithm can generate the secret key $\mathsf{sk}_{\mathsf{id}}$ via the efficient trapdoor of [43] if the function value $F(K, \mathsf{id}) = 1$, and can embed the R-DLWE challenges into the challenge ciphertext of id^* if $F(K, \mathsf{id}^*) = 0$. The work of [55] generalized the above property of $F(K, \mathsf{id})$ as follows:

$$F\left(K, id^{(1)}\right) = 1 \wedge \cdots \wedge F\left(K, id^{Q)}\right) = 1 \wedge F\left(K, id^*\right) = 0,$$

where $\mathsf{id}^{(1)} \cdots \mathsf{id}^{(Q)}$ are identities that the adversary asked for key extraction, and id^* is the challenge identity. As shown in [55], a lattice-based IBE scheme can be constructed by (1) designing a partition function with the above property[7], and (2) designing efficient homomorphic evaluation algorithms for the partition function. Following this idea, The follow-up work [2] showed that pairwise independent hash functions suffice for partitioning, and constructed a partition function from error correcting code (ECC) by repeating the basic hash function $H_{\alpha,\beta}(\mathsf{id}) := \mathsf{ECC}(\mathsf{id})[\alpha]+\beta \in \mathbb{Z}_p$ for $t = \omega(1)$ times[8] .Then they considered the whole design in the ring setting, and presented an efficient homomorphic evaluation of the partition function. Although the recent work of [34] constructed a partitioning function from LWE and an additional so called TCRHF, the IBE scheme of [34] still contains $\omega(\log(\lambda))$ matrices in the mpk (ring vectors in the ring setting). Thus, the state-of-the-art IBE constructions still ubable to construct compact IBE even using LWE and TCRHF.

Our Techniques. We observe that the previous mainstream designs [1,3,9,16,36,37,54–56] mainly based on the a partition function with an *explicitly* given function key. But the recent construction of the partitioning function given in [34] can be seen as the composition of pairwise independent key-less public hash function Hash (TCRHF in the construction) and another keyed function $f_K(*)$. Namely, $F(K, \mathsf{id}) := f_K(\mathsf{Hash}(\mathsf{id}))$. Another important observation is that the truncation collision resistant property of the TCRHF is exactly matches the prefixing idea of the tag function that we defined in Equation 1. Thus, combining our tag function with TCRHF, for a key $K = (t^*, i^*) \in \{0,1\}^d \times [d]$, we have a partition function $F(K, \cdot)$ defined as follows

$$F(K, \mathsf{id}) := \mathsf{Hash}(\mathsf{id})_{\le i^*}(x) - t^*_{\le i^*}(x) \in R_q,$$

where Hash is a truncation collision-resistant hash function [9], $t^*_{\le i^*}(x), \mathsf{Hash}(\mathsf{id})_{\le i^*}(x)$ are the binary coefficient ring elements. Note that $F(K, \mathsf{id})$ is trinary polynomial, and thus for some special quotient ring [10], it is invertible if it is non-zero. Furthermore, the partitioning property can be shown from the truncation collision resistance property of the hash function Hash.

Let $F_t(\cdot)$ be the tag function defined in equation (1), then we have the equality $F(K, \mathsf{id}) = F_{\mathsf{Hash}(\mathsf{id})}(K)$. Thus homomorphic evaluation algorithms for the above partition function can be obtained from the homomorphic evaluation algorithms for the tag function. As benefited from the compactness of our tag function, our partition function also shares compactness. We also believe that our newly designed tag function may find new applications in designing other cryptographic schemes, like ABE and others.

Remark 1.1 *Our IBE scheme is partition based, and the construction uses TCR hash function. Thus the signature scheme from our IBE via the Naor-transform also needs TCR hash*

[7] We further need the partition probability to be noticeable, and can be covered in a sufficiently small interval

[8] The error correcting code is defined as $\mathsf{ECC} : \{0,1\}^\ell \to \mathbb{Z}_p^{\mathsf{L}}$ for some prime p and integer L, $\mathsf{ECC}(\mathsf{id})[\alpha]$ denotes the α-th element of the vector $\mathsf{ECC}(\mathsf{id})$ for $\mathsf{id} \in \{0,1\}^\ell$.

[9] more details can be fined in the following sections or in the work [34]

[10] Let $R_q = \mathbb{Z}[x]/(x^n + 1)$ for n a power of 2, and q be a prime such that $q = 3 \mod 8$, then the polynomial $a \in R_q$ with $\|\mathsf{Coeffs}(a)\| \le \sqrt{q}$ is invertible in R_q

function, and the public key size still contains constant number of ring vectors. However, our direct design of signature scheme follows the idea of confined guessing, and is free of TCR hash function. We also remark that the smaller public keys do come at the cost of an increased time complexity. Our signature scheme needs more homomorphic computations compared with the previous works [7, 21].

2 Preliminaries

Notations. We use $[k]$ to denote the index set $\{1, 2, \cdots, k\}$. We use lower-case bold letters to denote the vectors (e.g., a), and use upper-case bold letters to denote the matrices (e.g., A). For a probability distribution χ, we use $x \leftarrow \chi$ to denote that x is sampled from the distribution χ. For a set S, $x \xleftarrow{\$} S$ denotes that x is uniformly sampled from the set S. The 2-norm of a vector a is denoted by $\|\mathsf{a}\|$. For a square matrix T, we use $\|\mathsf{T}\|_{\mathsf{GS}}$ to denote the longest column of Gram-Schmidt orthogonalization of T and use $s_1(\mathsf{T})$ to denote the largest spectral norm of T. The function $\mathsf{negl}(\cdot)$ denotes the negligible function, that is for any constant c, there is an λ_0 such that for all $\lambda > \lambda_0$ we have $\mathsf{negl}(\lambda) < \frac{1}{\lambda^c}$. We define the statistical distance between two random variables X and Y over the support Ω as $\Delta(X, Y) := \sum_{s \in \Omega} |\Pr[X = s] - \Pr[Y = s]|$. For two probability distributions $\mathcal{P}$ and $\mathcal{Q}$, we say $\mathcal{P}$ is statistically close to $\mathcal{Q}$ if $\Delta(\mathcal{P}, \mathcal{Q}) \leq \mathsf{negl}(\lambda)$.

Signatures, IBE, Hash Functions and Partition Function. In this paper, we use the standard syntaxes for the signatures and IBEs. The corresponding correctness and security definitions can be found in Supplementary Material A.1 and A.2.

We use the chameleon hash to transform a non-adaptive secure signature into an adaptive one. The definition and desired security syntaxes can be found in Supplementary Material A.3. We also use the truncation collision-resistant hash function from [34] in our construction of a compact IBE scheme, the relevant definition and security notions are presented in Supplementary Material A.4.

Partition function, also known as a balanced admissible hash function [4,55] is vital to the security proof of partition-based IBE schemes. Due to the space limit, we defer the related definition of the partition function [55] and the related results in Appendix A.5.

2.1 Gaussians, Rings, Ring SIS

Lattices. For a full rank matrix $\mathsf{B} = [\mathsf{b}_1, \cdots, \mathsf{b}_n] \in \mathbb{Z}^{m \times n}$, the lattice generated by B is the set $\Lambda = L(\mathsf{B}) := \{\mathsf{B} \cdot \mathsf{c} = \sum_{i=1}^{n} c_i \cdot \mathsf{b}_i | \mathsf{c} = (c_1, \cdots, c_n) \in \mathbb{Z}^n\}$. This paper focuses on a particular family of so-called q-ary integer lattices for some integer q. For the positive integers n, m, q, a matrix $\mathsf{A} \in \mathbb{Z}_q^{n \times m}$ defines two type of following full-rank m-dimensional q-ary lattices as follows:

$$\begin{aligned} \Lambda^{\perp}(\mathsf{A}) :=& \{\mathsf{e} | \mathsf{A} \cdot \mathsf{e} = 0 \mod q\}, \\ \Lambda(\mathsf{A}) :=& \{\mathsf{e} | \exists \mathsf{z} \in \mathbb{Z}_q^n \ s.t., \mathsf{A}^{\top} \cdot \mathsf{z} = \mathsf{e} \mod q\} \end{aligned}$$

For any vector $\mathsf{u} \in \mathbb{Z}_q^n$, the "shifted" lattice $\Lambda_{\mathsf{u}}^{\perp}(\mathsf{A})$ is defined as $\Lambda_{\mathsf{u}}^{\perp}(\mathsf{A}) := \{\mathsf{e} | \mathsf{A} \cdot \mathsf{e} = \mathsf{u} \mod q\}$.

Gaussians. For a real $s > 0$, the n-dimensional gaussian function with parameter s is defined as $\rho_s(\mathsf{x}) := \exp(-\pi \frac{\|\mathsf{x}\|^2}{s^2})$. For any n-dimensional vector c, define the shifted gaussian function by $\rho_{s,\mathsf{c}}(\mathsf{x}) := \exp(-\pi \frac{\|\mathsf{x}-\mathsf{c}\|^2}{s^2})$. The discrete gaussian distribution over a lattice coset $\Lambda + \mathsf{u}$ is defined as $D_{\Lambda+\mathsf{u},s}(\mathsf{x}) = \frac{\rho_s(\mathsf{x})}{\rho_s(\Lambda+\mathsf{u})}$. For any positive real $\epsilon_s > 0$, the smoothing parameter of a lattice Λ, denoted $\eta_{\epsilon_s}(\Lambda)$, is the smallest real $s > 0$ such that $\rho_{1/s}(\Lambda^* \setminus \{0\}) \leq \epsilon_s$. We have the following tail bound for the Gaussian distribution over a lattice.

Lemma 2.1 ([24,43]) *Let $\Lambda \subset \mathbb{R}^n$ be a lattice and $s > \eta_{\epsilon_s}(\Lambda)$ for some $\epsilon_s \in (0, 1/2)$. For any $\mathsf{c} \in \mathrm{span}(\Lambda)$, we have*

$$\mathsf{Pr}\left[\|D_{\Lambda+\mathsf{c},s}\| \geq s\sqrt{n}\right] \leq 2^{-n} \cdot \frac{1+\epsilon_s}{1-\epsilon_s}.$$

Furthermore, if $\mathsf{c} = 0$, the bound holds for any $r > 0$ with $\epsilon_s = 0$.

Rings. We use R to denote the quotient ring $\mathbb{Z}[x]/\Phi_m(x)$ for m-cyclotomic polynomial $\Phi_m(x)$, and use R_q to denote R/qR for some modulus q. In this paper, we set $m = 2n$ for some n a power of 2, then the cyclotomic polynomial $\Phi_m(x) = x^n + 1$. The following lemmas show the invertibility of the ring elements with small degree and trinary coefficients or with relatively small coefficients.

Lemma 2.2 ([21]) *let $n \geq 4$ be a power of 2, $q \geq 3$ a power of 3, and set $R_q = \mathbb{Z}[X]/(\Phi_{2n}(X), q)$. Then, any nonzero polynomial $t \in R_q$ of degree $d < n/2$ and coefficients in $\{0, \pm 1\}$ is invertible in R_q .*

Lemma 2.3 ([37]) *Let q be a prime such that $q \equiv 3 \mod 8$ and n be a power of 2. Let $R_q = \mathbb{Z}_q[x]/\Phi_{2n}(x)$. Then, all $u \in R_q$ satisfying $\|\mathsf{Coeffs}(u)\|_2 < \sqrt{q}$ are invertible, i.e., $u \in R_q^*$.*

We identify a ring element as a vector in $\mathbb{Z}^n$ by the natural coefficient embedding $\mathsf{Coeffs} : R \to \mathbb{Z}^n$ that maps a ring element $a = \sum_{i=0}^{n-1} a_i \cdot x^i$ to the integer vector $\mathsf{Coeffs}(a) := (a_0, \cdots, a_{n-1}) = \mathsf{a}$. Note that benefited from the special setting above, the ring homomorphism $\mathsf{rot} : R \to \mathbb{Z}^{n \times n}$ maps an element in R to an anti-circulant matrix in $\mathbb{Z}^{n \times n}$. Namely, for a ring element $a \in R$, $\mathsf{rot}(a) := [\mathsf{Coeffs}(a)|\mathsf{Coeffs}(a \cdot x^1 \mod \Phi(x))| \cdots |\mathsf{Coeffs}(a \cdot x^{n-1} \mod \Phi(x))]^\top$. Furthermore, we define the map rot over ring vectors or matrices naturally by applying it entry-wise. In this paper, we use the notion $\mathsf{a} \leftarrow D^{\mathsf{Coeffs}}_{R^k,s}$ to denote the coefficients of the ring vector $\mathsf{a} \in R^k$ is sampled from the Gaussian distribution $D_{\mathbb{Z},s}$. For the sake of simplicity, we may omit the subscript, and simply use $\mathsf{a} \leftarrow D_{R^k,s}$.

We identify the norms of a ring element or ring vectors by the norm of corresponding coefficient embeddings. For a matrix R, we define the largest singular value, denoted $s_1(\mathsf{R})$, by $s_1(\mathsf{R}) = \max_{\|\mathsf{x}\|=1} \|\mathsf{R} \cdot \mathsf{x}\|$. For a set S, we use notion $x \leftarrow S$ to denote x is uniformly sampled from the set S. The following results are from [21], which we use several times in the security reduction of the signature scheme.

Lemma 2.4 (Fact 6 of [21]) *If $\mathsf{R} \leftarrow D_{R^{w \times t},s}$, then the following holds*

$$\mathsf{Pr}\left[s_1(\mathsf{R}) \geq s\sqrt{n} \cdot \mathcal{O}\left(\sqrt{w} + \sqrt{t} + \omega(\sqrt{\log(n)})\right)\right] \leq \mathsf{negl}(\lambda).$$

Lemma 2.5 (Lemma 7 and Corollary 8 of [21]) *Let $R_q = \mathbb{Z}_q[x]/(\Phi_{2n}(x))$ for $n \geq 4$ a power of 2 and $q = 3^k$ a power of 3. Let $k \geq 2\lceil \log(q) \rceil + 2$ and $s \geq \omega(\sqrt{\log(nk)})$. With overwhelming probability over the choice of $\mathsf{a} \leftarrow R_q^k$, if $\mathsf{x}_i \leftarrow D_{R,s}$ (for $i = 1, \cdots k$) are chosen independently at random, then the following holds:*

- *The sum $\sum_{i\in[k]} \mathsf{a}_i \mathsf{x}_i$ is within negligible statistical distance from uniform over R.*
- *For any nonzero vector $\mathsf{v} \in R^k$, the conditional min-entropy of $\sum_{i\in[k]} \mathsf{v}_i \mathsf{x}_i$ given $\sum_{i\in[k]} \mathsf{a}_i \mathsf{x}_i$ is at least $\Omega(n)$*

For a set $S \in \mathbb{Z}$, we use S_R (or R_S) to denote the set of ring elements in which the coefficients are in the set S. The notion $x \leftarrow R_S$ means that x is uniformly sampled from the set R_S. In particular, for an integer $\rho > 0$, we can bound the singular value of a random matrix chosen from $[-\rho, \rho]_R^{s\times t}$ as follows.

Lemma 2.6 ([37] Lemma 2) *For a positive integer ρ and a matrix R uniformly chosen from $[-\rho, \rho]_R^{s\times t}$, there exists is a constant C ($C \approx 1/2\pi$) such that for any positive number ω, we have*

$$\mathsf{Pr}\left[s_1(\mathsf{R}) \geq C\rho\sqrt{n}\left(\sqrt{s} + \sqrt{t} + \omega(\sqrt{\log(n)})\right)\right] \leq \mathsf{negl}(\lambda).$$

The following lemma is a ring analogue of the standard lattice regularity lemma, and it plays an important role in the security proof of our signature scheme.

Lemma 2.7 (Regularity Lemma [37]) *Let n be a power of two, q be a prime $q \equiv 3 \mod 8$. Let ℓ, k', ρ be positive integers that $\ell, k' \geq 1$ and $\rho < \frac{1}{2}\sqrt{q/n}$. Then for $\mathsf{A} \xleftarrow{\$} R_q^{k'\times k}$ and $\mathsf{X} \xleftarrow{\$} [-\rho, \rho]_R^{k\times \ell}$, we have*

$$\Delta\left((\mathsf{A}, \mathsf{AX}), (\mathsf{A}, U(R_q^{k'\times \ell}))\right) \leq \frac{\ell}{2} \cdot \left(\frac{q^{k'}}{(2\rho+1)^k}\right)^{n/2}.$$

Ring LWE and Ring SIS. The ring Learning With Errors (RLWE) problem, introduced in [42], is the ring version of the LWE problem [47]. Let R_q be the ring defined above, χ be a distribution over R, and $s \xleftarrow{\$} R_q$, then the R-$\mathsf{LWE}_{n,q,\chi}$ distribution is the distribution of the pair $(a, b) \in R_q \times R_q$, where a is uniformly sampled from R_q and $b = a \cdot s + e$ for some $e \leftarrow \chi$. Through this R-$\mathsf{LWE}_{n,q,\chi}$ distribution, given the bounded number of samples, the decision version of the RLWE problem, denoted R-$\mathsf{DLWE}_{n,\ell,q,\chi}$, is defined as follows.

Definition 2.8 *Given ℓ independent samples $(a_i, b_i)_{i\in[\ell]} \in (R_q \times R_q)^\ell$, the R-$\mathsf{DLWE}_{n,\ell,q,\chi}$ problem asks to decide if the samples are from the R-$\mathsf{LWE}_{n,q,\chi}$ distribution or from the uniform distribution over $(R_q \times R_q)^\ell$.*

Although there are several works [42, 45] on the hardness of RLWE problems for special kinds of distributions χ, yet in this paper, we consider the case in which χ is Gaussian distribution. In this paper, we use the R-DLWE hardness results and re-randomization results from from [37]. Due to the space limit, we defer them to Supplementary Material A.6.

Ring SIS. Lattice-based signature schemes are mostly based on the Small Integer Solution (SIS) problems or its ring version (R-SIS). We recall the ring SIS problem as follows.

Definition 2.9 (R-SIS) *Given a ring matrix $\mathsf{A} \in R^{1\times m}$, the R-$\mathsf{SIS}_{q,n,m,\beta}$ problem asks to find a nonzero vector $\mathsf{x} \in \Lambda^{\perp}(\mathsf{rot}(\mathsf{A}))$ such that $\|\mathsf{x}\| \leq \beta$.*

Trapdoors for Rings. For positive integers b and $k > k' \geq \lceil \log(q) \rceil$, let $\mathsf{g}_b^{\top} = [1|b|b^2|...|b^{k'}|0] \in R^k$ be the gadget matrix. As stated in the work of [43], this gadget matrix has a public trapdoor $\mathsf{T}_{\mathbf{g}_b}$ with small norm, i.e., $\|\mathsf{T}_{\mathbf{g}_b}\| \leq \sqrt{b^2+1}$. Using this trapdoor $\mathsf{T}_{\mathbf{g}_b}$, there are several useful lattice sampling algorithms [37, 43]. Due to the space limit, we defer these lattice sampling algorithms to Supplementary Material A.7.

2.2 Homomorphic Computations

We defined the encoding of a ring element α as $\mathsf{Encode}(\alpha) := \mathsf{a}^{\top} \cdot \mathsf{R} + \alpha \cdot \mathsf{g}_b^{\top}$, where the ring vector a is uniform over R_q^k and $\mathsf{R} \in R_q^{k\times k}$ is a small norm ring matrix. It is easy to see the homomorphic additivity and multiplicativity of this encoding. For a function $f : R^{\kappa} \to R$, the above encoding enables us to obtain an encoding of the function value $f(\boldsymbol{\alpha})$ for $\boldsymbol{\alpha} \in R^{\kappa}$. To evaluate the quality of a homomorphic evaluation procedure, we usually use the syntax of δ-expanding, the definition of which can be found in Supplementary Material A.8

Furthermore, the equality test function defined below plays an important role in our design of the tag function and its homomorphic evaluations.

Definition 2.10 (Equality Test Function) *Define function $\mathsf{Equal}_{\beta}(\cdot)$ parameterized by $\beta \in [m]$ as follows: on input $\alpha \in \mathbb{Z}$, the function outputs 1 if $\alpha \equiv \beta \mod m$ and 0 otherwise.*

Although the definition of the above equality test function takes input an integer α rather than the monomial x^{α} as in the work of [2]. Since the monomial x^{α} can be get from the integer α, the following theorem from [2] still works.

Lemma 2.11 ([2]) *There are algorithms $(\mathsf{PubEval}, \mathsf{TrapEval})$ such that they are $mn(kb)^2$-expanding in the plain model and $m\sqrt{nk}b^2k$- expanding in the CRS model with respect to the function family $\{\mathsf{Equal}_{\beta}(\cdot)\}_{\beta\in[m]}$.*

3 Tag Function and Its Homomorphic Evaluation

In this section, we first present our construction of the tag function, which is an important tool for our compact design of the signature. Then we show its homomorphic evaluations and related results.

3.1 Tag Function

Tag. Let $\mathcal{T} := \{0,1\}^d$ be the full tag space, and for the sake of description simplicity, we identify each tag $t = (t_0, t_1, \cdots, t_{d-1}) \in \mathcal{T}$ as a polynomial of degree $d-1 < \frac{n}{2}$ with binary coefficients from corresponding tag bits, such as $t(x) = \sum_{i=0}^{d-1} t_i \cdot x^i$, and we may use the notion interchangeably. We use $\mathcal{T}_i$ to denote the tag space in which the tags are of length i, namely $\mathcal{T}_i := \{0,1\}^i$. For a full tag $t = (t_0, t_1, \cdots, t_{d-1}) \in \mathcal{T}$, the prefix of t of length $i \in [d]$ is denoted by $t_{\leq i}$, and for simplicity we identify $t_{\leq i}$ as a ring element in R_q, e.g., $t_{\leq i}(x) := \sum_{j=0}^{i-1} t_j \cdot x^j$

Now, we formally present the definition of tag function which plays a significant role in our signature scheme construction and proof of security.

Definition 3.1 *For a full tag $t \in \mathcal{T}$, we define the tag function $F_t : \mathcal{T} \times [d] \to R_q$ as $F_t(t^*, i^*) := t_{\leq i^*}(x) - t^*_{\leq i^*}(x)$. The tag function family $\mathcal{F}$ is defined naturally as $\mathcal{F} := \{F_t | t \in \mathcal{T}\}$*

Although the definition of tag function is exceptionally simple, it possesses extremely useful properties as follows.

Lemma 3.2 *For any tag $t \in \mathcal{T}$ and the tag function $F_t \in \mathcal{F}$ defined as in Definition 3.1, we have the following two properties.*

- *For any tag $t^* \in \mathcal{T}$ and any index $i^* \in [d]$ such that $t^*_{\leq i^*} = t_{\leq i^*}$, we have $F_t(t^*, i^*) = 0$.*
- *For any tag $t^* \in \mathcal{T}$ and any index $i^* \in [d/2]$ such that $t^*_{\leq i^*} \neq t_{\leq i^*}$, the function value $F_t(t^*, i^*)$ is invertible in R_q.*

The first property is obvious from the definition of the tag function. The second property is not hard to see from Lemma 2.2 and the fact that two tag polynomials $t_{\leq i^*}(x)$ and $t^*_{\leq i^*}(x)$ are both with binary coefficients of degree $d < n/2$.

The description of the tag function is exceptionally simple, on the other hand, the homomorphic evaluation procedure is relatively involved. Although the work of [7] presented a possible homomorphic evaluation of tag function, the construction described in [7] is not as efficient as our constructions as we elaborate below.

3.2 Homomorphic Evaluation of Tag Function

Here we show the homomorphic evaluation of tag function $F_t(t^*, i^*)$ given the encodings of $t^* \in \mathcal{T}$ and $i^* \in [d]$. To achieve this aim, we reformulate the tag function $F_t(t^*, i^*) := t_{\leq i^*}(x) - t^*_{\leq i^*}(x)$ as follows:

$$F_t(t^*, i^*) = -t^*_{\leq i^*}(x) + \sum_{j=1}^{d} \mathsf{Equal}_j(i^*) \cdot t_{\leq j}(x),$$

where the equality test function $\mathsf{Equal}_j(i^*)$ outputs 1 if $j = i^*$, and outputs 0 otherwise. From the above reformulation, it is easy to see that we can homomorphically evaluate the tag function if we can homomorphically evaluate the equality test function. Fortunately, given the GSW type encoding of x^{i^*} as [2], there is an compact and efficient homomorphic evaluation algorithms $(\mathsf{PubEval}_{ET}, \mathsf{TrapEval}_{ET})$ given by Abla et.al [2] for the equality test function $\mathsf{Equal}_j(i^*)$ for any $j \in [n]$. Therefore, given the GSW type encodings of $t^*_{\leq i^*}(x)$ and x^{i^*}, we are capable of homomorphically evaluating the tag function via the homomorphic evaluation algorithms $(\mathsf{PubEval}_{ET}, \mathsf{TrapEval}_{ET})$ and homomorphic additivity and multiplicativity of GSW type encoding. The concrete description of the homomorphic evaluation procedure is presented below.

Construction 3.3 *For any full tag $t^* \in \mathcal{T}$ and any index $i^* \in [d]$, let b_{t^*} and b_{i^*} be the GSW type encodings of $t^*_{\leq i^*}(x)$ and x^{i^*}, let $(\mathsf{PubEval}_{ET}, \mathsf{TrapEval}_{ET})$ be the homomorphic evaluation algorithms for $\left\{\mathsf{Equal}_\beta\right\}_{\beta\in[m]}$, then the homomorphic evaluation algorithms $(\mathsf{PubEval}, \mathsf{TrapEval})$ for any function $F_t \in \mathcal{F}$ are described as follows:*

$\mathsf{PubEval}(\mathsf{b}_{t^*}, \mathsf{b}_{i^*}, t) \to \mathsf{b}_F$:

1 *For $j = 1$ to d compute* $\mathsf{b}_{j,i^*} := \mathsf{PubEval}_{ET}(\mathsf{b}_{i^*}, \mathsf{Equal}_j)$.

2 *Compute* $\mathsf{b}_F := -\mathsf{b}_{t^*} + \sum_{j=1}^{d} \mathsf{b}_{j,i^*} \cdot t_{\leq j}(x)$, *and output* b_F.

$\mathsf{TrapEval}(\mathsf{b}_{t^*}, \mathsf{b}_{i^*}, \mathsf{R}_{t^*}, \mathsf{R}_{i^*}, t, F) \to \mathsf{R}_F$

1 *For $j = 1$ to d compute* $\mathsf{R}_{j,i^*} := \mathsf{TrapEval}_{ET}(\mathsf{R}_{i^*}, \mathsf{Equal}_j)$.

2 *Compute* $\mathsf{R}_F := -\mathsf{R}_{t^*} + \sum_{j=1}^{d} \mathsf{R}_{j,i^*} \cdot t_{\leq j}(x)$, *and output* R_F.

It is easy to see that at the 1-st step of $\mathsf{PubEval}$, the ring vector b_{j,i^*} represents the encoding of 1 if $j = i^*$ or represents the encoding of 0 otherwise. In other words $\mathsf{b}_{i^*,i^*} = \mathsf{Encode}(1)$ and $\mathsf{b}_{j,i^*} = \mathsf{Encode}(0)$ for $j \neq i^*$. Therefore, the sum $\sum_{j=1}^{d} \mathsf{b}_{j,i^*} \cdot t_{\leq j}$ is the encoding of $t_{\leq i^*}$ and thus the ring vector b_F is the homomorphic encoding of the function F_t. Similarly, the deterministic algorithm $\mathsf{TrapEval}$ computes the trapdoor information for this public evaluation procedure.

The following theorem shows the quality of the above homomorphic evaluation procedure. Due to the space limit, we defer the proof in Supplementary Material B.1

Theorem 3.4 *For the tag function family defined in Definition 3.1, if the pair $(\mathsf{PubEval}_{ET}, \mathsf{TrapEval}_{ET})$ is a δ-expanding homomorphic evaluation algorithms for the function family $\left\{\mathsf{Equal}_\beta\right\}_{\beta\in[n]}$, then the algorithm pair $(\mathsf{PubEval}, \mathsf{TrapEval})$ in Construction 3.3 is $d^2 \cdot \delta$-expanding respect to the function family $\{F_t\}_{t\in\mathcal{T}}$*

From the work of [2], we know that there is $mn(kb)^2$-expanding homomorphic evaluation algorithm pair $(\mathsf{PubEval}_{ET}, \mathsf{TrapEval}_{ET})$ for the equality test function. Therefore, the algorithm tuples $(\mathsf{PubEval}, \mathsf{TrapEval})$ described in the Construction 3.3 is $O(nm \cdot d^2 \cdot (kb)^2)$-expanding for the function family $\{F_t\}_{t\in\mathcal{T}}$. For the special parameter sets as [21, 37] that $b = 2$ and $k \approx \log(q)$, the above expanding factor can be reduced to $O(n^2 \cdot d^2 \cdot \log^2(q))$. We also note that the work of [2] showed the existence of a $(mb^2k\sqrt{nk})$-expanding homomorphic evaluation algorithm for the equality test function family $\{\mathsf{Equal}_j\}_{j\in[m]}$ in the Common Reference String (CRS) model. Based upon the above facts, we have the following corollary for the tag function family $\{F_t\}_{t\in\mathcal{T}}$.

Corollary 3.5 *For the tag function family $\mathcal{F}$ defined in Definition 3.1, there exist $O(n^2(kb)^2 \cdot d^2)$-expanding deterministic homomorphic evaluation algorithms $(\mathsf{PubEval}, \mathsf{TrapEval})$ in the plain model, and $O(mb^2k\sqrt{nk} \cdot d^2)$-expanding in the CRS model for any function $F_t \in \mathcal{F}$.*

Following the previous literature, we can set the parameters b and k with different values. If set $b = O(1)$, $k \approx O(\log(q))$ and length of tags $d = O(\log(n) \cdot \log\log(n))$, then the above corollary shows that there is an algorithm pair
$(\mathsf{PubEval}, \mathsf{TrapEval})$ which is $O\left(n^2 \cdot \log^2(q) \cdot (\log(n)\log\log(n))^2\right)$-expanding in the plain model for the function family $\mathcal{F}$. Note that this expanding value is asymptotically better than the previous construction of [7]. If consider the above setting in the CSR model, then we have $O\left(n^{1.5} \cdot \log^{1.5}(q) \cdot (\log(n)\log\log(n))^2\right)$-expanding algorithm pair $(\mathsf{PubEval}, \mathsf{TrapEval})$ in the CRS model. More parameter settings will be discussed in the following sections.

4 Compact Partitioning via Tag function

In this section, we show our construction of partitioning function from the tag function defined in the previous section. The key space of our partition function is $\mathcal{K} := \mathbb{Z}^d \times [d]$ for some integer $d > 0$. In other words, any key $K := (t, i)$ is a tuple consists of a λ-bit string tag (an element in $\mathbb{Z}_2[X]$) in $\mathcal{T}$ and an index in $[\lambda]$. We keep identifying the bits strings as binary coefficient ring elements.For example, for a integer d, a bit string $t = (t_0, t_1, \cdots, t_{d-1}) \in \{0,1\}^d$ can be identified as a ring element in R_q, e.g., $t(x) := \sum_{i=0}^{d-1} t_i \cdot x^i$.

Definition 4.1 *Let q be a prime such that $q \equiv 3 \mod 8$, n be a power of 2. Let $R_q = \mathbb{Z}_q[x]/\Phi_{2n}(x)$ and Hash be a hash function, and $d = O(\lambda)$ be a positive integer and $\mathcal{K}_\lambda := \{0,1\}^d \times [d]$, then for a key $K = (t^*, i^*) \in \mathcal{K}_\lambda$, we define the keyed function $F : \{0,1\}^\lambda \to R_q$ as follows:*

$$F(K, X) := \mathsf{Hash}(X)_{\leq i^*}(x) - t^*_{\leq i^*}(x).$$

The function family $\mathcal{F}$ is defined naturally as $\mathcal{F} := \{F(K,X) | K \in \mathcal{K}_\lambda\}$.

The following theorem shows that the partition property of above function. The full proof can be found in Supplementary Material C.1

Theorem 4.2 *The function defined in Definition 4.1 is partition function if Hash is truncation collision resistant hash function.*

4.1 Homomorphic Evaluation of Partition Function

Note that for a tag $t \in [d]$, the tag function is defined as $F_t(t^*, i^*) = t_{\leq i^*}(x) - t^*_{\leq i^*}(x)$, and from our definition of partition function in Definition 4.1, it is not hard to see that $F(K, X) = F_{\mathsf{Hash}(x)}(K)$. Furthermore, the Homomorphic evaluation algorithms for the tag Functions can be applied to publicly evaluate the partition functions. The public evaluation procedure is described as follows.

Construction 4.3 *For $K = (t^*, i^*) \in \{0,1\}^d \times [d]$, let the vectors $\mathsf{b}_{t^*}, \mathsf{b}_{i^*}$ are GSW type encodings of $t_{\leq i^*}$, and x^{i^*}, $\mathsf{Hash} : \{0,1\}^* \to \{0,1\}^d$ be a hash function, $(\mathsf{PubEval}, \mathsf{TrapEval})$ be the homomorphic evaluation algorithms for any tag function F_t, then the public evaluation algorithms for the partition function $F(K, X)$ are as follows:*

$\mathsf{PubEval}_{\mathsf{IBE}}(\mathsf{b}_{t^*}, \mathsf{b}_{i^*}, X) \to \mathsf{b}_{\mathsf{IBE}}$:

1 *Computes the hash value* $t := \mathsf{Hash}(X)$.
2 *Computes* $\mathsf{b}_{\mathsf{IBE}} := \mathsf{PubEval}(\mathsf{b}_{t^*}, \mathsf{b}_{i^*}, t)$, *and outputs* $\mathsf{b}_{\mathsf{IBE}}$

$\mathsf{TrapEval}_{\mathsf{IBE}}(\mathsf{R}_{t^*}, \mathsf{R}_{i^*}, X) \to \mathsf{R}_{\mathsf{IBE}}$:
1 *Computes the hash value* $t := \mathsf{Hash}(X)$.
2 *Computes* $\mathsf{R}_{\mathsf{IBE}} := \mathsf{TrapEval}(\mathsf{R}_{t^*}, \mathsf{R}_{i^*}, t)$, *and outputs* $\mathsf{R}_{\mathsf{IBE}}$

Note that public evaluation algorithms for the partition Functions are almost the same as the homomorphic evaluation algorithms for the tag functions. Thus, we have the following corollary form Corollary 3.5.

Corollary 4.4 *For the partition function $f \in \mathcal{F}$ defined in Definition 4.1, there exists deterministic public evaluation algorithms* $(\mathsf{PubEval}_{\mathsf{IBE}}, \mathsf{TrapEval}_{\mathsf{IBE}})$, *which are $O\left(n^2(kb)^2 \cdot d^2\right)$-expanding in the plain model, and $O\left(nb^2\sqrt{nk} \cdot d^2\right)$-expanding in the crs model.*

5 The Signature Scheme

In this section, we first present our construction of a non-adaptive lattice-based signature scheme, then we show its security analysis. The construction and corresponding security results of the adaptively secure signature scheme can be found in Supplementary Material D.5. In this section, we work on a power of 3 ring modulus $q \geq 3$.

5.1 Non-Adaptive Signature Scheme

In this section, we show our construction of a lattice-based signature scheme in the standard model with more compact public parameters and smaller security reduction losses. In particular, we show a more fine-grained construction of a tag-based signature scheme with smaller public keys.

Correctness. The correctness of the scheme is from the fact that the ring vector e is sampled from a distribution which is statistically close to $D_{\Lambda^{\perp}_{\mathsf{Coeffs}(u)}\mathsf{rot}([\mathsf{a}^\top|\mathsf{a}_t^\top]),\sigma}$, and thus $\|\mathsf{e}\| \leq \sigma \cdot \sqrt{2nk}$ holds with overwhelming probability. The proof of the following theorem can be found in Supplementary Material D.1

Theorem 5.1 *The signature scheme* $\mathsf{naCMA\text{-}Sign}$ *presented in Figure 1 is correct if the Gaussian parameter $\sigma > O\left(b\rho\sqrt{n\log_\rho(q)} \cdot \sqrt{\frac{\log(2n(1+1/\epsilon_s))}{\pi}}\right)$.*

5.2 Security

We call an algorithm Sim is (T, ϵ)-algorithm if it achieves its goal in time T with probability at least ϵ. Now we show the non-adaptive security of the signature scheme in Figure 1.

Theorem 5.2 *If there exist a (T, ϵ)-adversary $\mathcal{A}$ breaks the signature scheme* $\mathsf{naCMA\text{-}Sign}$ *in Figure 1 via making at most polynomially bounded Q signature queries and $\epsilon \in (0, 1/2)$, then there is a $(T + \mathsf{poly}(n), \frac{\epsilon}{2} \cdot \frac{\epsilon}{2Q^2})$-algorithm $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$ which solves R-$\mathsf{SIS}_{n,k,q,\beta}$ problem for $\beta = \tilde{O}\left((nkb)^5 \cdot \rho^2 \cdot \sqrt{nk}\right)$.*

$\mathsf{naCMA} \cdot \mathsf{KeyGen}(\lambda) \rightarrow (\mathsf{sk}, \mathsf{vk})$:

1: $(\mathsf{a}, \mathsf{T_a}) \leftarrow \mathsf{TrapGen}(n, k, \rho, q)$
2: $\mathsf{b}_0 \leftarrow R_q^k$
3: $\mathsf{b}_1 \leftarrow R_q^k$
4: $\mathsf{u} \leftarrow R_q^k$
5: $v \leftarrow R_q$
6: $\mathsf{sk} := (\mathsf{T_a})$
7: $\mathsf{vk} = (\mathsf{a}, \mathsf{b}_0, \mathsf{b}_1, \mathsf{u}, v)$
8: return $(\mathsf{sk}, \mathsf{vk})$

$\mathsf{naCMA} \cdot \mathsf{Sign}(\mathsf{sk}, \mathsf{vk}, \boldsymbol{\mu} \in R_2^k) \rightarrow (\Sigma)$:

1: $t \leftarrow \mathcal{T}$
2: $\mathsf{a}_t := \mathsf{PubEval}(\mathsf{b}_0, \mathsf{b}_1, t) \in R_q^k$
3: $u = \mathsf{u}^\top \boldsymbol{\mu} + v$
4: $\mathsf{e} \leftarrow \mathsf{SampleLeft}(\mathsf{a}, \mathsf{a}_t, \mathsf{T_a}, u, \sigma)$
5: $\Sigma = (t, \mathsf{e})$
6: return Σ

$\mathsf{naCMA} \cdot \mathsf{Verify}(\mathsf{vk}, \Sigma = (t, \mathsf{e}), \boldsymbol{\mu}) \rightarrow \top/\bot$:

1: $\mathsf{a}_t := \mathsf{PubEval}(\mathsf{b}_0, \mathsf{b}_1, t)$
2: $u := \mathsf{u}^\top \boldsymbol{\mu} + v$
3: **if** $\|\mathsf{e}\| \leq \sigma\sqrt{2nk}$ and $[\mathsf{a}^\top | \mathsf{a}_t^\top]\mathsf{e} = u$ **then**
4: Return $\top$ ▷ Valid signature
5: **else**
6: Return $\bot$ ▷ Invalid signature

Fig. 1. Construction of naCMA-Sign Scheme

Proof. To show the theorem, it suffices to construct an algorithm $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$ which invokes $\mathcal{A}$ as a subroutine to solve the R-SIS problem with probability at least $\frac{\epsilon}{2} \cdot \frac{\epsilon}{2Q^2}$, and running time of $T + \mathsf{poly}(n)$. Before giving the concrete description of $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$, we first analyze the relations between the following sequence of hybrid games, from which we come up with the construction of $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$.

Game_0 : This is the original signature forgery game between $\mathcal{A}$ and the challenger. Namely, in this game, the adversary $\mathcal{A}$ claims the messages $\{\boldsymbol{\mu}_j\}_{j\in[Q]}$ as its signature query before seeing the public parameters and verification keys of the signature scheme.

Game_1 : In this game, the challenger randomly chooses tags $\{t^1, \cdots, t^Q\}$ from the tag space $\mathcal{T}$. Let $i^* \in [d]$ be the smallest index such that $\frac{2Q^2}{\epsilon} \leq 2^{i^*}$ (such i^* exists by setting $d = \omega(\log(n))$, if there exist $i \neq j$ such that $t^i_{\leq i^*} = t^j_{\leq i^*}$, the challenger send $\bot$ to $\mathcal{A}$, and $\mathcal{A}$ aborts the game on receiving $\bot$. Otherwise challenger chooses a random tag prefix $t^*_{\leq i^*} \in \mathcal{T}_{i^*}$, and checks if there is a tag t^j such that $t^*_{\leq i^*} = t^j_{\leq i^*}$. If so, the challenger set $t^* := t^j$ and $\boldsymbol{\mu}^* := \boldsymbol{\mu}_j$, otherwise it randomly choose a dummy message $\boldsymbol{\mu}^*$ and extend $t^*_{\leq i^*}$ randomly to get a full tag $t^* \in \mathcal{T}$. In the rest of the game, the challenger generates the signatures corresponding to the pair $(t^j, \boldsymbol{\mu}_j)_{j\in[Q]}$ as in Game_0.

Game_2 : The only difference of this game from Game_1 is that, in this game the challenger changes the way of generating key pairs $(\mathsf{sk}, \mathsf{vk})$ and signing algorithm. In this game, the challenger generates a randomly instead of running $\mathsf{TrapGen}(n, k, \rho, q)$ as in Game_1. Additionally, the challenger randomly chooses matrices $\mathsf{R}_0, \mathsf{R}_1, \mathsf{R}_u \leftarrow D_{R,s}^{k\times k}$ for some Gausian parameter $s \geq \omega(\sqrt{\log(nk)})$ and $\mathsf{e}^* \leftarrow D_{R^{2k},\sigma}^{\mathsf{Coeffs}}$, and generate $\mathsf{b}_0, \mathsf{b}_1, \mathsf{u}, v$ as follows:

$$\mathsf{b}_0^\top := \mathsf{a}^\top \cdot \mathsf{R}_0 + t^*_{\leq i^*} \cdot \mathsf{g}_b^\top, \quad \mathsf{b}_1^\top := \mathsf{a}^\top \cdot \mathsf{R}_1 + x^{i^*} \cdot \mathsf{g}_b^\top,$$
$$\mathsf{u}^\top := \mathsf{a}^\top \cdot \mathsf{R}_u, \text{ and } v := [\mathsf{a}^\top, \mathsf{a}_{t^*}^\top]\mathsf{e}^* - \mathsf{u}^\top \boldsymbol{\mu}^*,$$

where $\mathsf{a}_{t^*} = \mathsf{PubEval}(\mathsf{b}_0, \mathsf{b}_1, F_{t^*})$. To generate a signature for the message $\boldsymbol{\mu}^j \neq \boldsymbol{\mu}^*$, the challenger computes $\mathsf{R}_t = \mathsf{TrapEval}(\mathsf{b}_0, \mathsf{b}_1, \mathsf{R}_0, \mathsf{R}_1, (t^*_{\leq i^*}, i^*), F_{t^j})$ and $u = \mathsf{u}^\top \boldsymbol{\mu}^j + v$. Then, it samples a ring vector $\mathsf{e}_j \in R_q^{2k}$ by running $\mathsf{SampleRight}$ as

$$\mathsf{e}_j \leftarrow \mathsf{SampleRight}(\mathsf{a}, \mathsf{R}_t, u, F_{t^j}(t^*, i^*), \mathsf{g}_b, \mathsf{T}_{\mathsf{g}_b}, \sigma).$$

Finally, it lets $\Sigma_j := (t^j, \mathsf{e}_j)$ be the corresponding signature of $\boldsymbol{\mu}^j$. For the message $\boldsymbol{\mu}^*$, it simply lets $\Sigma^* := (t^*, \mathsf{e}^*)$ be the corresponding signature of $\boldsymbol{\mu}^*$.

For $i \in \{0, 1, 2\}$, let E_i be the event that $\mathcal{A}$ wins in the Game_i, and let abort be the event that the challenger aborts in Game_1. Then, from the theorem assumption, we have that $\mathsf{Pr}[E_0] = \epsilon$ and the following lemmas. The proofs can be found in Supplementary Material D.2 and D.3

Lemma 5.3 $|\mathsf{Pr}[E_0] - \mathsf{Pr}[E_1]| \leq \frac{\epsilon}{2}$.

Lemma 5.4 $|\mathsf{Pr}[E_1] - \mathsf{Pr}[E_2]| \leq \mathsf{negl}(\lambda)$.

Now we show the description of algorithm $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$. The algorithm $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$ solves the $R\text{-}\mathsf{SIS}_\beta$ problem by invoking $\mathcal{A}$ as follow:

$\mathsf{Sim}_{R\text{-}\mathsf{SIS}}(\mathsf{a})$:

On input the $R\text{-}\mathsf{SIS}_\beta$ challenge $\mathsf{a} \in R^k$, $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$ simulates everything for $\mathcal{A}$ as the challenger did in Game_2. Since a is uniformly distributed, $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$ can perfectly simulates the challenger in the Game_2. After receiving adversary's output forgery $\left(\boldsymbol{\mu}^f, \Sigma^f = (t^f, \mathsf{e}^f)\right)$, $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$ splits e^f into two ring vectors as $\mathsf{e}^f = (\mathsf{e}_1^f, \mathsf{e}_2^f) \in R^k \times R^k$ and let $\mathsf{e}^* = (\mathsf{e}_1^*, \mathsf{e}_2^*) \in R^k \times R^k$, computes $\mathsf{R}_{t^*} = \mathsf{TrapEval}(\mathsf{b}_0, \mathsf{b}_1, \mathsf{R}_0, \mathsf{R}_1, (t^*_{\leq i^*}, i^*), F_{t^*})$ and then outputs

$$\mathsf{s} := [(\mathsf{e}_1^f - \mathsf{e}_1^*) + (\mathsf{R}_{t^f}\mathsf{e}_2^f - \mathsf{R}_{t^*}\mathsf{e}_2^*) + \mathsf{R}_u(\boldsymbol{\mu}^f - \boldsymbol{\mu}_1^*)]$$

as its solution to the $R\text{-}\mathsf{SIS}_\beta$ problem.

Analysis of s. For the output ring vector s from above algorithm $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$, we have following results. The full proof of this lemma can be found in Supplementary Material D.4

Lemma 5.5 *If the output* $\left(\boldsymbol{\mu}^f, \Sigma^f\right)$ *by the adversary is a valid forgery, then we have following results (1)* $\mathsf{Pr}[\mathsf{a}^\top\mathsf{s} = 0] = 2^{-i^*}$ *, (2)* $\mathsf{Pr}[\mathsf{s} = 0] \leq \mathsf{negl}(\lambda)$*, and (3)* $\mathsf{Pr}[\|\mathsf{s}\| \leq \beta] \leq 1 - \mathsf{negl}(\lambda)$.

Analysis of $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$. It's obvious from the choice of i^* that $2^{i^*} \leq \frac{4Q^2}{\epsilon}$. Since i^* is independent from $\mathcal{A}$, therefore the probability of $t^f_{\leq i^*} = t^*_{\leq i^*}$ is at least 2^{-i^*}. Let **coll** be the event that there is a collision for the randomly chosen tags $(t^0, \cdots, t^{Q-1})$, then we have following

$$\begin{aligned}
\epsilon_{\mathsf{Sim}} &= \Pr\left[\mathsf{a}\cdot\mathsf{s} = 0 \ \wedge \ (\mathsf{s} \neq 0) \ \wedge \|\mathsf{s}\| \leq \beta\right] \\
&\geq \Pr\left[\mathsf{a}\cdot\mathsf{s} = 0 \ \wedge \ (\mathsf{s} \neq 0) \ \wedge \|\mathsf{s}\| \leq \beta \ \wedge \ \overline{\mathbf{coll}}\right] \\
&\geq \Pr\left[\|\mathsf{s}\| \leq \beta | \mathsf{a}\cdot\mathsf{s} = 0 \ \wedge \ (\mathsf{s} \neq 0) \ \wedge \ \overline{\mathbf{coll}}\right] \cdot \Pr\left[\mathsf{s} \neq 0 | \mathsf{a}\cdot\mathsf{s} = 0 \wedge \overline{\mathbf{coll}}\right] \\
&\cdot \Pr\left[\mathsf{a}\cdot\mathsf{s} = 0 | \overline{\mathbf{coll}}\right] \cdot \Pr\left[\overline{\mathbf{coll}}\right] \\
&\geq (1 - \mathsf{negl}(\lambda)) \cdot (1 - \mathsf{negl}(\lambda)) \cdot \frac{1}{2^{i^*}} \cdot \frac{\epsilon}{2} \\
&\geq \frac{\epsilon}{2} \cdot \frac{\epsilon}{4Q^2} - \mathsf{negl}(\lambda) = O(\frac{\epsilon^2}{Q^2}),
\end{aligned}$$

where the third inequality is from Lemma 5.5, Lemma 5.3 and Lemma 5.4; we used the fact $2^{i^*} \leq \frac{4Q^2}{\epsilon}$ for the last inequality[11]. This completes the proof of the theorem. □

Note that it is common to set $\rho = 1, b = 2$ and $k = \log(q)$ in the lattice-based works of literature, following this setting we will have the upper bound $\tilde{O}(n^{5.5})$ for the R-SIS parameter β. Although this bound is comparable with the previous compact design of [7], we can have a smaller bound in the CRS setting via the techniques of [2]. Considering the CRS model and applying the results from Lemma 2.11 to the Lemma 5.5, we have the following corollary.

Corollary 5.6 *If there exist a (T, ϵ)-adversary $\mathcal{A}$ breaks the signature scheme* $\mathsf{naCMA\text{-}Sign}$ *in Construction 1 via making at most Q signature queries, then there is a $(T+\mathsf{poly}(n), \frac{\epsilon}{2}\cdot\frac{\epsilon}{2Q^2})$-algorithm* $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$ *which solves R-$\mathsf{SIS}_{n,k,q,\beta}$ problem for $\beta = \tilde{O}\left((nkb)^4 \cdot \rho^2 \cdot \sqrt{nk}\right)$.*

From above corollary, if set set $\rho = 1, b = 2$ and $k = \log(q)$, we have the upper bound $\tilde{O}\left(n^{4.5}\right)$, which is better that the previous result [7]. As a result, our construction of a lattice-based signature scheme construction not only enjoys better security analysis but also has relatively small parameters.

5.3 The Revised Signature Construction

Let the tag class $\mathcal{T}_{\hat{t}}$ be the set of tags of which d-prefixes equal to $\hat{t}_{\leq d}$, that is $\mathcal{T}_{\hat{t}_{\leq d}} := \{t | t_{\leq d} = \hat{t}_{\leq d}, t \in \mathcal{T}\}$. The revised construction is as follows.

Construction 5.7 *Let σ be a Gaussian parameter, let R_2^k be the message space, and $\mathcal{T}$ be a tag space and $\mathcal{F}$ be the tag function family. The construction of* $\mathsf{naCMA'\text{-}Sign} = (\mathsf{naCMA'}\cdot\mathsf{KeyGen}, \mathsf{naCMA'}\cdot\mathsf{Sign}, \mathsf{naCMA'}\cdot\mathsf{Verify})$ *is as follows:*

$\mathsf{naCMA'}\cdot\mathsf{KeyGen}(\lambda,) \to (\mathsf{sk}, \mathsf{vk})$*: On input security parameter λ, it runs as follow:*

1. *Run* $\mathsf{TrapGen}(n, k, \rho, q)$ *to get* $(\mathsf{a} \in R_q^k, \mathsf{T_a} \in R_q^{k\times k})$*.*

[11] Note that, from the context, the term $\mathsf{negl}(\lambda)$ is statistically small, and thus much smaller than the advantage ϵ of the adversary.

2. *Choose random vectors* $\mathsf{b}_0, \mathsf{b}_1, \mathsf{u}$ *from* R_q^k.
3. *Randomly select a set of ring elements* $S_v := \{v_i\}_{i\in[\mathsf{L}]}$ *from* R_q.
4. *Output the secret key* $\mathsf{sk} = (\mathsf{T_a})$ *and the verification key* $\mathsf{vk} = (\mathsf{a}, \mathsf{b}_0, \mathsf{b}_1, \mathsf{u}, S_v)$

$\mathsf{naCMA}' \cdot \mathsf{Sign}(\mathsf{sk}, \mathsf{vk}, \boldsymbol{\mu} \in R_2^k) \to (\Sigma)$: *On input* vk, sk *and a message* $\boldsymbol{\mu}$, *this randomized algorithm runs as follows:*

1. *Choose a random tag* $t \in \mathcal{T}$ *to obtain a tag function* F_t, *and compute* $\mathsf{a}_t := \mathsf{PubEval}(\mathsf{b}_0, \mathsf{b}_1, t) \in R_q^k$.
2. *Let* $\tau := |\mathcal{T}_{t_{\le d}}|$ *and compute* $u_\tau := \mathsf{u}^\top \boldsymbol{\mu} + v_\tau$, *and then* $\mathcal{T}_{t_{\le d}} := \mathcal{T}_{t_{\le d}} \cup \{t\}$
3. *Run* $\mathsf{SampleLeft}(\mathsf{a}, \mathsf{a}_t, \mathsf{T_a}, u_\tau, \sigma)$ *to get a ring vector* $\mathsf{e} \in R_q^{2k}$ *such that* $[\mathsf{a}^\top | \mathsf{a}_t^\top] \cdot \mathsf{e} = u_\tau$.
4. *Output the pair* $\Sigma = (\tau, t, \mathsf{e})$ *as a signature for the message* $\boldsymbol{\mu}$.

$\mathsf{naCMA}' \cdot \mathsf{Verify}(\mathsf{vk}, \Sigma = (\tau, t, \mathsf{e}), \boldsymbol{\mu}) \to \top/\bot$: *On input a signature* Σ, *message* $\boldsymbol{\mu}$ *and verification key* vk, *this deterministic verification algorithm runs as follows:*

1. *Compute* $\mathsf{a}_t := \mathsf{PubEval}(\mathsf{b}_0, \mathsf{b}_1, t)$.
2. *Compute* $u_\tau := \mathsf{u}^\top \boldsymbol{\mu} + v_\tau$.
3. *Check if* $\|\mathsf{e}\| \le \sigma\sqrt{2nk}$ *and* $[\mathsf{a}^\top | \mathsf{a}_t^\top]\mathsf{e} = u_\tau$. *If so, output* $\top$ *means that* Σ *is a valid signature for the message* $\boldsymbol{\mu}$; *otherwise, output* $\bot$ *implies that* Σ *is invalid.*

5.4 Analysis of Revised Signature Scheme.

The correctness of the signature scheme naCMA′-Sign is similar to the correctness signature scheme presented in the previous section. Thus, we omit the proof. However, the security proof of the revised signature scheme is subtly different, the full proof is presented in Supplementary Material D.6.

Theorem 5.8 *If there exist a* (T, ϵ)*-adversary* $\mathcal{A}$ *breaks the signature scheme* naCMA-Sign *in Construction 5.7 via making at most* Q *signature queries, then there is a* $(T+\mathsf{poly}(n), \left(\epsilon - \frac{1}{2^{\mathsf{w}}}\right) \cdot \frac{\mathsf{L}+1}{2eQ2^{\frac{\mathsf{w}+\log(Q)}{\mathsf{L}}}})$*-algorithm* $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$ *which solves* $R\text{-}\mathsf{SIS}_{n,k,q,\beta}$ *problem, where* $\beta = \tilde{O}\left((nkb)^5 \cdot \sqrt{nk}\right)$ *, and* $\mathsf{w} < \log(\frac{1}{\epsilon})$.

Note that if the parameter w is large enough, let's say $\mathsf{w} = \omega(\log(n))$, then the term $\frac{1}{2^{\mathsf{w}}}$ is negligible. Furthermore, if the parameter $\mathsf{L} \ge \mathsf{w} + \log(Q)$, then we can rewrite the theorem 5.8 simply as follows.

Corollary 5.9 *If there exist a* (T, ϵ)*-adversary* $\mathcal{A}$ *breaks the signature scheme* naCMA-Sign *in Construction 5.7 via making at most* Q *signature queries, then for* $\mathsf{L} \ge \omega(log(n)) + \log(Q)$, *there is a* $(T + \mathsf{poly}(n), \frac{\mathsf{L}\cdot\epsilon}{4eQ})$*-algorithm* $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$ *which solves* $R\text{-}\mathsf{SIS}_{n,k,q,\beta}$ *problem for* $\beta = \tilde{O}\left((nkb)^4 \cdot \sqrt{nk}\right)$.

The above corollary shows that, if use the asymptotic notations, then the advantage of the simulator $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$ becomes $O(\frac{\mathsf{L}\cdot\epsilon}{Q})$, which much better than the scheme given in the previous section. However, the revised signature scheme needs L ring elements in S_v while it contains exactly one element in the signature scheme presented in the previous section.

Table 3. Asymptotic parameterization of our signature scheme.

Parameters	Type I : $\rho = 1, b = 4$	Type II : $\rho = b = n^{\nu^*}$
Q	$\mathsf{poly}(n)$	$\mathsf{poly}(n)$
n	n	n
β	$\tilde{O}(n^{4.5})$	$\tilde{O}(n^{4.5} \cdot b^7)$
q	$\tilde{O}(n^5)$	$\tilde{O}(n^5 \cdot b^7)$
k	$O(\log(n))$	O(1)
σ	$\tilde{O}(n^2)$	$\tilde{O}(n^2 \cdot b^4)$
d	$\omega(\log(n))$	$\omega(\log(n))$
L	$\omega(\log(n))$	$\omega(\log(n))$

$(*)$ $\nu > 1$ is a small positive number.

5.5 Parameters for the Signature Schemes

Parameter Constraints. To meet the correctness and adaptive security, the parameters in our schemes should satisfy some desired constraints, the detailed description of constraints for signatures listed in Supplementary Material D.7 .

Asymptotic Parameterization. To show a more precise advantage of our scheme, we asymptotically determine other parameters in terms of the security parameter λ and n. To do this, we consider two type settings for ρ and b as shown in Table 3. Note that Type I is expected in the literature, yet we consider that Type II may bring smaller ring vector dimensions and may have further help for software implementations.

From the parameterizations in Table 3, we have following result for the signature scheme naCMA-Sign in Figure 1.

Corollary 5.10 *If there exist a (T, ϵ)-adversary $\mathcal{A}$ breaks the signature scheme* naCMA-Sign *in Figure 1 with the parameters as Table 3 (Type I) via making at most Q signature queries, then there is a $(T + \mathsf{poly}(n), \tilde{O}\left(\frac{\epsilon^2}{Q^2}\right))$-algorithm $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$ which solves R-$\mathsf{SIS}_{n,k,q,\beta}$ problem for $\beta = \tilde{O}\left(n^{4.5}\right)$.*

Note that Q is polynomially bounded, thus we have $2^{\omega(\log(n))} > Q$, and thus by setting $\mathsf{L} := \omega(\log(n))$ From the parameterizations in Table 3, we have following result for the signature scheme naCMA-Sign in Construction 5.7.

Corollary 5.11 *If there exist a (T, ϵ)-adversary $\mathcal{A}$ breaks the signature scheme* naCMA-Sign *in Construction 5.7 with the parameters as Table 3 (Type I) via making at most Q signature queries, then there is a $(T + \mathsf{poly}(n), \tilde{O}\left(\frac{\epsilon \cdot \omega(1)}{Q}\right))$-algorithm $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$ which solves R-$\mathsf{SIS}_{n,k,q,\beta}$ problem for $\beta = \tilde{O}\left(n^{4.5}\right)$.*

The above result shows that the reduction loss is relatively small, yet the verification key contains $\omega(\log(n))$ ring elements in R_q plus two ring vectors in R_q^k for some $k = O(\log(n))$. Therefore, we can roughly say the verification key of naCMA′-Sign contains $\omega(1)$ ring vectors in R_q^k. Thus, the resulting signature is a non-compact scheme. However, this scheme is almost compact and enjoys remarkably less reduction loss.

6 Our IBE Scheme

In this section, we first present our construction of a ring-based IBE scheme. Thereafter, we show the correctness and the security of our scheme. In this section, we work on a prime ring modulus q such that $q \equiv 3 \bmod 8$.

6.1 The Construction

Here we present our construction of an IBE scheme. Let ID be the identity space of the scheme, and Hash be a hash function, the deterministic algorithms $(\mathsf{PubEval}_{\mathsf{IBE}}, \mathsf{TrapEval}_{\mathsf{IBE}})$ are the public evaluation algorithm for the partition function, then the IBE scheme is described as follows.

Construction 6.1

$\mathsf{Setup}(1^\lambda) \to (\mathsf{msk}, \mathsf{mpk})$: *On input the security parameter λ, it first sets the parameters $n, k, q, \rho, \sigma_0, \sigma_1, \sigma$, and then runs as follows:*

- *It runs* $\mathsf{TrapGen}(n, k, \rho, q)$ *to get* $\left(\mathsf{a} \in R_q^k, \mathsf{T}_\mathsf{a} \in R_q^{k\times k}\right)$.
- *Chooses random vectors* $\mathsf{b}_0, \mathsf{b}_1$ *from* R_q^k, *and a random ring element* $u \xleftarrow{\$} R_q$.
- *It outputs the master secrete key* $\mathsf{msk} = (\mathsf{T}_\mathsf{a})$ *and the master public key* $\mathsf{mpk} = (\mathsf{a}, \mathsf{b}_0, \mathsf{b}_1, u)$.

$\mathsf{KeyGen}(\mathsf{msk}, \mathsf{mpk}, \mathsf{id}) \to (\mathsf{sk}_\mathsf{id})$: *On input the master key pair* (msk, mpk) *and an identity* $\mathsf{id} \in \mathsf{ID}$, *it runs as follows:*

- *Computes* $\mathsf{b}_\mathsf{id} = \mathsf{PubEval}_{\mathsf{IBE}}(\mathsf{b}_0, \mathsf{b}_1, \mathsf{id})$.
- *Runs* $\mathsf{SampleLeft}(\mathsf{a}, \mathsf{b}_\mathsf{id}, \mathsf{T}_\mathsf{a}, u, \sigma)$ *to get a ring vector* $\mathsf{e} \in R_q^{2k}$ *such that* $[\mathsf{a}^\top | \mathsf{b}_\mathsf{id}^\top] \cdot \mathsf{e} = u$.
- *Outputs the* $sk_\mathsf{id} = \mathsf{e}$ *as a secret key for the identity* id.

$\mathsf{Enc}(\mathsf{mpk}, \mathsf{id}, \mu) \to \mathsf{ct} = (c_0, c_1)$: *On input the master public key* mpk, *an identity* $\mathsf{id} \in \mathsf{ID}$ *and a message* $\mu \in R_2$, *it runs as follows:*

- *Computes* $\mathsf{b}_\mathsf{id} = \mathsf{PubEval}_{\mathsf{IBE}}(\mathsf{b}_0, \mathsf{b}_1, \mathsf{id})$.
- *Samples ring elements* $s \xleftarrow{\$} R_q, e_0 \leftarrow D^{\mathsf{Coeffs}}_{R_q,\sigma_0}$, *and ring vectors* $\mathsf{e}_1, \mathsf{e}_2 \leftarrow D^{\mathsf{Coeffs}}_{R_q^{2k},\sigma_1}$.
- *Computes* $c_0 = u \cdot s + e_0 + \left\lceil \frac{q}{2} \right\rceil \cdot \mu$, *and* $\mathsf{c}_1 = \begin{bmatrix} \mathsf{a} \\ \mathsf{b}_\mathsf{id} \end{bmatrix} \cdot s + \begin{bmatrix} \mathsf{e}_1 \\ \mathsf{e}_2 \end{bmatrix}$.
- *Outputs the ciphertext* $\mathsf{ct} = (c_0, \mathsf{c}_1)$.

$\mathsf{Dec}(\mathsf{sk}_\mathsf{id}, \mathsf{ct}) \to \mu$: *On input the secret key* $\mathsf{sk}_\mathsf{id} = \mathsf{e}$ *and the ciphertext* $\mathsf{ct} = (c_0, \mathsf{c}_1) \in R_q \times R_q^{2q}$, *it runs as follows:*

- *Computes* $\mu' = \left\lceil \frac{2}{q} \mathsf{Coeffs}\left(c_0 - \mathsf{e}^\top \cdot \mathsf{c}_1\right) \right\rfloor \bmod 2$, *where the rounding function* $\lceil * \rfloor$ *applies coefficient-wise.*
- *Outputs the* μ' *as the decrypted message.*

To ensure the IBE scheme in Construction 6.1 reaches the correctness and the adaptive security, the parameters should meet the corresponding requirements. Due to the space limit, we defer these requirements and the corresponding parameter settings in Supplementary Material E.1.

Correctness and Security. The correctness of the above IBE scheme is similar to the correctness of the Dual Regev encryption scheme, yet the security is more complex. The following theorems show the correctness and the adaptive security of our IBE scheme. Due to the space limit, we defer the proof in Supplementary Material E.2 and E.3

Theorem 6.2 *For the ring modulus q, and gaussian parameters $\sigma, \sigma_0, \sigma_1$ such that $q \geq O\left(\omega(\log(n)) \cdot \sigma_0 + \sigma\sigma_1 \cdot \omega(\log(n)) \cdot \sqrt{2nk}\right)$, the IBE scheme presented in Construction 6.1 is correct.*

Theorem 6.3 *The IBE scheme described in Construction 6.1 achieves adaptive security if the hash function Hash has TCR property and the underlying problem R-$\mathsf{DLWE}_{n,k+1,q,\sigma_0}$ is hard.*

Note that the ratio $1/\alpha = q/\sigma$ directly reflects the hardness of R-DLWE problems. Thus, from the parameter setting in the previous section, we have the following result for the IBE scheme in Construction 6.1.

Corollary 6.4 *If there exists a polynomial time adversary $\mathcal{A}$ who breaks the IBE scheme in Construction 6.1 under our recommended parameters with noticeable probability, then there is a polynomial time algorithm $\mathsf{Sim}_{R\text{-}\mathsf{DLWE}}$ who breaks the R-$\mathsf{DLWE}_{n,k+1,q,\sigma_0}$ problem for $\frac{1}{\alpha} = \tilde{O}\left(n^{5.5}\right)$.*

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Supplementary Material

A Supplementary Materials for Section 2

A.1 Signature Scheme

Definition A.1 *A signature scheme consists of three algorithms* $(\mathsf{KeyGen}, \mathsf{Sign}, \mathsf{Verify})$ *as follows:*

$\mathsf{KeyGen}(1^\lambda) \to (\mathsf{sk}, \mathsf{vk})$*: On input the security parameter* λ*,* KeyGen *generates the signing (secret) key* sk *and verification key* vk*, and output the key pair* $(\mathsf{sk}, \mathsf{vk})$*.*

$\mathsf{Sign}(\mathsf{sk}, \mathsf{vk}, \mu). \to \Sigma$*: On input the key pair* $(\mathsf{sk}, \mathsf{vk})$ *and a message* $\mu \in \mathsf{M}$*, the probabilistic polynomial time algorithm* Sign *use key pairs* $(\mathsf{sk}, \mathsf{vk})$ *to generate a signature* Σ *corresponding to the message* μ*, and output* Σ*.*

$\mathsf{Verify}(\Sigma, \mathsf{vk}, \mu)$*: On input the verification key* vk*, message* μ *and a signature* Σ*, the deterministic algorithm* Verify *outputs* $\top$ *if* Σ *is a valid signature for the message* μ*, ,and outputs* $\perp$ *otherwise.*

Definition A.2 *We say a signature scheme is* δ*-correct if following holds:*

$$\mathsf{Pr}\left[b = \top \middle| \begin{array}{l} (\mathsf{sk}, \mathsf{vk}) \leftarrow \mathsf{KeyGen}(1^\lambda); \\ \Sigma \leftarrow \mathsf{Sign}(\mathsf{sk}, \mathsf{vk}, \mu); \\ b \leftarrow \mathsf{Verify}(\Sigma, \mathsf{vk}, \mu) \end{array} \right] \geq 1 - \delta.$$

If the value of δ is negligible in the security parameter λ, then we simply call the signature scheme is correct.

For a signature scheme, we consider the Existential Unforgeability under non-adaptive chosen message attack, denoted $\mathsf{EUF\text{-}naCMA}$, security defined by the following game between the adversary $\mathcal{A}$ and the challenger.

$\mathsf{forge}_{\mathcal{A}}^{\mathsf{naCMA}}(\lambda)$**:**

Initi: At the begining of the game, the adversary $\mathcal{A}$ sends Q messages $\{\mu_1, \cdots, \mu_Q\}$ to the challenger as the signature queries.

Setup: After receiving the signature queries, the challenger runs $\mathsf{KeyGen}(1^\lambda)$ to get the key pair $(\mathsf{sk}, \mathsf{vk})$, and sends vk to the adversary.

Phase: The challenger runs $\mathsf{Sign}(\mathsf{sk}, \mathsf{vk}, \mu_i)$ to obtain a signature Σ_i for message μ_i for $i \in [Q]$. Then, the challenger sends $\{\mu_i, \Sigma\}_{i\in[Q]}$ back to the adversary $\mathcal{A}$.

forge: . After receiving the message-signature pairs $\{\mu_i, \Sigma_i\}_{i\in[Q]}$, $\mathcal{A}$ outputs a pair (μ^*, Σ^*) as its forgery.

We say the adversary $\mathcal{A}$ wins in the game $\mathsf{forge}_{\mathcal{A}}^{\mathsf{naCMA}}(\lambda)$ if (1) $\mathsf{Verify}(\Sigma^*, \mathsf{vk}, \mu^*) = \top$, and (2) $\mu^* \notin \{\mu_i\}_{i\in[q]}$. The formal definition of EUF-naCMA security as follows.

Definition A.3 (EUF-naCMA) *A signature scheme is called* $\mathsf{EUF\text{-}naCMA}$ *secure if for all* PPT *adversary* $\mathcal{A}$*, the following probability inequality holds.*

$$\mathsf{Pr}\left[\mathcal{A} \text{ wins } \mathsf{forge}_{\mathcal{A}}^{\mathsf{naCMA}}(\lambda)\right] \leq \mathsf{negl}(\lambda),$$

where $\mathsf{negl}(\lambda)$ *is the negligible function in security parameter* λ*.*

In the existentially unforgeable under adaptive chosen message attacks(EUF-CMA) security of a signature scheme, we allow the adversary to adaptively make signature queries. In other words, in the $\mathsf{EUF\text{-}CMA}$ security of a signature scheme, the adversary's signature queries may depend on the verification key vk and the previous signature queries. Note that there are two approaches that can be used to obtain a EUF-CMA secure signature scheme from a EUF-naCMA secure signature scheme: EUF-naCMA secure signature + One Time Signatures (OTS), and EUF-naCMA secure signature + chameleon hashing. Fortunately, there are lattice-based constructions [24, 41] for both chameleon hash and OTS. In this paper, we use the later approach that uses lattice-based secure chameleon hash.

A.2 Identity Based Encryption

IBE Definition. An IBE scheme consists of algorithm tuples $(\mathsf{Setup}, \mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec})$ as follows.

Definition A.4

$\mathsf{Setup}(\lambda) \to (\mathsf{mpk}, \mathsf{msk})$: *on input the security parameter λ, it outputs the master public key mpk and master secret key msk.*

$\mathsf{KeyGen}(\mathsf{msk}, \mathsf{mpk}, \mathsf{id}) \to \mathsf{sk}_{\mathsf{id}}$: *on input the master key pair $(\mathsf{mpk}, \mathsf{msk})$ and an identity id, it outputs the secret key $\mathsf{sk}_{\mathsf{id}}$ for the corresponding identity id.*

$\mathsf{Enc}(\mathsf{mpk}, \mathsf{id}, \mu) \to \mathsf{ct}$: *on input the master public key mpk, an identity id and a message μ, it outputs the ciphertext ct of the message μ under the identity id.*

$\mathsf{Dec}(\mathsf{mpk}, \mathsf{ct}, \mathsf{sk}_{\mathsf{id}}) \to \mu/\perp$: *on input the master public key mpk, ciphertext ct and the secret key $\mathsf{sk}_{\mathsf{id}}$ of identity id, it outputs a message μ if ct is the encryption of μ under the identity id, and outputs $\perp$ otherwise.*

Correctness. We say an IBE scheme Π_{IBE} is δ-correct if for the security parameter λ, identity $\mathsf{id} \in \mathsf{ID}$ and any message μ, the following holds

$$\mathsf{Pr}\left[\mu' \neq \mu \middle| \begin{array}{cl} (\mathsf{mpk}, \mathsf{msk}) & \leftarrow \mathsf{Setup}(\lambda); \\ \mathsf{sk}_{\mathsf{id}} & \leftarrow \mathsf{KeyGen}(\mathsf{msk}, \mathsf{mpk}, \mathsf{id}); \\ \mathsf{ct} & \leftarrow \mathsf{Enc}(\mathsf{mpk}, \mathsf{id}, \mu); \\ \mu' & \leftarrow \mathsf{Dec}(\mathsf{mpk}, \mathsf{ct}, \mathsf{sk}_{\mathsf{id}}) \end{array} \right] < \delta,$$

where the probability is taken over the randomness of the algorithms $\mathsf{Setup}, \mathsf{KeyGen}$ and the encryption algorithm Enc. If the value of δ is negligible in the security parameter λ, then we simly call the IBE scheme is correct.

Adaptive Security. The adaptive security of an IBE scheme is described by the following experiment $\mathsf{EXPT}^{\mathsf{ID\text{-}CPA}}_{\mathcal{A},\pi}(\lambda)$ between a challenger and an adversary $\mathcal{A}$ as follows:

$\mathsf{EXPT}^{\mathsf{ID\text{-}CPA}}_{\mathcal{A},\mathsf{IBE}}(\lambda) \to \{0, 1\}$:

Setup: At the beginning of the game, the challenger runs $\mathsf{Setup}(\lambda)$ to obtain master key pair $(\mathsf{mpk}, \mathsf{msk})$, and then it sends public master key mpk to the adversary $\mathcal{A}$.

Phase 1: The adversary may adaptively make secret key queries for the identities in ID. Receiving an identity $\mathsf{id} \in \mathsf{ID}$ as a key query fron the adversary, the challenger runs $\mathsf{KeyGen}(\mathsf{msk}, \mathsf{mpk}, \mathsf{id})$ to get a secret key $\mathsf{sk}_{\mathsf{id}}$ for the identity id, and sends it to the adversary.

Chall: After finishing Phase 1, the adversary sends out two equal-length messages μ_0, μ_1 and an identity $\mathsf{id}^* \in \mathsf{ID}$, which is never made a secret key query in previous Phase 1, to the challenger as its challenge. The challenger chooses a random bit $b \in \{0, 1\}$, and obtains cahllenge ciphertext ct^* by encrypting μ_b under the challenge identity id^*, and then sends ct^* to the adversary.

Phase 2: After the challenge query, the adversary $\mathcal{A}$ continue to make adaptive secret key query for the any identity $\mathsf{id} \neq \mathsf{id}^*$. The challenger responds as in Phase 1.

Guess: After all, the adversary outputs an guess b' of b. The experiment outputs 1 if $b = b'$, and 0 otherwise.

Definition A.5 *We say an* IBE *scheme achieves adaptive security under choosen message security attack (simply, adaptive security) if for any probabilistic polynomial time (PPT) adversary* $\mathcal{A}$ *and the experiment* $\mathsf{EXPT}^{\mathsf{ID\text{-}CPA}}_{\mathcal{A},\mathsf{IBE}}(\lambda)$ *defined above, the following holds:*

$$\left| \Pr\left[\mathsf{EXPT}^{\mathsf{ID\text{-}CPA}}_{\mathcal{A},\mathsf{IBE}}(\lambda) \to 1\right] - \frac{1}{2} \right| \leq \mathsf{negl}(\lambda),$$

where $\mathsf{negl}(\lambda)$ *is the negligible function in security parameter* λ*.*

A.3 Hash Functions

We recall the definition of chameleon hash from the work of [21] as follows.

Definition A.6 (Chameleon Hash) *For a message space* M*, and two distributions* $\mathcal{X}, \mathcal{Y}$ *over the sample space* X, Y*, A chameleon hash consists of three algorithms* $\mathsf{CH} = (\mathsf{Gen}, \mathsf{Hash}, \mathsf{Hash}^{-1})$ *as follows:*

$\mathsf{Gen}(\lambda) \to (\mathsf{ek}, \mathsf{sk})$: *On input the security parameter, it outputs the evaluation key* ek *and the secret key* sk*.*

$\mathsf{Hash}(ek, x \leftarrow \mathcal{X}, \mu \in \mathsf{M} \to y)$: *On input the evaluation key* ek*, an element* $x \in \mathsf{X}$ *and a message* $\mu \in \mathsf{M}$*, it outputs* $y \in \mathsf{Y}$ *such that* $y = hash(\mathsf{ek}, x, \mu)$*.*

$\mathsf{Hash}^{-1}(\mathsf{sk}, y \leftarrow \mathcal{Y}, \mu \in \mathsf{M}) \to x$: *On input the secret key* sk*, an element* $y \in \mathsf{Y}$ *and a message* $\mu \in \mathsf{M}$*, it outputs* $x \in \mathsf{X}$ *such that* $y = hash(\mathsf{ek}, x, \mu)$*.*

To construct adaptively secure lattice based signature scheme, we need following security from the chameleon hash in Definition A.6.

Definition A.7 *We say a chameleon hash* $\mathsf{CH} = (\mathsf{Gen}, \mathsf{Hash}, \mathsf{Hash}^{-1})$ *is secure if it satisfies following properties except with negligible probability over the randomness of* Gen*.*

Uniformity: For a message $\mu \in \mathsf{M}$ *and* $x \leftarrow \mathcal{X}$*, the distribution* $\mathsf{Hash}(\mathsf{ek}, x, \mu)$ *is statistically close to the distribution* $\mathcal{Y}$*.*

Preimage Sampling Correctness: For a message $\mu \in \mathsf{M}$ *and* $y \in \mathsf{Y}$*, the ditribution of* $\mathsf{Hash}^{-1}(\mathsf{sk}, y, \mu)$ *is statistically close to the conditional distribution* $\{x \leftarrow \mathcal{X} | \mathsf{Hash}(\mathsf{ek}, x, \mu) = y\}$*.*

Collision Resistance: For any probabilistic polynomial time algorithm $\mathcal{A}$, the following probability holds.

$$\Pr\left[\begin{array}{c}(\mu, x) \neq (\mu', x')\wedge \\ \mathsf{Hash}(\mathsf{ek}, x, \mu) = \mathsf{Hash}(\mathsf{ek}, x', \mu');\end{array}\middle|\begin{array}{l}(\mathsf{sk}, \mathsf{vk}) \leftarrow \mathsf{KeyGen}(\lambda); \\ (\mu, x, \mu', x') \leftarrow \mathcal{A}(\mathsf{ek});\end{array}\right] \leq \mathsf{negl}(\lambda).$$

A.4 Truncation Collision-Resistant Hash

Here we define the truncation collision-resistant hash function. For an integer n, let $H : \{0,1\}^* \to \{0,1\}^n$ be a hash function, then, for an index $i \in [n]$, we identify the i-th truncation of the hash functions H, denoted H_i, as a function which outputs the prefix of the output of the hash function H of length i. In particular, the n-th truncated hash function H_n is identical to H. Note that a hash function family can be constructed from a hash function by introducing an evaluation key. Namely, we can construct a hash function family $\mathcal{H}$ from a hash function H as follows:

$$\mathcal{H} := \left\{H_{\mathsf{ek}}, \mathsf{ek} \leftarrow \{0,1\}^\lambda | H_{\mathsf{ek}}(x) := H(x|\mathsf{ek}) \text{ for any } x \in \{0,1\}^*\right\}$$

We formally define the Truncation Collision Resistance (TCR) security of a hash function as follows.

Definition A.8 *We say a hash function family $\mathcal{H}$ is i-th truncation collision resistant if for any PPT algorithm $\mathcal{A}$, the following probability holds:*

$$\Pr_{H \leftarrow \mathcal{H}}\left[\begin{array}{c}(x_1, x_2, \cdots, x_Q) \leftarrow \mathcal{A}(H) \\ \exists j, k \in [Q], s.t. H_i(x_j) = H_i(x_k) \wedge x_j \neq x_k\end{array}\right] \leq \frac{t_{\mathcal{A}}^2}{2^{i+1}},$$

where H_i is the i-th truncation of H, Q is the number of elements that the algorithm $\mathcal{A}$ able to output, and $t_{\mathcal{A}}$ is the running time of the algorithm $\mathcal{A}$.

It is obvious that, for a small constant index $i \in [n]$, the i-th truncation collision resistance of any hash function trivially can be broken. This is the case, since the PPT algorithm $\mathcal{A}$ can output more than 2^i elements for any constant integer i, there exists an collision in the first i bits. Hence, to make use of the truncation collision property of a hash function, we need to set the index to a super-constant number. More detailed discussion about the rationality of the above definition can be found in [33].

A.5 Partition Functions

Partition function, also known as a balanced admissible hash function [32], is vital to the security proof of partition-based IBE schemes. Our following definition of partition function slightly generalizes the definition in [55]. We observe that, instead of outputting a binary value, the invertibility of the output of the partition function in R_q is sufficient for the security proof of ring lattice-based IBE schemes. Thus we re-write the definition of the partition function in [55] as follows.

Definition A.9 (Partition Function[[55]) *Let $\mathbf{F} = \{\mathbf{F}_\lambda : \mathcal{K}_\lambda \times \mathcal{X}_\lambda \to R_2\}$ be an ensemble of function family, we say that $\mathbf{F}$ is a partition function, if there exists an efficient algorithm $\mathsf{PrtSmp}(1^\lambda)$, which takes as input polynomial bounded $Q = Q(\lambda) \in \mathbb{N}$ and noticeable $\epsilon = \epsilon(\lambda) \in (0, 1/2]$ and outputs K such that :*

- *There exists $\lambda_0 \in N$ such that $\mathsf{Pr}\left[K \in \mathcal{K}_\lambda : K \xleftarrow{\$} \mathsf{PrtSmp}(1^\lambda, Q(\lambda), \epsilon(\lambda))\right] = 1$ for all $\lambda > \lambda_0$, where λ_0 may depend on $Q(\lambda)$ and $\epsilon(\lambda)$.*
- *For $\lambda > \lambda_0$, there exists $\gamma_{\max}(\lambda)$ and $\gamma_{\min}(\lambda)$ that depends on $Q(\lambda)$ and $\epsilon(\lambda)$ such that for all $X^{(1)}, \cdots, X^{(Q)}, X^* \in \mathcal{X}_\lambda$ with $X^* \notin \left\{X^{(1)}, \cdots, X^{(Q)}\right\}$*

$$\gamma_{\max}(\lambda) \geq \mathsf{Pr}\left[\mathbf{F}\left(X^{(1)}\right) \in R_q^*, \cdots, \mathbf{F}\left(X^{(Q)}\right) \in R_q^* \wedge \mathbf{F}\left(X^*\right) = 0\right] \geq \gamma_{\min}(\lambda)$$

  holds and the function $\tau(\lambda)$ defined as $\tau(\lambda) := \gamma_{\min} \cdot \epsilon(\lambda) - \frac{\gamma_{\max}(\lambda) - \gamma_{\min}(\lambda)}{2}$ is noticeable. The probability is taken over the random choice of the function key $K \xleftarrow{\$} \mathsf{PrtSmp}(1^\lambda, Q(\lambda), \epsilon(\lambda))$.

We call K the partition key and $\tau(\lambda)$ the quality of the partition function.

The proof of following lemma can be found in the paper of [3, 37]

Lemma A.10 (Lemma 8 in [37]) *Let us consider an IBE scheme and an adversary A that breaks the adaptively-anonymous security (resp. pseudorandomness) with advantage λ. Let the identity space (resp. input space) be X and consider a map γ that maps a sequence of elements in X to a value in [0,1]. We consider the following experiment. We first execute the security game for $\mathcal{A}$. Let X^* be the challenge identity (resp. challenge input) and $X_1, \cdots, X_Q$ be the identities (resp. inputs) for which key extraction queries (resp. evaluation queries) were made. We denote $\mathbb{X} = (X^*, X_1, \cdots, X_Q)$. At the end of the game, we set $coin' \in \{0, 1\}$ as $\hat{b} = b'$ with probability $\gamma(\mathbb{X})$ and $\hat{b} \leftarrow \{0, 1\}$ with probability $1 - \gamma(\mathbb{X})$. Then, the following holds*

$$\left|\mathsf{Pr}[\hat{b} == b] - \frac{1}{2}\right| \geq \gamma_{\min} \cdot \epsilon - \frac{\gamma_{\max} - \gamma_{\min}}{2}$$

where $\gamma_{\max}$ and $\gamma_{\min}$ are the maximum and the minimum of $\gamma(\mathbb{X})$ taken over all possible $\mathbb{X}$, respectively.

A.6 Supplementary Materials for Gaussians, Rings, and LWE

Lemma A.11 (Noise Rerandomization [37]) *Let q, ℓ, m be positive integers and r a positive real satisfying $r > \max\{\eta_{\epsilon_s}(\mathbb{Z}^m), \eta_{\epsilon_s}(\mathbb{Z}^\ell)\}$. Let $\mathsf{b} \in \mathbb{Z}_q^m$ be arbitrary and x chosen from $D_{\mathbb{Z}^m, r}$. Then for any $\mathsf{V} \in \mathbb{Z}^{m \times \ell}$ and positive real $\sigma > \mathsf{s}_1(\mathsf{V})$, there exists a PPT algorithm $\mathsf{ReRand}(\mathsf{V}, \mathsf{b} + \mathsf{x}, r, \sigma)$ that outputs $\mathsf{b}'^\top = \mathsf{b}^\top \mathsf{V} + \mathsf{x}'^\top \in \mathbb{Z}_q^\ell$ where the statistical distance of the discrete Gaussian $D_{\mathbb{Z}^\ell, 2r\sigma}$ and the distribution of x' is within $8\epsilon_s$.*

Lemma A.12 (Theorem 1 of [37]) *Let α be the positive real, m be a power of 2, ℓ be an integer, $\Phi(x) = x^n + 1$ be the mth cyclotomic polynomial where $m = 2n$, and $R = \mathbb{Z}[x]/(\Phi(x))$. Let $q \equiv 3 \mod 8$ be a prime such that there is another prime $p \equiv 1 \mod m$ satisfying $p \leq q \leq 2p$. Let $\sigma_{R\text{-}\mathsf{DLWE}} := \alpha q \geq n^{3/2}\ell^{1/4}\omega(\log^{9/4}(n))$. Then, there is a PPT quantum reduction from $\tilde{O}(n/\alpha)$-approximate SIVP(or SVP) to R-$\mathsf{DLWE}_{n,\ell,q,\chi}$ with $\chi = D^{\mathsf{Coeffs}}_{\mathbb{Z}^n, \sigma_{R\text{-}\mathsf{DLWE}}}$.*

A.7 Supplementary Materials for Lattice Trapdoors

Lemma A.13 ([37]) *Let n be a power of 2, q be prime larger than $4n$ such that $q \equiv 3 \mod 8$, b, ρ be positive integers satisfying $\rho < \frac{1}{2}\sqrt{q/n}$, and $\epsilon_s \in (0,1)$ be a small real regarding the smoothing parameter. Furthermore, define $\log_1(\cdot) := \log_2(\cdot)$. There are efficient algorithms such that:*

- $\mathsf{TrapGen}(n, k, \rho, q) \to (\mathsf{a}, \mathsf{T_a})$ ([43], Lemma 5.3): *A randomized algorithm that, when $k \geq 2\log_\rho(q)$, outputs a ring vector $\mathsf{a} \in R^k$ and a matrix $\mathsf{T_a} \in R^{k\times k}$, where $\mathsf{rot}(\mathsf{a}^\top) \in \mathbb{Z}^{n\times nk}$ is full-rank matrix and $\mathsf{rot}(\mathsf{T_a}) \in \mathbb{Z}^{nk\times nk}$ is a basis for $\Lambda^\perp(\mathsf{rot}(\mathsf{a}^\top))$ such that a is statistically close to uniform and $\|\mathsf{rot}(\mathsf{T_a})\|_{\mathsf{GS}} < O\left(b\rho\sqrt{n\log_\rho(q)}\right)$.*
- $\mathsf{SampleLeft}(\mathsf{a}, \mathsf{b}, \mathsf{T_a}, u, \sigma) \to \mathsf{e}$ ([16]): *A randomized algorithm that, on input the vectors $\mathsf{a}, \mathsf{b} \in R^k$, where $\mathsf{rot}(\mathsf{a}^\top), \mathsf{rot}(\mathsf{b}^\top) \in \mathbb{Z}^{n\times nk}$ are full-rank, an element $u \in R_q$, a matrix $\mathsf{T_a}$ such that $\mathsf{rot}(\mathsf{T_a}) \in \mathbb{Z}^{nk\times nk}$ is a basis for $\Lambda^\perp(\mathsf{rot}(\mathsf{a}^\top))$ and Gaussian parameter $\sigma > \|\mathsf{rot}(\mathsf{T_a})\|_{\mathsf{GS}} \cdot \sqrt{\log(2n(1+1/\epsilon_s))/\pi}$, outputs a vector $\mathsf{e} \in R^{2k}$ sampled from a distribution which is statistically close to $D_{\Lambda^\perp_{\mathsf{Coeffs}(u)}\left(\mathsf{rot}([\mathsf{a}^\top|\mathsf{b}^\top])\right),\sigma}$, i.e., $[\mathsf{a}^\top|\mathsf{b}^\top]\cdot\mathsf{e} = u$, and $\mathsf{Coeffs}(\mathsf{e})$ is distributed according to $D_{\Lambda^\perp_{\mathsf{Coeffs}(u)}\left(\mathsf{rot}([\mathsf{a}^\top|\mathsf{b}^\top])\right),\sigma}$.*
- $\mathsf{SampleRight}(\mathsf{a}, \mathsf{R}, u, y, \mathsf{g}_b, \mathsf{T}_{\mathsf{g}_b}, \sigma) \to \mathsf{e}$ *where $\mathsf{b}^\top = \mathsf{a}^\top\mathsf{R} + y\cdot\mathsf{g}_b^\top$* ([3]): *A randomized algorithm that, on input the ring vectors $\mathsf{a}, \mathsf{g}_b \in R^k$ such that $\mathsf{rot}(\mathsf{a}^\top), \mathsf{rot}(\mathsf{g}_b^\top) \in \mathbb{Z}^{n\times nk}$ are full-rank, elements $y \in R^*, u \in R$, a matrix $\mathsf{R} \in R^{k\times k}$, a matrix $\mathsf{T}_{\mathsf{g}_b} \in R^{k\times k}$ such that $\mathsf{rot}(\mathsf{T}_{\mathsf{g}_b})$ is a basis for the lattice $\Lambda^\perp(\mathsf{rot}(\mathsf{g}_b))$, and a Gaussian parameter $\sigma > s_1(\mathsf{R}) \cdot \|\mathsf{rot}(\mathsf{T}_{\mathsf{g}_b})\|_{\mathsf{GS}} \cdot \sqrt{\log(2n(1+1/\epsilon_s))/\pi}$, outputs a vector $\mathsf{e} \in R^{2k}$ sampled from a distribution which is statistically close to $D_{\Lambda^\perp_{\mathsf{Coeffs}(u)}\left(\mathsf{rot}([\mathsf{a}^\top|\mathsf{b}^\top])\right),\sigma}$, i.e., $[\mathsf{a}^\top|\mathsf{b}^\top]\cdot\mathsf{e} = u$, and $\mathsf{Coeffs}(\mathsf{e})$ is distributed according to $D_{\Lambda^\perp_{\mathsf{Coeffs}(u)}\left(\mathsf{rot}([\mathsf{a}^\top|\mathsf{b}^\top])\right),\sigma}$.*
- ([43]) *Let $k \geq \lceil\log_b(q)\rceil$. There is a publicly known matrix $\mathsf{T}_{\mathsf{g}_b}$ such that $\mathsf{rot}(\mathsf{T}_{\mathsf{g}_b})$ is a basis for the lattice $\Lambda^\perp(\mathsf{rot}(\mathsf{g}_b^\top))$ and $\|\mathsf{rot}(\mathsf{T}_{\mathsf{g}_b})\|_{\mathsf{GS}} \leq \sqrt{b^2+1}$. Furthermore, there exists a deterministic polynomial time algorithm g_b^{-1} which takes input $\mathsf{u} \in R_q^k$, and outputs $\mathsf{R} = \mathsf{g}_b^{-1}(\mathsf{u}^\top)$ such that $\mathsf{R} \in [-b, b]_{R^{k\times k}}$, $\mathsf{g}_b^\top \cdot \mathsf{R} = \mathsf{u}^\top$, and $s_1(\mathsf{R}) \leq nkb$. Similarly, there exists a randomized polynomial time algorithm $\widehat{\mathsf{g}_b}^{-1}$ which takes input $\mathsf{u} \in R_q^k$, and outputs $\mathsf{R} \leftarrow \widehat{\mathsf{g}_b}^{-1}(\mathsf{u}^\top)$ such that $\mathsf{g}_b^\top \cdot \mathsf{R} = \mathsf{u}^\top$. Each coefficient in any entry of R follows a sub-Gaussian centered 0 with parameter $O(1)$, implying $s_1(\mathsf{R}) \leq \tilde{O}(b\sqrt{nk})$ with an overwhelming probability.*

A.8 Supplementary Materials for Homomorphic Computations

Recall that we homomorphically evaluate the tag function (which will be described in the next section) in our design of signature scheme, to evaluate the quality of homomorphic evaluation of tag functions, we use the notion of δ-expanding from the previous literatures [2, 9, 55].

Definition A.14 *For the function $f_K : \mathcal{X}^2 \to \mathcal{Y} \in \mathbb{Z}$ and a function family $\mathcal{F}$ defined as $\mathcal{F} := \{f_K : K \in \mathcal{K}\}$, we say that the two deterministic algorithms* ($\mathsf{PubEval}, \mathsf{TrapEval}$) *are δ-expanding with respect to the function $f_K(\cdot) \in \mathcal{F}$ if they have following properties:*

- $\mathsf{PubEval}(\mathsf{b}_0, \mathsf{b}_1, f) \to \mathsf{b}_f \in R_q^k$ **:** *on input a encoding vectors* $\mathsf{b}_0, \mathsf{b}_1 \in R_q^k$ *and a function* f*, it outputs a vector* $\mathsf{b}_f \in R_q^k$

$$\mathsf{b}_f \leftarrow \mathsf{PubEval}(\mathsf{b}_0, \mathsf{b}_1, f)$$

- $\mathsf{TrapEval}(\mathsf{b}_0, \mathsf{b}_1, \mathsf{R}_0, \mathsf{R}_1, x \in \mathcal{X}^2, f) \to \mathsf{R}_f \in R_q^{k\times k}$ **:** *on input a encoding vectors* $\mathsf{b}_0, \mathsf{b}_1 \in R_q^k$*, trapdoor informations* $\mathsf{R}_0, \mathsf{R}_1 \in R_q^{k\times k}$*, an element* $x \in \mathcal{X}$ *such that* $\mathsf{b}_0 = \mathsf{a}^\top \cdot \mathsf{R}_0 + \mathsf{x}_0 \mathsf{g}_b^\top$ *and* $\mathsf{b}_1 = \mathsf{a}^\top \cdot \mathsf{R}_1 + \mathsf{x}_1 \mathsf{g}_b^\top$*, and the function* $f \in \mathcal{F}$*, this trapdoor evaluation algorithm outputs a matrix* $\mathsf{R}_f \in \mathbb{Z}_q^{k\times k}$ *such that*

$$\mathsf{PubEval}(\mathsf{b}_0, \mathsf{b}_1, f) = \mathsf{a}^\top \cdot \mathsf{R}_f + f(\mathsf{x}) \cdot \mathsf{g}_b^\top,$$

and $\|\mathsf{R}_f\| \leq \delta \cdot \|\mathsf{R}\|$.

B Supplementary Materials for Section 3

B.1 Proof of Theorem 3.4

Proof. To show the theorem, we nee to show $\|\mathsf{R}_F\| \leq d^2 \cdot \delta \cdot \max\{\|\mathsf{R}_{t^*}\|, \|\mathsf{R}_{i^*}\|\}$. In order to show this, we have followings for the 2-norm of R_F

$$\begin{aligned}\|\mathsf{R}_F\| \leq \|\mathsf{R}_{t^*}\| + \|\sum_{j=1}^{d} \mathsf{R}_{j,i^*} \cdot t_{\leq j}\| &\leq \|\mathsf{R}_{t^*}\| + \sum_{j=1}^{d} \|\mathsf{R}_{j,i^*}\| \cdot \|t_{\leq j}\| \\ &\leq d^2 \cdot \delta \cdot \max\{\|\mathsf{R}_{t^*}\|, \|\mathsf{R}_{i^*}\|\}.\end{aligned}$$

where we used the triangular inequality property of the norm and the fact that $deg(t_{\leq j}(x)) < d$. This completes the proof. ☐

C Supplementary Materials for Section 4

C.1 Proof of Theorem 4.2

Proof. To show the theorem, we first to define a PrtSmp and then need to show that the PrtSmp satisfies two properties of partition function in Definition A.9. Our definition of PrtSmp as follows:

$\mathsf{PrtSmp}(1^\lambda, t_{\mathcal{A}}, \epsilon)$ **:** let $i^* \in [d]$ be the integer such that

$$\frac{1}{2^{i^*+1}} \leq \frac{\epsilon}{2t_{\mathcal{A}}^2} \leq \frac{1}{2^{i^*}},$$

then it selects a random $t^* \xleftarrow{\$} \{0,1\}^d$, and outputs $K = (t^*, i^*)$.

We now show that PrtSmp satisfies the first property in Definition A.9. When $d = O(\lambda)$, $\epsilon(\lambda)$ is noticeable, and $t_{\mathcal{A}}(\lambda)$ is polynomially bounded, then we have $2^d > 2t_{\mathcal{A}}^2/\epsilon(\lambda)$ for sufficiently large λ, and thus there exists an integer $i^* := \left\lfloor \log\left(\frac{2t_{\mathcal{A}}^2}{\epsilon(\lambda)}\right)\right\rfloor <= d$ such that

$$2^{i^*} = 2^{\left\lfloor \log\left(\frac{2t_{\mathcal{A}}^2}{\epsilon(\lambda)}\right)\right\rfloor} \leq 2^{\log\left(\frac{2t_{\mathcal{A}}^2}{\epsilon(\lambda)}\right)} = \left(\frac{2t_{\mathcal{A}}^2}{\epsilon(\lambda)}\right) \leq 2^{\left\lfloor \log\left(\frac{2t_{\mathcal{A}}^2}{\epsilon(\lambda)}\right)\right\rfloor+1} = 2^{i^*+1}.$$

Therefore, we have that $K = (t^*, i^*) \in \{0,1\}^d \times [d] = \mathcal{K}$ for sufficiently large parameter λ.

Next, we show that PrtSmp satisfies the second property in Definition A.9. Let $\gamma = \gamma(X^{(1)}, X^{(2)}, \cdots, X^{(Q)}, X^*) := \Pr_K\left[\bigwedge_{i=1}^{Q} \mathbf{F}\left(X^{(i)}\right) \in R_q^* \wedge \mathbf{F}\left(X^*\right) = 0\right]$, we first show (1) $\gamma_{\max} = \frac{1}{2^{i^*}}$ is an upper bound of $\gamma(X^{(1)}, X^{(2)}, \cdots, X^{(Q)}, X^*)$, and (2) $\gamma_{\min} = \left(1 - \frac{t_{\mathcal{A}}^2}{2^{i^*+1}}\right) \cdot \frac{1}{2^{i^*}}$ is an lower bound of $\gamma(X^{(1)}, X^{(2)}, \cdots, X^{(Q)}, X^*)$.

To show (1), we have that

$$\begin{aligned}
\gamma(X^{(1)}, X^{(2)}, \cdots, X^{(Q)}, X^*) &\leq \Pr_K\left[\mathbf{F}\left(X^*\right) = 0\right] \\
&= \Pr_K\left[\mathsf{Hash}(X)_{\leq i^*}(x) - t^*_{\leq i^*}(x) = 0\right] \\
&= \Pr_K\left[\mathsf{Hash}(X)_{\leq i^*} = t^*_{\leq i^*}\right] \\
&= \frac{1}{2^{i^*}},
\end{aligned}$$

where the last equality is from the randomness of $t \in \{0,1\}^d$.

To show (2), we have followings:

$$\begin{aligned}
\gamma(X^{(1)}, X^{(2)}, \cdots, X^{(Q)}, X^*) &= \Pr_K\left[\bigwedge_{i=1}^{Q} \mathbf{F}\left(X^{(i)}\right) \in R_q^* | \mathbf{F}\left(X^*\right) = 0\right] \cdot \Pr_K\left[\mathbf{F}\left(X^*\right) = 0\right] \\
&= \Pr_K\left[\exists j \in [Q], s.t., \mathbf{F}\left(X^{(j)}\right) \notin R_q^* | \mathbf{F}\left(X^*\right) = 0\right] \cdot \frac{1}{2^{i^*}} \\
&= \left(1 - \Pr_K\left[\exists j \in [Q], s.t., \mathbf{F}\left(X^{(j)}\right) \notin R_q^* | \mathbf{F}\left(X^*\right) = 0\right]\right) \cdot \frac{1}{2^{i^*}}
\end{aligned}$$

Note that the coefficients of $\mathbf{F}\left(X^{(j)}\right)$, for any $j \in [Q]$, are in $\{-1, 0, 1\}$. Additionally, Lemma 2.3 shows that the polynomial $\mathbf{F}\left(X^{(j)}\right)$ is invertible if it is non-zero. Therefore, we have

$$\begin{aligned}
\gamma(X^{(1)}, X^{(2)}, \cdots, X^{(Q)}, X^*) &= \left(1 - \Pr_K\left[\exists j \in [Q], s.t., \mathbf{F}\left(X^{(j)}\right) = 0 | \mathbf{F}\left(X^*\right) = 0\right]\right) \cdot \frac{1}{2^{i^*}} \\
&\geq \left(1 - \Pr_K\left[\exists j \in [Q], s.t., \mathbf{F}\left(X^{(j)}\right) = \mathbf{F}\left(X^*\right)\right]\right) \cdot \frac{1}{2^{i^*}} \\
&\geq \left(1 - \Pr_K\left[\exists j \in [Q], s.t., \mathsf{Hash}_{i^*}\left(X^{(j)}\right) = \mathsf{Hash}_{i^*}\left(X^*\right)\right]\right) \cdot \frac{1}{2^{i^*}} \\
&\geq \left(1 - \frac{t_{\mathcal{A}}^2}{2^{i^*+1}}\right) \cdot \frac{1}{2^{i^*}},
\end{aligned}$$

where the last inequality is from the TCR property of Hash.

Next we show that $\tau(\lambda) = \gamma_{\min} \cdot \epsilon - \frac{\gamma_{\max} - \gamma_{\min}}{2}$ is noticeable. From above results, we have that

$$\tau = \left(1 - \frac{t_{\mathcal{A}}^2}{2^{i^*+1}}\right) \cdot \frac{1}{2^{i^*}} \cdot \epsilon - \left(\frac{1}{2^{i^*}} - \left(1 - \frac{t_{\mathcal{A}}^2}{2^{i^*+1}}\right) \cdot \frac{1}{2^{i^*}}\right) \cdot \frac{1}{2} \tag{2}$$

$$= \left(1 - \frac{t_{\mathcal{A}}^2}{2^{i^*+1}}\right) \cdot \frac{1}{2^{i^*}} \cdot \epsilon - \frac{1}{2^{i^*}} \cdot \frac{t_{\mathcal{A}}^2}{2^{i^*+1}} \cdot \frac{1}{2} \tag{3}$$

$$\geq \frac{1}{2^{i^*}} \cdot \frac{\epsilon}{2} - \frac{1}{2^{i^*}} \cdot \frac{\epsilon}{4} \tag{4}$$

$$\geq \frac{\epsilon}{2t_{\mathcal{A}}^2} \cdot \frac{\epsilon}{4} = \frac{\epsilon^2}{8t_{\mathcal{A}}^2}, \tag{5}$$

where the inequalities are from the setting of i^*. Since ϵ is noticeable and $t_{\mathcal{A}}$ is polynomially bounded, thus $\tau(\lambda)$ is noticeable. This completes the proof. $\square$

D Supplementary Materials for Section 5

D.1 Proof of Theorem 5.1

Proof. To show the theorem, we need to show (1) $[\mathsf{a}^\top | \mathsf{a}_t^\top]\mathsf{e} = u$, and (2) $\mathsf{Pr}[\|\mathsf{e}\| \geq \sigma\sqrt{2nk}] \leq \mathsf{negl}(\lambda)$. The first statement (1) is obvious from the signing procedure. Now we show (2). Since $\mathsf{T_a}$ is generated by $\mathsf{TrapGen}$, thus $\|\mathsf{T_a}\|_{GS} \leq O\left(b\rho\sqrt{n \log_\rho(q)}\right)$ from the first property of Lemma A.13. Furthermore, from the theorem assumption we know that the gaussian parameter σ used in $\mathsf{SampleLeft}$ satisfies $\sigma > \|\mathsf{T_a}\|_{GS} \cdot \sqrt{\frac{\log(2n(1+1/\epsilon_s))}{\pi}}$, and thus the second property of Lemma A.13 shows that e is statistically close to $D_{\Lambda^\perp_{\mathsf{Coeffs}(u)}(\mathsf{rot}([\mathsf{a}^\top | \mathsf{a}_t^\top])),\sigma}$. Therefore, from Lemma 2.1 we have that $\mathsf{Pr}[\|\mathsf{e}\| > \sigma\sqrt{2nk}] \leq \mathsf{negl}(\lambda)$. This completes the proof. $\square$

D.2 Proof of Lemma 5.3

Proof. Since the choice of the tags in Game_1 are independent of the view of $\mathcal{A}$, the event abort is independent of $\mathcal{A}$, it is not hard to see that Game_1 is identical to Game_0 if abort not happens. In other words, we have that $|\mathsf{Pr}[E_0] - \mathsf{Pr}[E_1]| \leq \mathsf{Pr}[\overline{\mathsf{abort}}]$. Additionally, the choice of i^* guarantees that the event abort happens with probability at most $\frac{\epsilon}{2}$. $\square$

D.3 Proof of Lemma 5.4

Proof. In Game_2, the challenger changed the way of generating the key pairs $(\mathsf{sk}, \mathsf{vk})$ and signing procedure. Therefore, in order to show this lemma, it suffices to show: (1) the difference between the key pair generating procedure of the two games is bounded by $\mathsf{negl}(\lambda)$, and (2) the difference between the signing procedure of the two games is also bounded by $\mathsf{negl}(\lambda)$. Applying the union bound, we complete the proof of this lemma.

Now we show the statement (1): Note that the verification key components $\mathsf{b}_0, \mathsf{b}_1, \mathsf{u}$ in Game_2 are generated by setting $\mathsf{b}_0 = \mathsf{a}^\top \cdot \mathsf{R}_0 + t^*_{\leq i^*} \cdot \mathsf{g}_b, \mathsf{b}_1 = \mathsf{a}^\top \cdot \mathsf{R}_1 + x^{i^*} \cdot \mathsf{g}_b$, and $\mathsf{u} = \mathsf{a}^\top \cdot \mathsf{R}_u$ for randomly chosen $\mathsf{a} \in R^{k \times k}$, $\mathsf{R}_0, \mathsf{R}_1, \mathsf{R}_u \in R^{k \times k}_{[-\rho,\rho]}$. Therefore, from lemma 2.5, $\mathsf{a}, \mathsf{b}_0, \mathsf{b}_1$ and u are uniformly distributed except with probability $\mathsf{negl}(\lambda)$. Note that the ring element v is set as

$$v = [\mathsf{a}^\top, \mathsf{a}^\top_{t^*}]\mathsf{e}^* - \mathsf{u}^\top \boldsymbol{\mu}^* = \mathsf{a}^\top \cdot \mathsf{e}^*_1 + \mathsf{a}^\top_{t^*} \cdot \mathsf{e}^*_2 - \mathsf{u}^\top \boldsymbol{\mu}^*,$$

where $\mathsf{e}^* = (e^*_1, e^*_2) \leftarrow D^{\mathsf{Coeffs}}_{R^{2k},\sigma}$, and thus v is uniformly distributed except with probability $\mathsf{negl}(\lambda)$ by lemma 2.5 (here we implicitly set $\sigma \geq \omega(\sqrt{nk})$). Furthermore, note that the ring vector a is sampled uniformly form R^k_q in Game_2 and it is generated by $\mathsf{TrapGen}$ in Game_1. From the first property of Lemma A.13, we also have that the difference between the two cases is negligible in security parameter λ. Thus, from the union bound, we have statement (1).

Next we show statement (2): In spite of the fact that the challenger in Game_2 have no trapdoor for the random ring vector a, yet the challenger able to generate samples, which are statistically close to $D^{\mathsf{Coeffs}}_{[\mathsf{a}^\top,\mathsf{a}^\top_t],u,\sigma}$ for large enough σ. Since the challenger have the trapdoor R_t for the matrix a_t for the messages $\boldsymbol{\mu}_j$ such that $\boldsymbol{\mu}_j \neq \boldsymbol{\mu}^*$, from third property of lemma A.13, except with negligible probability in security parameter λ, the challenger is able to generate a ring vector $\mathsf{e} \in R^{2k}$ from a distribution which statistically close to $D^{\mathsf{Coeffs}}_{[\mathsf{a}^\top,\mathsf{a}^\top_t],u,\sigma_1}$ by running $\mathsf{SampleRight}(\mathsf{a}, \mathsf{R}_t, u, F_{t^*,i^*}(t^j), \mathsf{g}_b, \mathsf{T}_{\mathsf{g}_b}, \sigma)$. For the challenge message $\boldsymbol{\mu}^*$, we note that the ring vector e^* is a valid signature for μ^* except with probability $\mathsf{negl}(\lambda)$ from the properties of $\mathsf{SampleLeft}$. Thus the statement (2) holds by the union bound.

This completes the proof of the lemma.

□

D.4 Proof of Lemma 5.5

Proof. We show the lemma by proving above three statements separately.

First we show statement (1): Note that the pre-sampled tag t^* has random prefix of length i^* independent from the view of adversary, thus we have $\mathsf{Pr}[t^*_{\leq i^*} = t^f_{\leq i^*}] = \frac{1}{2^{i^*}}$. Therefore, from the definition of F_t, we have $F_{t^f}(t^*, i^*) = 0$ with probability $\frac{1}{2^{i^*}}$. Since e^f is a valid signature for μ^f, we have $\mathsf{a}^\top_{t^f} \cdot \mathsf{e}^f = u = \mathsf{u}^\top \cdot \boldsymbol{\mu}^f + v$, and thus with probability $\frac{1}{2^{i^*}}$ we further have following

$$v = \mathsf{a}^\top_{t^f} \cdot \mathsf{e}^f - \mathsf{u}^\top \cdot \boldsymbol{\mu}^f = [\mathsf{a}^\top | \mathsf{a}^\top \cdot \mathsf{R}_{t^f}] \cdot \begin{bmatrix} \mathsf{e}^f_1 \\ \mathsf{e}^f_2 \end{bmatrix} - \mathsf{a}^\top \cdot \mathsf{R}_u \cdot \mu^f \tag{6}$$

$$= \mathsf{a}^\top [\mathsf{e}^f_1 + \mathsf{R}_{t^f} \cdot \mathsf{e}^f_2 - \mathsf{R}_u \cdot \mu^f] \tag{7}$$

$$= \mathsf{a}^\top \left[I_k | \mathsf{R}_{t^*} | \mathsf{R}_{t^f} | \mathsf{R}_u \right] \cdot \begin{bmatrix} \mathsf{e}^f_1 \\ 0 \\ \mathsf{e}^f_2 \\ \boldsymbol{\mu}^f \end{bmatrix} \tag{8}$$

where we used the fact that $F_{t^f}(t^*, i^*) = 0$ in the second equality.

In addition, from the fact that e^* is a valid signature for the message $\boldsymbol{\mu}^*$, thus we have a similar result as above that

$$v = \mathsf{a}_{t^*}^\top \cdot \mathsf{e}^* - \mathsf{u}^\top \cdot \boldsymbol{\mu}^* = [\mathsf{a}^\top | \mathsf{a}^\top \cdot \mathsf{R}_{t^*}] \cdot \begin{bmatrix} \mathsf{e}_1^* \\ \mathsf{e}_2^* \end{bmatrix} - \mathsf{a}^\top \cdot \mathsf{R}_u \cdot \mu^* \tag{9}$$

$$= \mathsf{a}^\top \left[I_k | \mathsf{R}_{t^*} | \mathsf{R}_{t^f} | \mathsf{R}_u \right] \cdot \begin{bmatrix} \mathsf{e}_1^* \\ \mathsf{e}_2^* \\ 0 \\ \boldsymbol{\mu}^* \end{bmatrix} \tag{10}$$

where we used the fact that $F_{t^*}(t^*, i^*) = 0$. Above expression (13) and expression (15) shows that we have $\mathsf{a} \cdot \mathsf{s} = 0$ with probability 2^{-i^*}. This completes the prove of statement (1).

Next we show statement (2): Since $\mathcal{A}$'s output $\left(\boldsymbol{\mu}^f, \Sigma^f\right)$ is a valid forgery, thus with overwhelming probability we have the *case* that $(\boldsymbol{\mu}^f, t^f, \mathsf{e}^f) \neq (\boldsymbol{\mu}^*, t^*, \mathsf{e}^*)$. We further split it into the following subcases such that the union of the subcases is the *case*. We prove the statement (2) by showing $\Pr[\mathsf{s} = 0] \leq \mathsf{negl}(\lambda)$ for each subcase as follows.

If $\boldsymbol{\mu}^* - \boldsymbol{\mu}^f \neq 0$: From Lemma 2.5, the vector $\mathsf{R}_u \cdot (\boldsymbol{\mu}^* - \boldsymbol{\mu}^f)$ conditioned on a and $\mathsf{a}^\top \cdot \mathsf{R}_u$ has at least $\Omega(n)$ bits of min-entropy [12] , and thus $\Pr[\mathsf{s} = 0] \leq 2^{\Omega(-n)} \leq \mathsf{negl}(\lambda)$.

If $\mathsf{e}_2^* - \mathsf{e}_2^f \neq 0$: From the description of $\mathsf{TrapEval}$, the following holds.

$$\mathsf{R}_{t^*} = -\mathsf{R}_0 + \sum_{j=1}^{d} \mathsf{TrapEval}_{ET}(\mathsf{R}_1, \mathsf{Equal}_j) \cdot t^*_{\leq j}(x)$$

$$\mathsf{R}_{t^f} = -\mathsf{R}_0 + \sum_{j=1}^{d} \mathsf{TrapEval}_{ET}(\mathsf{R}_1, \mathsf{Equal}_j) \cdot t^f_{\leq j}(x)$$

Therefore, we have

$$\begin{aligned} \mathsf{R}_{t^f} \cdot \mathsf{s}_2^f - \mathsf{R}_{t^*} \cdot \mathsf{s}_2^* &= \mathsf{R}_0 \cdot (\mathsf{s}_2^* - \mathsf{s}_2^f) \\ &+ \left(\sum_{j=1}^{d} \mathsf{TrapEval}_{ET}(\mathsf{R}_1, \mathsf{Equal}_j) \cdot \left(t^f_{\leq j}(x) - t^*_{\leq j}(x) \right) \right) \cdot (\mathsf{s}_2^* - \mathsf{s}_2^f). \end{aligned}$$

Thus even revealing R_1 and R_u, Lemma 2.5 shows that $\mathsf{R}_0 \cdot (\mathsf{s}_2^* - \mathsf{s}_2^f)$ has at least $\Omega(n)$ bits of min-entropy conditioned on a and $\mathsf{a}^\top \cdot \mathsf{R}_0$, and thus $\Pr[\mathsf{s} = 0] \leq 2^{\Omega(-n)} \leq \mathsf{negl}(\lambda)$.

If $\mathsf{e}_1^* - \mathsf{e}_1^f = 0$ and $t^* \neq t^f$: From the description of $\mathsf{TrapEval}_{ET}$, we know that there is an additive term $\mathsf{R}_1 \cdot x^\kappa$ for some integer κ, and thus we have an additive term $\mathsf{R}_1 \cdot x^\kappa \cdot \left(t^f_{\leq j}(x) - t^*_{\leq j}(x) \right) \cdot \mathsf{e}_2^*$ in the above expression of $\mathsf{R}_{t^f} \cdot \mathsf{s}_2^f - \mathsf{R}_{t^*} \cdot \mathsf{s}_2^*$. Additionally, the

[12] Assume the first μ' of the vector $\boldsymbol{\mu}^* - \boldsymbol{\mu}^f$ is non-zero. From the first property of Lemma 2.4, we know that $a^\top \cdot \mathsf{R}_u$ is statistically close to a uniform sample over R_q. Thus $a^\top \cdot \mathsf{R}_u$ reveals negligible bits of information about the ring element $\mathsf{R}_{u,00}$. From a similar analysis as the proof of Corollary 8 in [21], we can have $\mathsf{R}_{u,00}\mu'$ has at least $\Omega(n)$ bits of entropy, and thus $\mathsf{R}_u \cdot (\boldsymbol{\mu}^* - \boldsymbol{\mu}^f)$ has at least $\Omega(n)$ bits of entropy.

Lemma 2.5 shows that $\mathsf{R}_1 \cdot \mathsf{s}_2^*$ has at least $\Omega(n)$ bits of min-entropy conditioned on a and $\mathsf{a}^\top \cdot \mathsf{R}_1$. Since $t^* \neq t^f$, there is some index $j \in [d]$ such that $\left(t^f_{\leq j}(x) - t^*_{\leq j}(x)\right) \neq 0$, and thus $x^\kappa \cdot \left(t^f_{\leq j}(x) - t^*_{\leq j}(x)\right)$ is invertible from Lemma 2.2. As a result, the ring vector $\mathsf{R}_1 \cdot x^\kappa \cdot \left(t^f_{\leq j}(x) - t^*_{\leq j}(x)\right) \cdot \mathsf{e}_2^*$ also has at least $\Omega(n)$ bits of min-entropy conditioned on a and $\mathsf{a}^\top \cdot \mathsf{R}_1$, and thus $\Pr[\mathsf{s} = 0] \leq 2^{\Omega(-n)} \leq \mathsf{negl}(\lambda)$.

If $\boldsymbol{\mu}^* - \boldsymbol{\mu}^f = 0$, $\mathsf{e}_2^* - \mathsf{e}_2^f = 0$, $t^* = t^f$, and $\mathsf{e}_1^* - \mathsf{e}_1^f \neq 0$: In this case, it is easy to see that $\mathsf{s} = \mathsf{e}_1^* - \mathsf{e}_1^f$, and thus obviously $\Pr[\mathsf{s} = 0] \leq 2^{\Omega(-n)} \leq \mathsf{negl}(\lambda)$.

Now we show statement (3): Since $\mu^*,\ \mu^f \in R_2^k$, we have $\|\mu^* - \mu^f\| \leq \sqrt{nk}$. As the challenger randomly sampled $\mathsf{R}_u \leftarrow R^{k\times k}_{[-\rho,\rho]}$, we have $\|\mathsf{R}_u\| \leq O\left(\rho \cdot \sqrt{nk}\right)$ from Lemma 2.6. Thus we obtain that $\|\mathsf{R}_u \cdot (\mathsf{e}^f - \mathsf{e}^*)\| \leq O\left(\rho \cdot nk\right)$.

From the description of $\mathsf{TrapEval}$, we have that $\|\mathsf{R}_{t^*}\| \leq O\left(n^2 \cdot (kb)^2\right) \cdot \max\{\|\mathsf{R}_0\|, \|\mathsf{R}_1\|\}$. Furthermore, form Lemma 2.6, we have that

$$\|\mathsf{R}_{t^*}\| \leq O\left(n^2 \cdot (kb)^2 \cdot \rho \cdot \sqrt{nk}\right).$$

We also have the same norm bound for R_{t^f}.

In order to use $\mathsf{SampleRight}$, from Lemma A.13, we need $\sigma > \|\mathsf{R}_{t^*}\| \cdot \sqrt{b^2+1} \cdot \omega(\sqrt{\log(n)})$. Since e^* is a valid signature , except with negligible probability, we have $\|\mathsf{e}^*\| \leq \sigma\sqrt{nk}$ from Lemma 2.1, and thus

$$\|\mathsf{e}^*\| \leq \tilde{O}\left(n^2 \cdot (kb)^2 \cdot \rho \cdot \sqrt{nk} \cdot \sqrt{b^2+1} \cdot \sqrt{nk}\right)$$

except with negligible probability. We have same norm bound for the ring vector e^f. From above arguments we have following upper bound for s except with negligible probability.

$$\begin{aligned}\|\mathsf{s}\| &\leq \|\mathsf{e}_1^f - \mathsf{e}_1^*\| + \|\mathsf{R}_{t^*} \cdot \mathsf{e}_2^*\| + \|\mathsf{R}_{t^f} \cdot \mathsf{e}_2^f\| + \|\mathsf{R}_u \cdot (\boldsymbol{\mu}^f - \boldsymbol{\mu}^*)\| \\ &\leq \tilde{O}\left(n^2 \cdot (kb)^2 \cdot \rho \cdot nk \cdot \sqrt{b^2+1}\right) O\left(\cdot n^2 \cdot (kb)^2 \cdot \rho \cdot \sqrt{nk}\right) \\ &\leq \tilde{O}\left((nkb)^5 \cdot \rho^2 \cdot \sqrt{nk}\right) = \beta\end{aligned}$$

This complete the proof of lemma. □

D.5 Adaptively Secure Signature Scheme

Note that the signature scheme presented in Construction 1 is non-adaptively secure. In this section, we present the construction of a lattice-based adaptively secure signature scheme in the standard model. In order to obtain adaptive security, we use a chameleon hash. Fortunately, the work of [21] showed the existence of a lattice-based secure chameleon hash (see section B.3 of [21]). The construction of an adaptively secure signature scheme is as follows.

Construction D.1 *Let* $\mathsf{CH} = (\mathsf{Gen}, \mathsf{Hash}, \mathsf{Hash}^{-1})$ *be a chameleon hash and let* $\mathsf{naCMA} \cdot \mathsf{Sign} = (\mathsf{naCMA} \cdot \mathsf{KeyGen}, \mathsf{naCMA} \cdot \mathsf{Sign}, \mathsf{naCMA} \cdot \mathsf{Verify})$ *be a signature scheme, the description of* $\mathsf{CMA} \cdot \mathsf{Sign} = (\mathsf{CMA} \cdot \mathsf{KeyGen}, \mathsf{CMA} \cdot \mathsf{Sign}, \mathsf{CMA} \cdot \mathsf{Verify})$ *as follows;*

$\mathsf{CMA} \cdot \mathsf{KeyGen}(\lambda) \to (\mathsf{sk}, \mathsf{vk})$: *On input the security parameter* λ*, it runs as follows:*

1. *Runs* $\mathsf{Gen}(\lambda)$ *to get* $(\mathsf{ek}_{\mathsf{Hash}}, \mathsf{sk}_{\mathsf{Hash}})$ *and runs* $\mathsf{naCMA} \cdot \mathsf{KeyGen}$ *to get* $(\mathsf{sk}_{\mathsf{naCMA}}, \mathsf{vk}_{\mathsf{naCMA}})$.
2. *Output the secret key* $\mathsf{sk} = (\mathsf{sk}_{\mathsf{Hash}}, \mathsf{sk}_{\mathsf{naCMA}})$ *and the verification key* $\mathsf{vk} = (\mathsf{ek}_{\mathsf{Hash}}, \mathsf{vk}_{\mathsf{naCMA}})$

$\mathsf{CMA}\text{-}\mathsf{Sign}(\mathsf{sk}, \mathsf{vk}, \mu) \to \Sigma$: *On input the secret key* sk*, the verification key* vk *and a message* μ*, it runs as follows:*

1. *Samples* $x \leftarrow \mathcal{X}$ *and computes* $y = \mathsf{Hash}(\mathsf{ek}_{\mathsf{Hash}}, x, \mu)$.
2. *Generates a signature* σ *for* y *by running* $\sigma \leftarrow \mathsf{naCMA} \cdot \mathsf{Sign}(\mathsf{sk}_{\mathsf{naCMA}}, \mathsf{vk}_{\mathsf{naCMA}}, y)$.
3. *Outputs* $\Sigma = (x, \sigma)$ *as a signature for* μ.

$\mathsf{CMA}\text{-}\mathsf{Verify}(\mathsf{vk}, \Sigma, \mu) \to \top/\bot$: *On input the verification key* vk*, a signature* Σ *and a message* $\mu \in \mathsf{M}$*, it runs as follows:*

1. *Splits* Σ *into two pieces as* $(x, \sigma) := \Sigma$ *and computes* $y = \mathsf{Hash}(\mathsf{ek}_{\mathsf{Hash}}, x, \mu)$.
2. *Outputs* $\mathsf{naCMA} \cdot \mathsf{Verify}(\mathsf{vk}_{\mathsf{naCMA}}, \sigma, y)$.

The correctness of the above construction can be easily obtained from the correctness of the chameleon hash and the correctness of the signature scheme $\mathsf{naCMA} \cdot \mathsf{Sign}$. The adaptive security of Construction D.1 can be obtained in a similar way as in the work of [21], thus we omit the proof here and simply present the result as follows.

Theorem D.2 *If* CH *is a secure chameleon hash, and the signature scheme* $\mathsf{naCMA} \cdot \mathsf{Sign}$ *is existentially unforgeable under non-adaptive chosen message attacks, then the signature scheme presented in ConstructionD.1 is existentially unforgeable under adaptive chosen message attacks.*

D.6 Proof of the Theorem 5.8

Proof. To show the theorem, it suffices to construct an algorithm $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$ which invokes $\mathcal{A}$ as a subroutine to solve the R-SIS problem with probability at least $\left(\epsilon - \frac{1}{2^{\mathsf{w}}}\right) \cdot \frac{\mathsf{L}+1}{2eQ2^{\frac{\mathsf{w}+\log(Q)}{\mathsf{L}}}}$, and running time of $T + \mathsf{poly}(n)$. Before giving the concrete description of $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$, we first analyse the relations between the following sequence of hybrids, from which we come up with the construction of $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$.

Game_0 : This is the original nan-adaptive signature forgery game between $\mathcal{A}$ and the challenger. Namely, in this game, the adversary $\mathcal{A}$ first claims the messages $\{\boldsymbol{\mu}_j\}_{j\in[Q]}$ as its signature query at the outset of the signature scheme.

Game_1 : In this game, the challenger randomly chooses tags $\left\{t^1, \cdots, t^Q\right\}$ from the tag space $\mathcal{T}$. Let $i^* \in [d]$ be the smallest index such that $\frac{e \cdot Q \cdot 2^{\frac{\mathsf{w}+\log(Q)}{\mathsf{L}}}}{\mathsf{L}+1} \leq 2^{i^*}$ (such i^* exists by setting $d = \Theta(n)$), if there is a tag class $\mathcal{T}_{\hat{t}_{\leq i^*}}$ of which cardinality is at least $\mathsf{L}+1$, then the challenger send $\perp$ to $\mathcal{A}$, and $\mathcal{A}$ aborts the game on receiving $\perp$. Otherwise the challenger chooses a random tag prefix $t^*_{\leq i^*} \in \mathcal{T}_{i^*}$, and checks if there is a *non-empty* tag class $\mathcal{T}_{t^*_{\leq i^*}}$.If so, the challenger sets $t^* := t^j$ for some tag $t^j \in \mathcal{T}_{t^*_{\leq i^*}}$, and sets $\boldsymbol{\mu}^* := \boldsymbol{\mu}_j$. Otherwise it randomly choose a dummy message $\boldsymbol{\mu}^*$ and extend $t^*_{\leq i^*}$ randomly to get a full tag $t^* \in \mathcal{T}$, and then add t^* to the class $\mathcal{T}_{t^*_{\leq i^*}}$. In the latter case, the class $\mathcal{T}_{t^*_{\leq i^*}}$ contains the single tag t^*. Then, the challenger randomly extends the tag prefix $t^*_{\leq i^*}$ for $\mathsf{L} - \|\mathcal{T}_{t^*_{\leq i^*}}\|$ times to obtain full tags $\{t^{*^j} \in \mathcal{T}\}_{j \in [\mathsf{L}] \setminus [\|\mathcal{T}_{t^*_{\leq i^*}}\|]}$, and add them to the class $\mathcal{T}_{t^*_{\leq i^*}}$, and generates corresponding dummy random messages $\{\mu^{*^j}\}_{j \in [\mathsf{L}] \setminus [\|\mathcal{T}_{t^*_{\leq i^*}}\|]}$. Now, there are L tags in $\mathcal{T}_{t^*_{\leq i^*}}$. In the rest of the game, the challenger generates the signatures corresponding to the pair $(t^j, \boldsymbol{\mu}_j)_{j \in [Q]}$ as in Game_0.

Game_2 : The only difference of this game from Game_1 is that, in this game, the challenger changes the way of generating verification key vk and signing algorithm. In this game, the challenger runs following algorithm $\mathsf{VerifyKey}$ to generate verification key vk.

$\mathsf{VerifyKey}\ (\lambda) \to \mathsf{vk}$:

$\mathsf{a} \leftarrow R_q^k$
$\mathsf{R}_0, \mathsf{R}_1, \mathsf{R}_u \leftarrow D_{R^{k \times k}, s}$ $\quad \triangleright\ s \geq \omega(\sqrt{\log(n)})$
Let $\mathsf{b}_0 := \mathsf{a}^\top \cdot \mathsf{R}_0 + t^*_{\leq i^*} \cdot \mathsf{g}_b$
Let $\mathsf{b}_1 := \mathsf{a}^\top \cdot \mathsf{R}_1 + x^{i^*} \cdot \mathsf{g}_b$
Let $\mathsf{u} := \mathsf{a}^\top \cdot \mathsf{R}_u$
$S_v := \emptyset$
for $j = 0$ to $\mathsf{L} - 1$ **do**
  $\mathsf{e}^{*^j} \leftarrow D^{\mathsf{Coeffs}}_{R^{2k}, \sigma}$
  $v_j := [\mathsf{a}^\top, \mathsf{a}^\top_{t^{*^j}}]\mathsf{e}^{*^j} - \mathsf{u}^\top \boldsymbol{\mu}^{*^j}$ $\quad \triangleright\ t^{*^j} \in \mathcal{T}_{t^*_{\leq i^*}}$
  $S_v := S_v \cup v_j$
Return $\mathsf{vk} := (\mathsf{a}, \mathsf{b}_0, \mathsf{b}_1, \mathsf{u}, S_v)$.

Let $\mathcal{M}_{t^j_{\leq i^*}}$ be the set of messages corresponding to the tags in the class $\mathcal{T}_{t^j_{\leq i^*}}$, to generate a signature for the message $\boldsymbol{\mu}^j$, the challenger do one of the followings: (1) if $\boldsymbol{\mu}^j \notin \mathcal{M}_{t^*_{\leq i^*}}$, it runs algorithm $\mathsf{Sign} \cdot \overline{\mathsf{Chall}}$, or (2) it runs algorithm $\mathsf{Sign} \cdot \mathsf{Chall}$ for $\boldsymbol{\mu}^j \in \mathcal{M}_{t^*_{\leq i^*}}$.

$\mathsf{Sign} \cdot \overline{\mathsf{Chall}}(\mathsf{sk}, \mathsf{vk}, \boldsymbol{\mu}^j, t^j) \to \Sigma$:

Let τ be the index of t^j in $\mathcal{T}_{t^j_{\leq i^*}}$
Compute $u_\tau = \mathsf{u}^\top \boldsymbol{\mu}^j + v_\tau$
$\mathsf{R}_t = \mathsf{TrapEval}(\mathsf{R}_0, \mathsf{R}_1, t^j)$
$\mathsf{e}_j \leftarrow \mathsf{SampleRight}(\mathsf{a}, \mathsf{R}_t, u_\tau, F_{t^j}(t^*, i^*), \mathsf{g}_b, \mathsf{T}_{\mathsf{g}_b}, \sigma)$
Return $\Sigma := (\tau, t^j, \mathsf{e}_j)$.

$\mathsf{Sign} \cdot \mathsf{Chall}(\mathsf{sk}, \mathsf{vk}, \boldsymbol{\mu}^j, t^j) \to \Sigma$:

Let τ be the index of t^j in $\mathcal{T}_{t^*_{\leq i^*}}$
Return $\Sigma := (\tau, t^j, \mathsf{e}^{*^\tau})$.

For $i \in \{0, 1, 2\}$, let E_i be the event that $\mathcal{A}$ wins in the Game_i, and let abort be the event that the challenger aborts in Game_1. Then, from the theorem assumption, we have that $\mathsf{Pr}[E_0] = \epsilon$ and the following lemmas.

Lemma D.3 $|\mathsf{Pr}[E_0] - \mathsf{Pr}[E_1]| \leq \frac{1}{2^{\mathsf{w}}}$.

Proof. Since the choice of the tags in Game_1 is independent of the view of $\mathcal{A}$, the event abort is independent of $\mathcal{A}$, it is not hard to see that Game_1 is identical to Game_0 if abort not happens. In other words, we have that $|\mathsf{Pr}[E_0] - \mathsf{Pr}[E_1]| \leq \mathsf{Pr}[\mathsf{abort}]$. Note that the abort happens if there are $\mathsf{L}+1$ tags such that their i^*-prefixes are matched. Therefore, we have the following for the probability of abort :

$$\begin{aligned}\mathsf{Pr}[\mathsf{abort}] = \binom{Q}{\mathsf{L}+1} \cdot \left(\frac{1}{2^{i^*}}\right)^{\mathsf{L}} &\leq \frac{Q(Q-1)\cdots(Q-\mathsf{L})}{(\mathsf{L}+1)!} \cdot \frac{(\mathsf{L}+1)^{\mathsf{L}}}{(eQ)^{\mathsf{L}} \cdot 2^{\mathsf{w}+\log(Q)}} \\ &\leq \frac{Q^{\mathsf{L}+1}}{(\mathsf{L}+1)!} \cdot \frac{(\mathsf{L}+1)^{\mathsf{L}}}{(eQ)^{\mathsf{L}} \cdot 2^{\mathsf{w}+\log(Q)}} = \frac{1}{(\mathsf{L}+1)!} \cdot \frac{(\mathsf{L}+1)^{\mathsf{L}}}{e^{\mathsf{L}} \cdot 2^{\mathsf{w}}} \\ &\leq \frac{e^{\mathsf{L}}}{(\mathsf{L}+1)^{\mathsf{L}+1}} \cdot \frac{(\mathsf{L}+1)^{\mathsf{L}}}{e^{\mathsf{L}} \cdot 2^{\mathsf{w}}} = \frac{1}{\mathsf{L}+1} \cdot \frac{1}{2^{\mathsf{w}}} \leq \frac{1}{2^{\mathsf{w}}},\end{aligned}$$

where the first inequality is from the setting of i^*; the third inequality is form the approximation that $\frac{n^n}{e^{n-1}} \leq n! \leq \frac{n^{n+1}}{e^{n-1}}$. Therefore, we have that

$$|\mathsf{Pr}[E_0] - \mathsf{Pr}[E_1]| \leq \mathsf{Pr}[\mathsf{abort}] \leq \frac{1}{2^{\mathsf{w}}}.$$

This completes the proof of lemma.

$\square$

Lemma D.4 $|\mathsf{Pr}[E_1] - \mathsf{Pr}[E_2]| \leq \mathsf{negl}(\lambda)$.

Proof. In Game_2, the challenger changed the way of generating the verification key vk and signing procedure. Therefore, to show this lemma, it suffices to show: (1) the difference between the verification key generating procedure of the two games is bounded by $\mathsf{negl}(\lambda)$, and (2) the difference between the signing procedure of the two games is also bounded by $\mathsf{negl}(\lambda)$. Using the union bound, we complete the proof of this lemma.

Now we show the statement (1): Note that the verification key components $\mathsf{b}_0, \mathsf{b}_1, \mathsf{u}$ in Game_2 are generated by setting $\mathsf{b}_0 = \mathsf{a}^\top \cdot \mathsf{R}_0 + t^*_{\leq i^*} \cdot \mathsf{g}_b, \mathsf{b}_1 = \mathsf{a}^\top \cdot \mathsf{R}_1 + x^{i^*} \cdot \mathsf{g}_b$, and $\mathsf{u} = \mathsf{a}^\top \cdot \mathsf{R}_u$ for randomly chosen $\mathsf{a} \in R^{k\times k}$, $\mathsf{R}_0, \mathsf{R}_1, \mathsf{R}_u \in R^{k\times k}_{[-\rho,\rho]}$. Therefore, from lemma 2.5, $\mathsf{a}, \mathsf{b}_0, \mathsf{b}_1$ and u are uniformly distributed except with probability $\mathsf{negl}(\lambda)$. Note that the elements v_j in S_v is set as $v_j = [\mathsf{a}^\top, \mathsf{a}^\top_{t^{*j}}]\mathsf{e}^{*j} - \mathsf{u}^\top \boldsymbol{\mu}^{*j}$ for $\mathsf{e}^{*j} \leftarrow D^{\mathsf{Coeffs}}_{R^{2k},\sigma}$, and thus v_j is uniformly distributed except with $\mathsf{negl}(\lambda)$ probability by lemma 2.5. Furthermore, note that the ring vector a is sampled uniformly from R_q^k in Game_2 and generated by $\mathsf{TrapGen}$ in Game_1. From the first property of Lemma A.13, we also have that the difference between the two cases is negligible in security parameter λ. Thus, from the union bound, we have statement (1).

Next, we show statement (2): Although the challenger in Game_2 has no trapdoor for the random ring vector a. However, the challenger can generate samples statistically close to $D^{\mathsf{Coeffs}}_{[\mathsf{a}^\top,\mathsf{a}_t^\top],u,\sigma}$ for large enough σ. Since the challenger have the trapdoor R_t for the matrix a_t corresponding to the messages $\boldsymbol{\mu}_j \notin \mathcal{M}_{t^*_{\leq i^*}}$, from third property of lemma A.13, except with negligible probability in security parameter λ, the challenger is able to generate a ring vector $\mathsf{e} \in R^{2k}$ from a distribution statistically close to $D^{\mathsf{Coeffs}}_{[\mathsf{a}^\top,\mathsf{a}_t^\top],u_\tau,\sigma_1}$ by running

$\mathsf{SampleRight}(\mathsf{a}, \mathsf{R}_t, u_\tau, F_{t^*,i^*}(t^j), \mathsf{g}_b, \mathsf{T}_{\mathsf{g}_b}, \sigma)$. For the challenge messages $\boldsymbol{\mu}^{*j} \in \mathcal{M}_{t^*_{\leq i^*}}$, we note that the ring vector e^{*j} is a valid signature for μ^{*j} except with probability $\mathsf{negl}(\lambda)$ from the properties of SampleLeft. Thus, statement (2) is held by the union bound.

This completes the proof of the lemma.

□

Now we show the description of algorithm $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$. The algorithm $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$ solves the R-SIS_β problem by invoking $\mathcal{A}$ as follow:

$\mathsf{Sim}_{R\text{-}\mathsf{SIS}}(\mathsf{a})$:

On input the R-SIS_β challenge $\mathsf{a} \in R^k$, the algorithm $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$ simulates the non-adaptive signature experiment $\mathsf{forge}^{\mathsf{naCMA}}_{\mathcal{A}}(\lambda)$ for $\mathcal{A}$ as the challenger did in Game_2. After receiving the signature queries $(\mu_1, \cdots, \mu_Q)$, the $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$ runs exactly as the challenger in Game_2 for a. Since a is uniformly distributed, $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$ can perfectly simulate the challenger in the Game_2. After receiving adversary's output forgery $\big(\boldsymbol{\mu}^f, \Sigma^f = (\tau^f, t^f, \mathsf{e}^f)\big)$, the $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$ splits e^f into two ring vectors as $\mathsf{e}^f = (\mathsf{e}^f_1, \mathsf{e}^f_2) \in R^k \times R^k$, and let $\mathsf{e}^{*^{\tau f}} = (\mathsf{e}^{*^{\tau f}}_1, \mathsf{e}^{*^{\tau f}}_2) \in R^k \times R^k$ be the signature for the message tag pair $(\boldsymbol{\mu}^{*^{\tau f}}, t^{*^{\tau f}})$. In what follows, we let $t^* := t^{*^{\tau f}}, \mu^* := \boldsymbol{\mu}^{*^{\tau f}}$ and $\mathsf{e}^* := \mathsf{e}^{*^{\tau f}}$ for notation simplicity. Finally, the $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$ outputs

$$\mathsf{s} := [(\mathsf{e}^f_1 - \mathsf{e}^*_1) + (\mathsf{R}_{t^f}\mathsf{e}^f_2 - \mathsf{R}_{t^*}\mathsf{e}^*_2) + \mathsf{R}_u(\boldsymbol{\mu}^f - \boldsymbol{\mu}^*)]$$

as its solution to the R-SIS_β problem.

Analysis of s. We have the following results for the output ring vector s from the above algorithm $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$. The full proof of this lemma can be found in Supplementary Material D.4

Lemma D.5 *If the output* $\big(\boldsymbol{\mu}^f, \Sigma^f\big)$ *by the adversary is a valid forgery, then we have the following results (1)* $\Pr[\mathsf{a}^\top \mathsf{s} = 0] \leq 2^{-i^*}$*, (2)* $\Pr[\mathsf{s} = 0] = \mathsf{negl}(\lambda)$*, and (3)* $\Pr[\|\mathsf{s}\| \leq \beta] \leq 1 - \mathsf{negl}(\lambda)$*.*

Proof. We show the lemma by proving the above three statements separately.

First, we show statement (1): Note that the pre-sampled tag t^* has a random prefix of length i^* independent from the view of adversary, thus we have $\Pr[t^*_{\leq i^*} = t^f_{\leq i^*}] = \frac{1}{2^{i^*}}$. Therefore, from the definition of F_t, we have $F_{t^f}(t^*, i^*) = 0$ with probability $\frac{1}{2^{i^*}}$. Since e^f is a valid signature for μ^f, we have $[\mathsf{a}^\top | \mathsf{a}^\top_{t^f}] \cdot \mathsf{e}^f = u_{\tau^f} = \mathsf{u}^\top \cdot \boldsymbol{\mu}^f + v_{\tau^f}$, and thus with probability $\frac{1}{2^{i^*}}$ we have

$$v_{\tau^f} = [\mathsf{a}^\top | \mathsf{a}^\top_{t^f}] \cdot \mathsf{e}^f - \mathsf{u}^\top \cdot \boldsymbol{\mu}^f = [\mathsf{a}^\top | \mathsf{a}^\top \cdot \mathsf{R}_{t^f}] \cdot \begin{bmatrix} \mathsf{e}^f_1 \\ \mathsf{e}^f_2 \end{bmatrix} - \mathsf{a}^\top \cdot \mathsf{R}_u \cdot \mu^f \tag{11}$$

$$= \mathsf{a}^\top [\mathsf{e}^f_1 + \mathsf{R}_{t^f} \cdot \mathsf{e}^f_2 - \mathsf{R}_u \cdot \mu^f] \tag{12}$$

$$= \mathsf{a}^\top \left[I_k | \mathsf{R}_{t^*} | \mathsf{R}_{t^f} | \mathsf{R}_u\right] \cdot \begin{bmatrix} \mathsf{e}^f_1 \\ 0 \\ \mathsf{e}^f_2 \\ \boldsymbol{\mu}^f \end{bmatrix} \tag{13}$$

where we used the fact that $F_{t^f}(t^*, i^*) = 0$ in the second equality.

In addition, from the fact that e^* is a valid signature for the message $\boldsymbol{\mu}^*$, thus we have a similar result as above that

$$v = \mathsf{a}_{t^*}^\top \cdot \mathsf{e}^* - \mathsf{u}^\top \cdot \boldsymbol{\mu}^* = [\mathsf{a}^\top | \mathsf{a}^\top \cdot \mathsf{R}_{t^*}] \cdot \begin{bmatrix} \mathsf{e}_1^* \\ \mathsf{e}_2^* \end{bmatrix} - \mathsf{a}^\top \cdot \mathsf{R}_u \cdot \mu^* \tag{14}$$

$$= \mathsf{a}^\top [I_k | \mathsf{R}_{t^*} | \mathsf{R}_{t^f} | \mathsf{R}_u] \cdot \begin{bmatrix} \mathsf{e}_1^* \\ \mathsf{e}_2^* \\ 0 \\ \boldsymbol{\mu}^* \end{bmatrix} \tag{15}$$

where we used the fact that $F_{t^{*\tau^f}}(t^*, i^*) = 0$. Above expression (13) and expression (15) shows that we have $\mathsf{a} \cdot \mathsf{s} = 0$ with probability 2^{-i^*}. This completes the proof of statement (1).

Next we show statement (2): Since $\mathcal{A}$'s output $\left(\boldsymbol{\mu}^f, \Sigma^f\right)$ is a valid forgery, thus with overwhelming probability we have the *case* that $(\boldsymbol{\mu}^f, \tau^f, t^f, \mathsf{e}^f) \neq (\boldsymbol{\mu}^*, \tau^f, t^*, \mathsf{e}^*)$. We further split it into the following subcases such that the union of the subcases is the *case*. We prove the statement (2) by showing $\mathsf{Pr}[\mathsf{s} = 0] \leq \mathsf{negl}(\lambda)$ for each subcase as follows.

If $\boldsymbol{\mu}^* - \boldsymbol{\mu}^f \neq 0$ **:** From Lemma 2.5, the vector $\mathsf{R}_u \cdot (\boldsymbol{\mu}^* - \boldsymbol{\mu}^f)$ conditioned on a and $\mathsf{a}^\top \cdot \mathsf{R}_u$ has at least $\Omega(n)$ bits of min-entropy, and thus $\mathsf{Pr}[\mathsf{s} = 0] \leq 2^{\Omega(-n)} \leq \mathsf{negl}(\lambda)$.

If $\mathsf{e}_2^* - \mathsf{e}_2^f \neq 0$ **:** From the description of $\mathsf{TrapEval}$, the following holds.

$$\mathsf{R}_{t^*} = -\mathsf{R}_0 + \sum_{j=1}^{d} \mathsf{TrapEval}_{ET}(\mathsf{R}_1, \mathsf{Equal}_j) \cdot t^*_{\leq j}(x)$$

$$\mathsf{R}_{t^f} = -\mathsf{R}_0 + \sum_{j=1}^{d} \mathsf{TrapEval}_{ET}(\mathsf{R}_1, \mathsf{Equal}_j) \cdot t^f_{\leq j}(x)$$

Therefore, we have

$$\begin{aligned} &\mathsf{R}_{t^f} \cdot \mathsf{s}_2^f - \mathsf{R}_{t^*} \cdot \mathsf{s}_2^* = \mathsf{R}_0 \cdot (\mathsf{s}_2^* - \mathsf{s}_2^f) \\ &+ \left(\sum_{j=1}^{d} \mathsf{TrapEval}_{ET}(\mathsf{R}_1, \mathsf{Equal}_j) \cdot \left(t^f_{\leq j}(x) - t^*_{\leq j}(x)\right)\right) \cdot (\mathsf{s}_2^* - \mathsf{s}_2^f). \end{aligned}$$

Thus even revealing R_1 and R_u, Lemma 2.5 shows that $\mathsf{R}_0 \cdot (\mathsf{s}_2^* - \mathsf{s}_2^f)$ has at least $\Omega(n)$ bits of min-entropy conditioned on a and $\mathsf{a}^\top \cdot \mathsf{R}_0$, and thus $\mathsf{Pr}[\mathsf{s} = 0] \leq 2^{\Omega(-n)} \leq \mathsf{negl}(\lambda)$.

If $\mathsf{e}_1^* - \mathsf{e}_1^f = 0$ **and** $t^* \neq t^f$ **:** From the description of $\mathsf{TrapEval}_{ET}$, we know that there is an additive term $\mathsf{R}_1 \cdot x^\kappa$ for some integer κ, and thus we have an additive term $\mathsf{R}_1 \cdot x^\kappa \cdot \left(t^f_{\leq j}(x) - t^*_{\leq j}(x)\right) \cdot \mathsf{e}_2^*$ in the above expression of $\mathsf{R}_{t^f} \cdot \mathsf{s}_2^f - \mathsf{R}_{t^*} \cdot \mathsf{s}_2^*$. Additionally, the Lemma 2.5 shows that $\mathsf{R}_1 \cdot \mathsf{s}_2^*$ has at least $\Omega(n)$ bits of min-entropy conditioned on a and $\mathsf{a}^\top \cdot \mathsf{R}_1$. Since $t^* \neq t^f$, there is some index $j \in [d]$ such that $\left(t^f_{\leq j}(x) - t^*_{\leq j}(x)\right) \neq 0$, and thus $x^\kappa \cdot \left(t^f_{\leq j}(x) - t^*_{\leq j}(x)\right)$ is invertible from Lemma 2.2. As a result, the ring vector $\mathsf{R}_1 \cdot x^\kappa \cdot \left(t^f_{\leq j}(x) - t^*_{\leq j}(x)\right) \cdot \mathsf{e}^*_2$ also has at least $\Omega(n)$ bits of min-entropy conditioned on a and $\mathsf{a}^\top \cdot \mathsf{R}_1$, and thus $\Pr[\mathsf{s} = 0] \leq 2^{\Omega(-n)} \leq \mathsf{negl}(\lambda)$.

If $\boldsymbol{\mu}^* - \boldsymbol{\mu}^f = 0$, $\mathsf{e}^*_2 - \mathsf{e}^f_2 = 0$, $t^* = t^f$, and $\mathsf{e}^*_1 - \mathsf{e}^f_1 \neq 0$: In this case, it is easy to see that $\mathsf{s} = \mathsf{e}^*_1 - \mathsf{e}^f_1$, and thus obviously $\Pr[\mathsf{s} = 0] \leq 2^{\Omega(-n)} \leq \mathsf{negl}(\lambda)$.

Now we show statement (3): Since μ^*, $\mu^f \in R^k_2$, we have $\|\mu^* - \mu^f\| \leq \sqrt{nk}$. As the challenger randomly sampled $\mathsf{R}_u \leftarrow D_{R^{k\times k},s}$, we have $\|\mathsf{R}_u\| \leq O\left(s \cdot \sqrt{nk}\right)$ from Lemma 2.6. Thus we obtain that $\|\mathsf{R}_u \cdot (\mathsf{e}^f - \mathsf{e}^*)\| \leq O\left(s\sigma \cdot nk\right)$.

From the description of $\mathsf{TrapEval}$, we have that $\|\mathsf{R}_{t^*}\| \leq O\left(n^2 \cdot (kb)^2\right) \cdot \max\{\|\mathsf{R}_0\|, \|\mathsf{R}_1\|\}$. Furthermore, form Lemma 2.6, we have that

$$\|\mathsf{R}_{t^*}\| \leq O\left(n^2 \cdot (kb)^2 \cdot s \cdot \sqrt{nk}\right).$$

We also have the same norm bound for R_{t^f}.

In order to use $\mathsf{SampleRight}$, from Lemma A.13, we need $\sigma > \|\mathsf{R}_{t^*}\| \cdot \sqrt{b^2+1} \cdot \omega(\sqrt{\log(n)})$. Since e^* is a valid signature, thus, except with negligible probability, we have $\|\mathsf{e}^*\| \leq \sigma\sqrt{nk}$ from Lemma 2.1, and thus

$$\|\mathsf{e}^*\| \leq \tilde{O}\left(n^2 \cdot (kb)^2 \cdot s \cdot \sqrt{nk} \cdot \sqrt{b^2+1} \cdot \sqrt{nk}\right)$$

except with negligible probability. We have the same norm bound for the ring vector e^f. From the above arguments and by setting $s = log(n)$, we have the following upper bound for s except with negligible probability.

$$\begin{aligned}
\|\mathsf{s}\| &\leq \|\mathsf{e}^f_1 - \mathsf{e}^*_1\| + \|\mathsf{R}_{t^*} \cdot \mathsf{e}^*_2\| + \|\mathsf{R}_{t^f} \cdot \mathsf{e}^f_2\| + \|\mathsf{R}_u \cdot (\boldsymbol{\mu}^f - \boldsymbol{\mu}^*)\| \\
&\leq \tilde{O}\left(n^2 \cdot (kb)^2 \cdot s \cdot nk \cdot \sqrt{b^2+1}\right) O\left(\cdot n^2 \cdot (kb)^2 \cdot s \cdot \sqrt{nk}\right) \\
&\leq \tilde{O}\left((nkb)^5 \cdot \sqrt{nk}\right) = \beta.
\end{aligned}$$

This completes the proof of lemma. ☐

Analysis of $\mathsf{Sim}_{R\text{-}\mathsf{SIS}}$. It's obvious from the choice of i^* that $2^{i^*} \leq 2^{\frac{eQ2^{\frac{\mathsf{w}+\log(Q)}{\mathsf{L}}}}{\mathsf{L}+1}}$. Let **coll** be the event that there is a $\mathsf{L}+1$-collision occurs for the randomly chosen tags $(t^0, \cdots, t^{Q-1})$, then we have followings

$$\begin{aligned}
\epsilon_{\mathsf{Sim}} &= \Pr\left[\mathsf{a} \cdot \mathsf{s} = 0 \ \wedge\ (\mathsf{s} \neq 0) \ \wedge \|\mathsf{s}\| \leq \beta\right] \\
&\geq \Pr\left[\mathsf{a} \cdot \mathsf{s} = 0 \ \wedge\ (\mathsf{s} \neq 0) \ \wedge \|\mathsf{s}\| \leq \beta \ \wedge\ \overline{\mathbf{coll}}\right] \\
&\geq \Pr\left[\|\mathsf{s}\| \leq \beta | \mathsf{a} \cdot \mathsf{s} = 0 \ \wedge\ (\mathsf{s} \neq 0) \ \wedge\ \overline{\mathbf{coll}}\right] \cdot \Pr\left[\mathsf{s} \neq 0 | \mathsf{a} \cdot \mathsf{s} = 0 \wedge \overline{\mathbf{coll}}\right] \\
&\quad \cdot \Pr\left[\mathsf{a} \cdot \mathsf{s} = 0 | \overline{\mathbf{coll}}\right] \cdot \Pr\left[\overline{\mathbf{coll}}\right] \\
&\geq (1 - \mathsf{negl}(\lambda)) \cdot (1 - \mathsf{negl}(\lambda)) \cdot \frac{1}{2^{i^*}} \cdot \left(\epsilon - \frac{1}{2^{\mathsf{w}}}\right) \\
&\geq \left(\epsilon - \frac{1}{2^{\mathsf{w}}}\right) \cdot \frac{\mathsf{L}+1}{2eQ2^{\frac{\mathsf{w}+\log(Q)}{\mathsf{L}}}} - \mathsf{negl}(\lambda),
\end{aligned}$$

where the third inequality is from Lemma 5.5, Lemma 5.3 and Lemma 5.4; we used the fact $\frac{1}{2^{i^*}} \geq \frac{\mathsf{L}+1}{2eQ2^{\frac{\mathsf{w}+\log(Q)}{\mathsf{L}}}}$ for the last inequality[13]. This completes the proof of the theorem. $\square$

D.7 Parameter Constraints for our signature scheme

Now we present the parameter constraints which ensure our signature scheme $\mathsf{naCMA} \cdot \mathsf{Sign}$ in Construction 1 reaches the correctness and the non-adaptive security. To meet the corresponding requirements, the parameters should satisfy the following conditions:

- To satisfies the requirements of $\mathsf{TrapGen}$ and gadget vector g_b, we need

$$\rho < \frac{1}{2}\sqrt{q/n}, \quad k \geq 2\lceil \log_\rho(q) \rceil, \quad k \geq \lceil \log_b(q) \rceil.$$

- To ensure the Lemma 5.3 and ensure the trinary ring elements are invertible, we need to set d such that $2^d \geq \frac{2Q^2}{\epsilon}$ and $d \leq \frac{n}{2}$, where ϵ is the advantage of the adversary.
- To meet the requirements of worst-case hardness reduction for SIS problem, we need $q \geq \beta \cdot \sqrt{n} \cdot \omega(\sqrt{\log(n)})$.
- To meet the condition of Theorem 5.2, we need β satisfy

$$\beta \geq \sigma \cdot O(\sqrt{nk}) \cdot s_1(\mathsf{R}_{t^*}).$$

- To meet the requirements of $\mathsf{SampleLeft}$ algorithm in Lemma A.13, we need

$$\sigma > \|\mathsf{rot}(\mathsf{T}_\mathsf{a})\|_{\mathsf{GS}} \cdot \sqrt{\log(2n(1 + 1/\epsilon_s))/\pi},$$

  where $\|\mathsf{rot}(\mathsf{T}_\mathsf{a})\|_{\mathsf{GS}} < O\left(b\rho\sqrt{n \log_\rho(q)}\right)$.
- To meet the requirements of $\mathsf{SampleRight}$ algorithm in Lemma A.13, we need

$$\sigma > s_1(\mathsf{R}_t) \cdot \|\mathsf{rot}(\mathsf{T}_{\mathsf{g}_b})\|_{\mathsf{GS}} \cdot \sqrt{\log(2n(1 + 1/\epsilon_s))/\pi}$$

  where $s_1(\mathsf{R}_t)$ satisfies following in the plain model

$$s_1(\mathsf{R}_t) < \tilde{O}\left(d^2 \cdot (nkb)^2\right) \cdot \|\mathsf{R}\|,$$

  or satisfies following in the CRS model

$$s_1(\mathsf{R}_t) < \tilde{O}\left(d^2 n\sqrt{nk} b^2 \cdot k\right) \cdot \|\mathsf{R}\|.$$

  Additionally, $\|\mathsf{R}\| \leq C\rho\sqrt{n}\left(2\sqrt{k} + \omega\right)$ with probability $2e^{-\pi\omega^2}$, and $\|\mathsf{rot}(\mathsf{T}_{\mathsf{g}_b})\|_{\mathsf{GS}} \leq \sqrt{b^2+1}$.

[13] Note that, from the context, the term $\mathsf{negl}(\lambda)$ is statistically small, and thus much smaller than the advantage ϵ of the adversary.

E Supplementary Materials for Section 6

E.1 Parameter Settings for the IBE Scheme in Construction 6.1

To ensure our IBE in Construction 6.1 reaches the correctness and the adaptive security, the parameters should meet the corresponding requirements as follows:

- For the requirements of $\mathsf{TrapGen}$ algorithm and gadget vector g_b, the parameters should holds $\rho < \frac{1}{2}\sqrt{q/n}, \quad k \geq 2\lceil \log_\rho(q) \rceil, \quad k \geq \lceil \log_b(q) \rceil$.
- To ensure the existence of such i^* in the definition of PrtSmp in Theorem 4.1, we need to set d such that $2^d > \frac{2t_{\mathcal{A}}^2}{\epsilon}$, where $t_{\mathcal{A}}$ is the running time of the adversary, and λ is the advantage of the adversary.
- To ensure the hardness of R-DLWE problem, we need σ_0 satisfies the condition of the reduction in Lemma A.12, that is $\sigma_0 \geq n^{3/2}(k+1)^{1/4}\omega(\log^{9/4}(n))$.
- To meet the requirement of ReRand algorithm, the parameters should satisfy: $\sigma_0 \geq \omega\left(\sqrt{\log(n)}\right)$ and $\sigma_1 \geq 2\sigma_0 \cdot \sqrt{s_1^2(\mathsf{R}_{\mathsf{id}^*}) + 1}$.
- To meet the requirement of $\mathsf{SampleLeft}$ algorithm in Lemma A.13, we need $\sigma > \|\mathsf{rot}(\mathsf{T}_{\mathsf{a}})\|_{\mathsf{GS}} \cdot \omega\left(\sqrt{\log(n)}\right)$, where $\|\mathsf{rot}(\mathsf{T}_{\mathsf{a}})\|_{\mathsf{GS}} < O\left(b\rho\sqrt{n \log_\rho(q)}\right)$.
- To meet the requirement of $\mathsf{SampleRight}$ algorithm in Lemma A.13, we need $\sigma > s_1(\mathsf{R}_{\mathsf{id}}) \cdot \|\mathsf{rot}(\mathsf{T}_{\mathsf{g}_b})\|_{\mathsf{GS}} \cdot \omega\left(\sqrt{\log(n)}\right)$, where $s_1(\mathsf{R}_t)$ satisfies following in the plain model $s_1(\mathsf{R}_t) < \tilde{O}\left(d^2 \cdot (nkb)^2\right) \cdot \|\mathsf{R}\|$, or satisfies the following in the CRS model $s_1(\mathsf{R}_t) < \tilde{O}\left(d^2 n\sqrt{nk} b^2 \cdot k\right) \cdot \|\mathsf{R}\|$.

  Additionally, $\|\mathsf{R}\| \leq C\rho\sqrt{n}\left(2\sqrt{k} + \omega\right)$ with probability $2e^{-\pi\omega^2}$, and we have $\|\mathsf{rot}(\mathsf{T}_{\mathsf{g}_b})\|_{\mathsf{GS}} \leq \sqrt{b^2+1}$.
- For the correctness of our IBE scheme, we need the modulus q satisfy $q \geq O\Big(\omega(\log(n)) \cdot \sigma_0 + \sigma\sigma_1 \cdot \omega(\log($
  To achieve overwhelming correctness.

Satisfying the above constraints, We set $b = 2$ and $\rho = 1$, and set other scheme parameters as follows:

$$
\begin{aligned}
&k = O(\log(n)) \qquad && d = \omega(\log(n)) \quad && \sigma_0 = \tilde{O}(n^{1.5}) \\
&\sigma_1 = \tilde{O}(n^4) && \sigma = \tilde{O}(n^{2.5}) && q = \tilde{O}(n^7)
\end{aligned}
$$

Below, we present the correctness and security of our IBE scheme under the above parameter setting.

E.2 Proof of Theorem 6.2

Proof. From the description of the encryption algorithm, we know that

$$\begin{aligned} c_0 - \mathsf{e} \cdot \mathsf{c}_1 =& u \cdot s + e_0 + \left\lceil \frac{q}{2} \right\rceil \cdot \mu - \mathsf{e}^\top \cdot \left(\begin{bmatrix} \mathsf{a} \\ \mathsf{b}_{\mathsf{id}} \end{bmatrix} \cdot s - \begin{bmatrix} \mathsf{e}_1 \\ \mathsf{e}_2 \end{bmatrix} \right) \\ =& \left\lceil \frac{q}{2} \right\rceil \cdot \mu + e_0 - \mathsf{e}^\top \cdot \begin{bmatrix} \mathsf{e}_1 \\ \mathsf{e}_2 \end{bmatrix} \end{aligned}$$

Note that e_0 is sampled from the gaussian distribution $D^{\mathsf{Coeffs}}_{R_q,\sigma_0}$, thus from the gaussian tail bound (Lemma 2.1), we have that $\mathsf{Pr}[\|e_0\|_\infty \leq \omega(\log(n)) \cdot \sigma_0] \leq \mathsf{negl}(\lambda)$. Furthermore, we have followings:

$$\begin{aligned} \left\| \mathsf{e}^\top \cdot \begin{bmatrix} \mathsf{e}_1 \\ \mathsf{e}_2 \end{bmatrix} \right\|_\infty &= \left\| \mathsf{Coeffs}(\mathsf{e}^\top) \cdot \mathsf{rot}\left(\begin{bmatrix} \mathsf{e}_1 \\ \mathsf{e}_2 \end{bmatrix} \right) \right\|_\infty = \max_{j \in [n]} \left\| \mathsf{Coeffs}(\mathsf{e}^\top) \cdot \mathsf{rot}\left(\begin{bmatrix} \mathsf{e}_1 \\ \mathsf{e}_2 \end{bmatrix} \right)_j \right\|_\infty \\ &\leq \max_{j \in [n]} \|\mathsf{Coeffs}(\mathsf{e}^\top)\| \cdot \left\| \mathsf{rot}\left(\begin{bmatrix} \mathsf{e}_1 \\ \mathsf{e}_2 \end{bmatrix} \right)_j \right\|_\infty \\ &\leq \sqrt{2nk}\sigma \cdot \omega(\log(n)) \cdot \sigma_1, \end{aligned}$$

where the last inequality holds except with negligible probability by Lemma 2.1. Thus from the above results, we have that

$$\mathsf{Pr}\left[\left\| e_0 - \mathsf{e}^\top \cdot \begin{bmatrix} \mathsf{e}_1 \\ \mathsf{e}_2 \end{bmatrix} \right\|_\infty \leq \sqrt{n}\sigma_0 + \sigma\sigma_1 \cdot \omega(\log(n)) \cdot \sqrt{2nk} \right] \leq \mathsf{negl}(\lambda).$$

Thus from the description of the decryption algorithm, it decrypts correctly except with negligible probability, and thus the IBE scheme is correct.

□

E.3 Proof of Theorem 6.3

Proof. To show the theorem, we prove that if there is an adversary $\mathcal{A}$ who breaks the IBE scheme in the Construction 6.1 with noticeable probability ϵ in a polynomial running time of $t_{\mathcal{A}}$, then there is an algorithm $\mathsf{Sim}_{R\text{-}\mathsf{DLWE}}$ which invokes $\mathcal{A}$ to solve the $R\text{-}\mathsf{DLWE}_{n,k+1,q,\sigma_0}$ problem with noticeable probability in polynomial time as well. This contradicts to the theorem assumption, and thus the theorem follows. Before presenting the concrete description of $\mathsf{Sim}_{R\text{-}\mathsf{DLWE}}$, we first define and analysis follows hybrid games, from which the design idea of $\mathsf{Sim}_{R\text{-}\mathsf{DLWE}}$ can be easily seen. The hybrid games are defined as follows.

Game_0 : This is the original adaptive IBE security game between the adversary $\mathcal{A}$ and the challenger Chall.

Game_1 : In this game, we change the Game_0 so that the challenger Chall procedure an conditional abort at the end of the game. Namely, this game is identical to the Game_0 except that at the end of the game, the challenger runs $\mathsf{PrtSmp}(1^\lambda, t_{\mathcal{A}}, \epsilon) \to K = (t^*, i^*)$ and *abort* the game if the following condition holds;

$$\bigwedge_{i=1}^{Q} \left(F\left(K, id^i\right) \in R_q^*\right) \bigwedge F\left(K, id^*\right) = 0,$$

where Q is the upper bound of the number of identities that the adversary asked for key extraction queries, and $\mathsf{id}^1, \cdots, \mathsf{id}^Q$ are the identities for which $\mathcal{A}$ has made key extraction queries, and id^* is the challenge identity.

Game_2 : In this game, we change the way that the ring vectors $\mathsf{b}_0, \mathsf{b}_1$ are generated. Here, instead of sampling $\mathsf{b}_0, \mathsf{b}_1$ uniformly over R_q^k, the challenger first samples $\mathsf{R}_0, \mathsf{R}_1 \leftarrow R_{[-\rho,\rho]}^{k\times k}$, and sets

$$\mathsf{b}_0 := \mathsf{a}^\top \cdot \mathsf{R}_0 + t^*_{\leq i^*}(x) \cdot \mathsf{g}_b,\ \mathsf{b}_1 := \mathsf{a}^\top \cdot \mathsf{R}_1 + x^{i^*} \cdot \mathsf{g}_b,$$

Game_3 : This game is the same as Game_2 except that we change way of generating public ring vector a and the key extraction procedure. Here, the challenger samples $\mathsf{a} \xleftarrow{\$} R_q^k$ uniformly instead of generating it with trapdoor by running the trapdoor generating algorithm $\mathsf{TrapGen}$. To extract a secret key for an identity id, the challenger first computes

$$\mathsf{R}_{\mathsf{id}} = \mathsf{TrapEval}_{\mathsf{id}}\left(\mathsf{R}_0, \mathsf{R}_1, \mathsf{id}\right).$$

Then sample a ring vector $\mathsf{e}_{\mathsf{id}} \in R_q^{2k}$ by running $\mathsf{SampleRight}$ as

$$\mathsf{e}_{\mathsf{id}} = \mathsf{SampleRight}\left(\mathsf{a}, \mathsf{R}_{\mathsf{id}}, u, F_{\mathsf{id}}(t^*, i^*), \mathsf{g}_b, \mathsf{T}_{\mathsf{g}_b}, \sigma\right).$$

Finally, the challenger outputs e_{id} as a corresponding key for the identity id.

Game_4 : In this game, we change the way that the challenge ciphertext $\mathsf{ct}^* = (c_0^*, \mathsf{c}_1^*)$ is generated. Here, the challenger first selects $s \xleftarrow{\$} R_q$ and $\mathsf{x} \leftarrow D^{\mathsf{Coeffs}}_{R^k,\sigma_0}$, and compute $\mathsf{v} = \mathsf{a} \cdot s + \mathsf{x}$. Then the challenger samples $e_0 \leftarrow D^{\mathsf{Coeffs}}_{R,\sigma_0}$, and generates the challenge ciphertext as follows:

$$c_0^* = u \cdot s + e_0 + \left\lceil \frac{q}{2} \right\rceil \cdot \mu,\ ,\ \mathsf{c}_1^* = \mathsf{ReRand}\left([I_k | \mathsf{R}_{\mathsf{id}^*}]^\top, \mathsf{v}, \sigma_0, \frac{\sigma_1}{2\sigma_0}\right),$$

where I_k is the identity matrix.

Game_5 : In this game, we change the way that the challenge ciphertext $\mathsf{ct}^* = (c_0^*, \mathsf{c}_1^*)$ is generated. Here, instead of masking the message with LWE samples, the challenger masks the message with uniformly random samples. Formally, the challenger first samples $u_0 \leftarrow R_q$, $\mathsf{v}' \leftarrow R_q^k$ and $\mathsf{x} \leftarrow D^{\mathsf{Coeffs}}_{R^k,\sigma_0}$, then set $\mathsf{v} = \mathsf{v}' + x$ and compute the challenger ciphertext as follows:

$$c_0^* = u_0 + \left\lceil \frac{q}{2} \right\rceil \cdot \mu,\ ,\ \mathsf{c}_1^* = \mathsf{ReRand}\left([I_k | \mathsf{R}_{\mathsf{id}^*}]^\top, \mathsf{v}, \sigma_0, \frac{\sigma_1}{2\sigma_0}\right),$$

For $i \in \{0,1,2,3,4,5\}$, let E_1 be the event that the adversary $\mathcal{A}$ wins in Game_i, and let abort be the event that challenger aborts in Game_1. From the assumption, we have that $\mathsf{Pr}[E_0] = \epsilon$. Additionally, from the description of Game_5, it is easy to see that c_0 is independent of c_1, and from the uniformity of u_0, the advantage of any adversary in Game_5 is exactly 0, that is $\mathsf{Pr}[E_5] = 0$.

Lemma E.1 $\mathsf{Pr}[E_1]$ *is noticeable if* ϵ *is noticeable.*

Proof. Since the $\mathcal{A}$'s advantage ϵ is noticeable and the running time $T_{\mathcal{A}}$ is polynomially bounded, the Lemma A.10 and the second property of the partition function implies that

$$\mathsf{Pr}[E_1] \geq \gamma_{\min} \cdot \epsilon - \frac{1}{2}\left(\gamma_{\max} - \gamma_{\min}\right) \geq \frac{\epsilon^2}{8t_{\mathcal{A}}}$$

, and the probability is noticeable. This completes the proof of LemmaE.1 ☐

Lemma E.2 $|\mathsf{Pr}[E_2] - \mathsf{Pr}[E_1]| \leq \mathsf{negl}(\lambda)$

Proof. Recall that the difference between the Game_1 and Game_2 is the way that they generate the public ring vectors b_0 and b_1. Namely, the two games generate the public ring vectors as follows:

$$\begin{aligned}&\mathsf{Game}_1 : \left(\mathsf{b}_0 \leftarrow R_q^k, \mathsf{b}_1 \leftarrow R_q^k\right)\\ &\mathsf{Game}_2 : \left(\mathsf{b}_0 = \mathsf{a}^\top \cdot \mathsf{R}_0 + t^*_{\leq i^*}(x) \cdot \mathsf{g}_b, \mathsf{b}_1 = \mathsf{a}^\top \cdot \mathsf{R}_1 + x^{i^*} \cdot \mathsf{g}_b\right)\end{aligned}$$

where $R_0, R_1 \leftarrow R^{k\times k}_{-\rho,\rho}$. Thus Lemma 2.7 shows that the difference between the above two games is negligible. This completes the proof of Lemma E.2. ☐

Lemma E.3 $|\mathsf{Pr}[E_3] - \mathsf{Pr}[E_2]| \leq \mathsf{negl}(\lambda)$

Proof. Note that, in Game_3, the challenger changed the way of generating the public ring vector a and the key extraction procedure. So, to show this lemma, it suffices to show that (1) the difference between the generating procedure of a is negligible, and (2) the difference between the key extraction procedure is negligible. Thus, applying the probability union bound, we complete the proof of this lemma.

Now we show the statement (1): Note that public ring vector a is sampled uniformly from the ring R_q^k in Game_3 instead of by running $\mathsf{TrapGen}$ as in Game_2. From the first property of Lemma A.13, the difference between the two cases is negligible in security parameter λ.

Next we show the statement (2):Note that in Game_2, the challenger uses the trapdoor information of a to extract the identity key by running $\mathsf{SampleLeft}$ algorithm of Lemma A.13 . However, in Game_3, the challenger has no trapdoor information about the ring vector a, yet it can use R_{id} to generate the corresponding secret identity key by running the $\mathsf{SampleRight}$ algorithm of Lemma A.13. The second and third property of Lemma A.13 shows that the outputs of above two sampling algorithms are statistically close to $D_{[\mathsf{a}^\top|\mathsf{b}_{\mathsf{id}}^\top],u,\sigma}$. Thus the difference between the above two procedure of generating identity keys is negligible.

☐

Lemma E.4 $|\mathsf{Pr}[E_4] - \mathsf{Pr}[E_3]| \leq \mathsf{negl}(\lambda)$

Proof. Note that the only difference between the Game_3 and Game_4 is the way the challenger generates the challenge ciphertext $\mathsf{ct} = (c_0, \mathsf{c}_1$. In Game_3, the challenge ciphertext is generated by the encryption algorithm, while in $game_4$, c_0 is generated by the same way as in Game_3, yet $\mathsf{c}_1 = \mathsf{ReRand}\left([I_k|\mathsf{R}_{\mathsf{id}^*}, \mathsf{v}, \sigma_0, \frac{\sigma}{2\sigma_0}]\right)$. From Lemma A.11, we have that the difference between the above two games is negligible.

□

Lemma E.5 $|\mathsf{Pr}[E_5] - \mathsf{Pr}[E_4]| \leq \mathsf{negl}(\lambda)$ *if the R-$\mathsf{DLWE}_{q,n,\sigma_0,k,k+1}$ problem is hard.*

Proof. To show this lemma, we construct a simulation algorithm $\mathsf{Sim}_{R\text{-}\mathsf{DLWE}}$ (which is desired in the Theorem 6.3) which tries to solve the R-DLWE problem by invoking an algorithm $\mathcal{B}$ who tries to distinguish Game_4 and Game_5 . The description if $\mathsf{Sim}_{R\text{-}\mathsf{DLWE}}$ is as follows:

$\mathsf{Sim}_{R\text{-}\mathsf{DLWE}}\left((\mathsf{a}^\top|u), (\mathsf{b}^\top, b_0)\right) \to \{0,1\}$:

On input the $(k+1)$ R-$\mathsf{DLWE}_{n,q,k_1,\sigma_0}$ challenge samples $\left([\mathsf{a}\top|u], [\mathsf{b}^\top, b_0]\right)$, algorithm $\mathsf{Sim}_{R\text{-}\mathsf{DLWE}}$ simulates the Game_4 for $\mathcal{B}$ except the challenge ciphertext respond. It first lets a as the part of the public key, and generates the two ring vectors $\mathsf{b}_0, \mathsf{b}_1$ as in Game_4, and let $\mathsf{mpk} = (\mathsf{a}, \mathsf{b}_0, \mathsf{b}_1, u)$ as the master public key of the IBE scheme. The identity key extraction procedure is the same as in Game_4, yet it generates the challenge ciphertext as follows:

$$c_0^* = b_0 + \lceil \frac{q}{2} \rceil \cdot \mu \ , \ \mathsf{c}_1^* = \mathsf{ReRand}\left([I_k|\mathsf{R}_{\mathsf{id}^*}], \mathsf{b}, \sigma_0, \frac{\sigma}{2\sigma_0}\right).$$

If the algorithm $\mathcal{B}$ outputs 1, meaning that the simulated game is Game_4, then the algorithm $\mathsf{Sim}_{R\text{-}\mathsf{DLWE}}$ outputs 1 implies the challenge samples are from $\mathsf{Sim}_{R\text{-}\mathsf{DLWE}}$ distribution. If $\mathcal{B}$ outputs 0, meaning that the simulated game is Game_5, then the algorithm $\mathsf{Sim}_{R\text{-}\mathsf{DLWE}}$ outputs 0 implies the challenge samples are from uniform distribution.

It is not hard to see that if the given challenge samples $\left([\mathsf{a}\top|u], \mathsf{b}^\top, b_0\right)$ are from R-DLWE distribution, then the algorithm $\mathsf{Sim}_{R\text{-}\mathsf{DLWE}}$ exactly simulates the Game_4; if the given challenge samples are from uniform distribution, then the algorithm $\mathsf{Sim}_{R\text{-}\mathsf{DLWE}}$ exactly simulates the Game_5. Thus the advantage of the simulation algorithm $\mathsf{Sim}_{R\text{-}\mathsf{DLWE}}$ is exactly the advantage of the distinguishing advantage of $\mathcal{B}$. From the hardness assumption of R-DLWE problem, advantage of $\mathsf{Sim}_{R\text{-}\mathsf{DLWE}}$ is negligible, and thus the distinguishing advantage of any algorithm $\mathcal{B}$ is negligible. This completes the proof of Lemma E.5 □

Put all together. Combining the above results, we have $\mathsf{Pr}[E_5] = 0$ and

$$\begin{aligned}\mathsf{Pr}[E_5] &= \sum_{i=1}^{4}(\mathsf{Pr}[E_{i+1}] - \mathsf{Pr}[E_i]) + \mathsf{Pr}[E_1] \geq \mathsf{Pr}[E_1] - \sum_{i=1}^{4}|(\mathsf{Pr}[E_{i+1}] - \mathsf{Pr}[E_i])| \\ &\geq \mathsf{Pr}[E_1] - \mathsf{negl}(\lambda) \geq \frac{\epsilon^2}{8T_{\mathcal{A}}^2} - \mathsf{negl}(\lambda),\end{aligned}$$

Which is contradicts to the assumption that ϵ is noticeable and $T_{\mathcal{A}}$ is polynomially bounded. This completes the proof of Theorem 6.3.

$\square$